Reasoning Without Certainty

An Interactive Exploration of Computational Argumentation



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Inria UNIVERSITÉ CÔTE D'AZUR



Online Handbook of Argumentation for AI

Thanks to Andreas Xydis, Daphne Odekerken, Madeleine Waller for contributions to the slides!

Overview

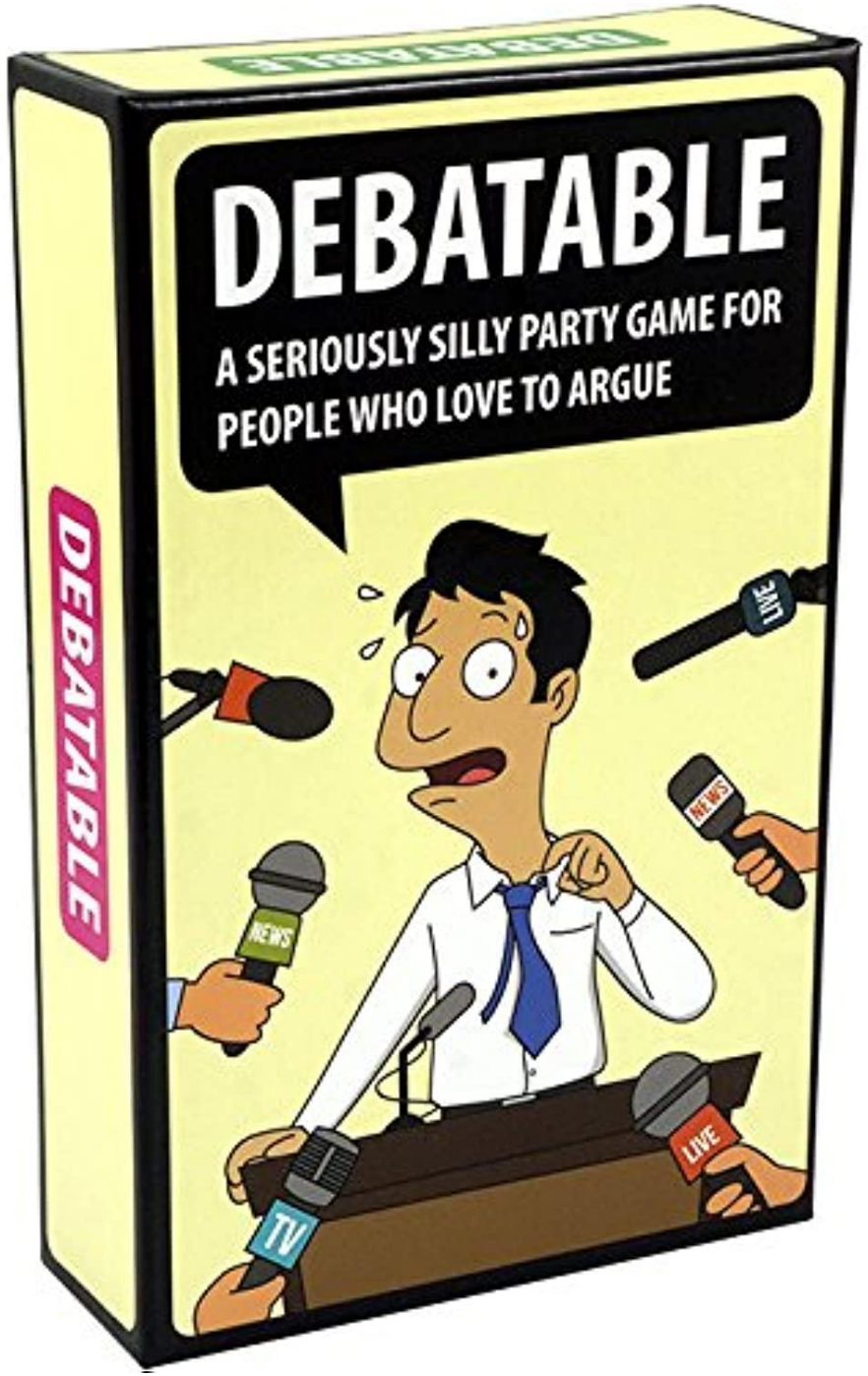
- Introduction to Argumentation
- Argumentation theory
- Abstract argumentation & variations
- Structured argumentation
- Enthymemes
- Applications of argumentation

What is argumentation?

https://www.menti.com/alsi3h2j2nab

Which words come to mind when you think of argumentation?

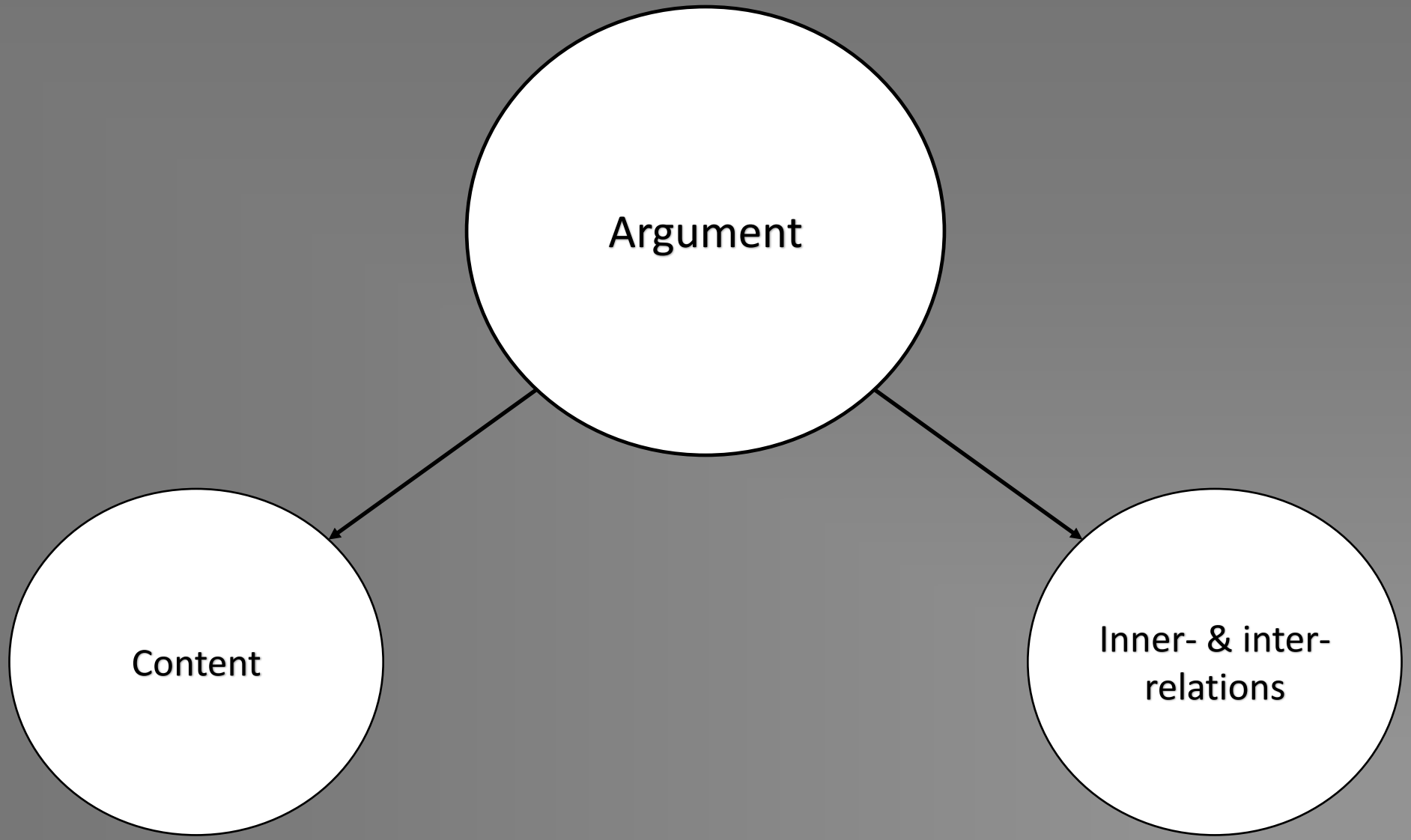


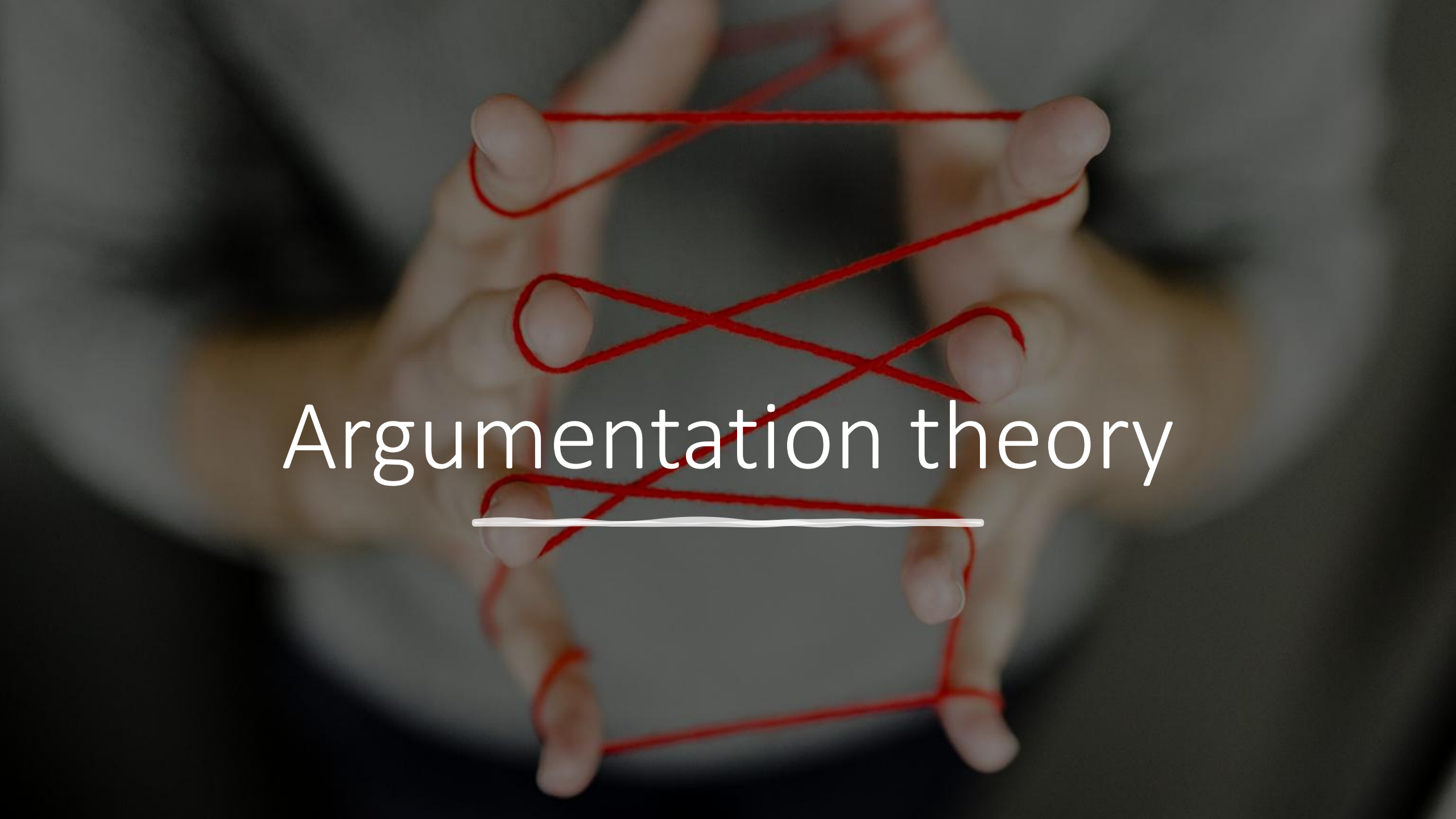


Let's play!

- Get into groups
- Alphabetically, by name:
  1. YES
  2. MODERATOR
  3. NO

1. Moderator:
   - Chairs debate
   - Decides winning side
2. First round:
   - 2 min to prep
   - 2 min to debate it
3. Second round:
   - Switch roles
   - 2min **per side**

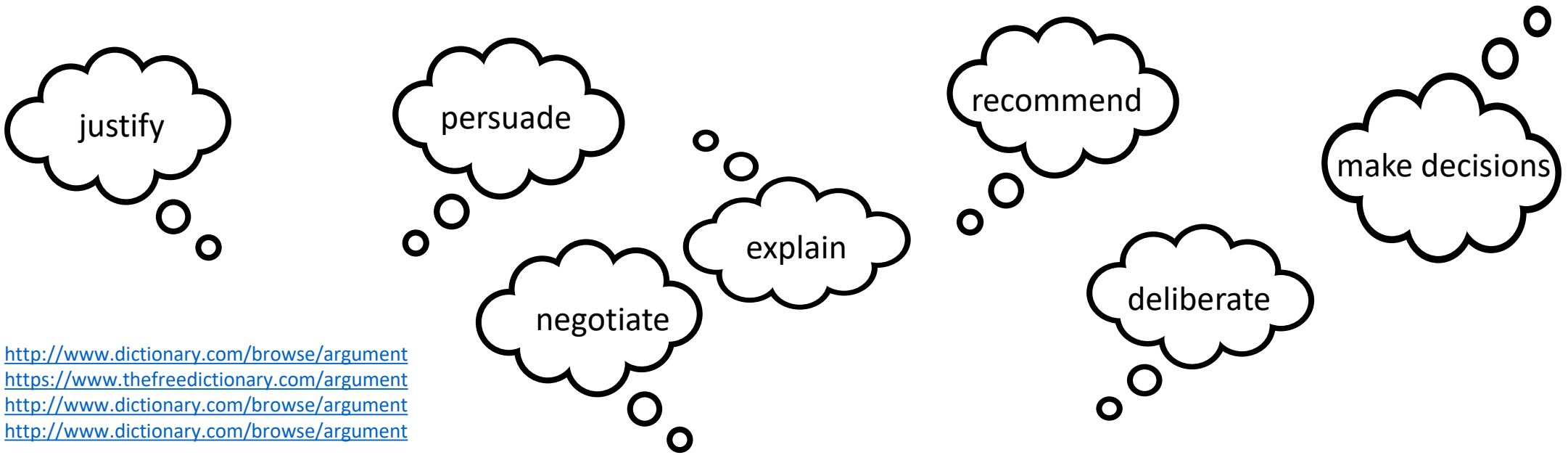

Argument
Content
Inner- & inter-relations



Argumentation theory

What is an argument? Why do we argue?

- "A statement, reason, or fact for or against a point" [1]
- "A course of reasoning aimed at demonstrating truth or falsehood" [2]
- "A discussion involving differing points of view" [3]
- "An address or composition intended to convince or persuade" [4]



1. http://www.dictionary.com/browse/argument
2. https://www.thefreedictionary.com/argument
3. http://www.dictionary.com/browse/argument
4. http://www.dictionary.com/browse/argument

Internal Reasoning

- Information processing
- Reasoning about beliefs, goals, intentions





I will go left because it is the fastest route.

p: Left is the fastest route.
c: I will go left.
r: I want to take the fastest route.

Commonsense reasoning: defeasible

- Inconsistent information
- Knowledge often uncertain or incomplete:
  - conclusions under certain assumptions
  - retract conclusions once learn an assumption is unwarranted

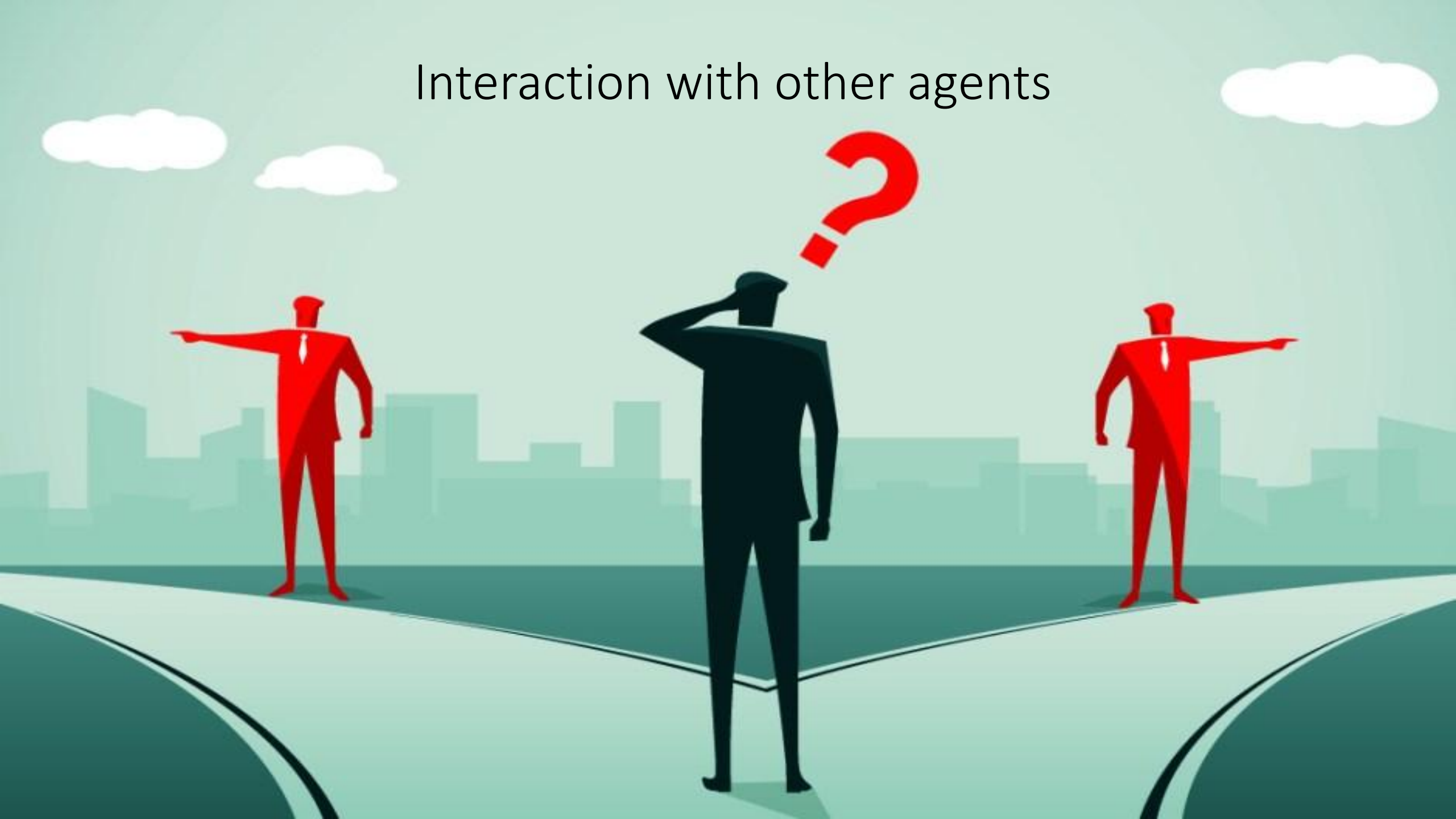
→Non monotonic logic





Actually, today, I will go right because there is an obstacle on the left.

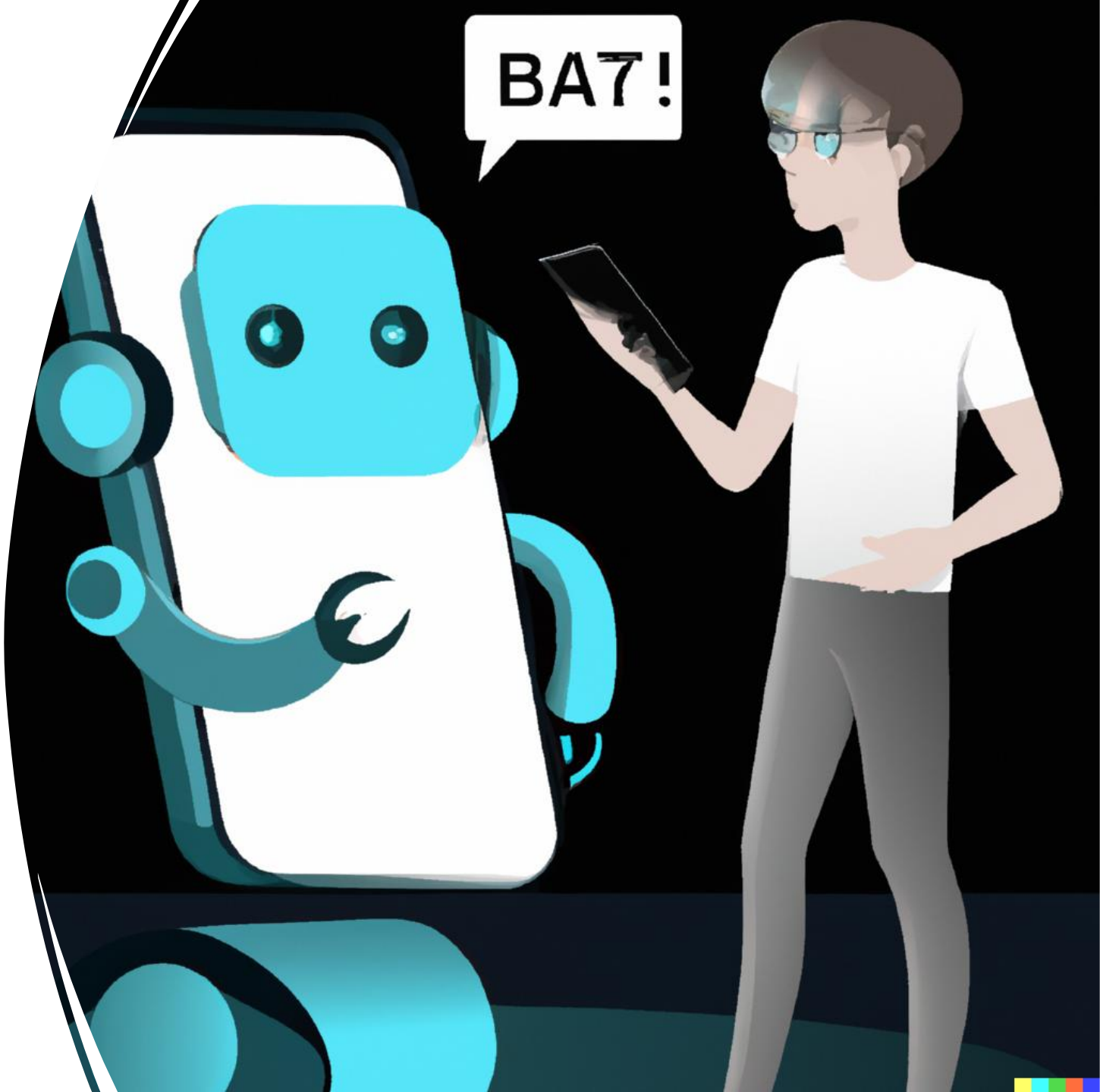
Interaction with other agents



Dialogue

- Tool of interaction & communication
- Enables understanding of both parties involved
  - Information
  - Reasoning exploration

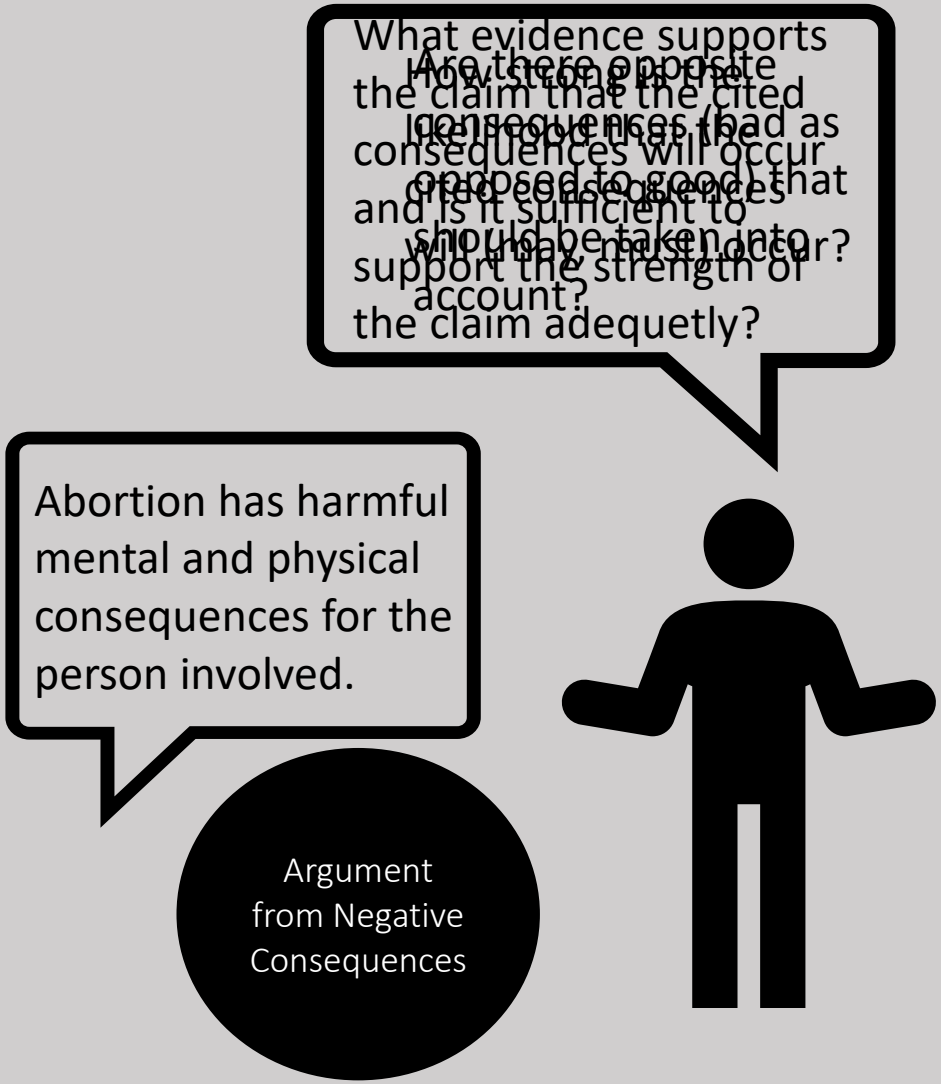
☞ Joint reasoning





Argument Schemes: guiding dialogue

Critical Questions

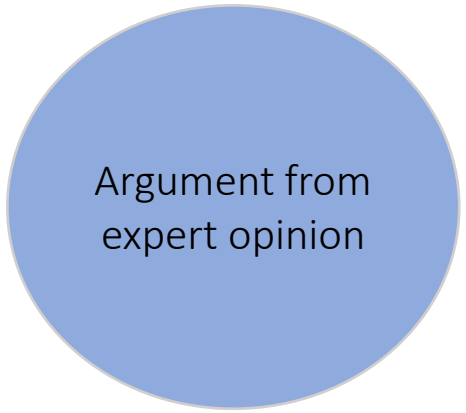


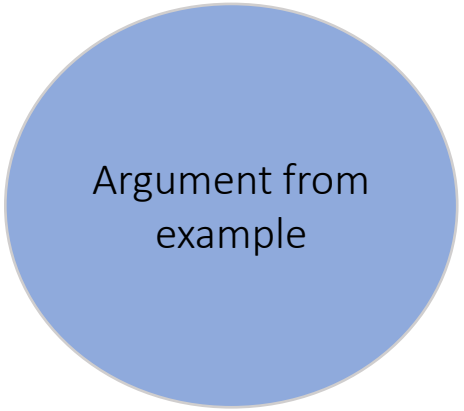
Walton's Argument Schemes

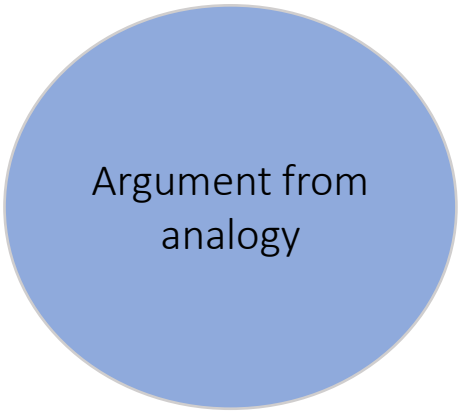
Access to legal abortion improves the health and safety of pregnant people so pregnant people should have the right to choose abortion.

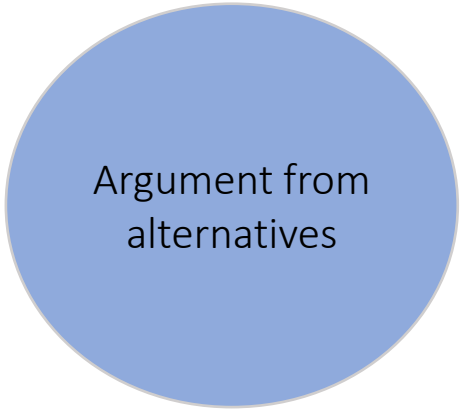
<u>Argument from Positive Consequences</u>

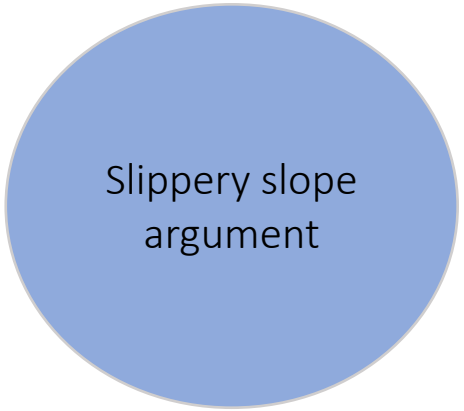
- Premise : If *A* is brought about, good consequences will occur.
- Conclusion : Therefore *A* should be brought about.

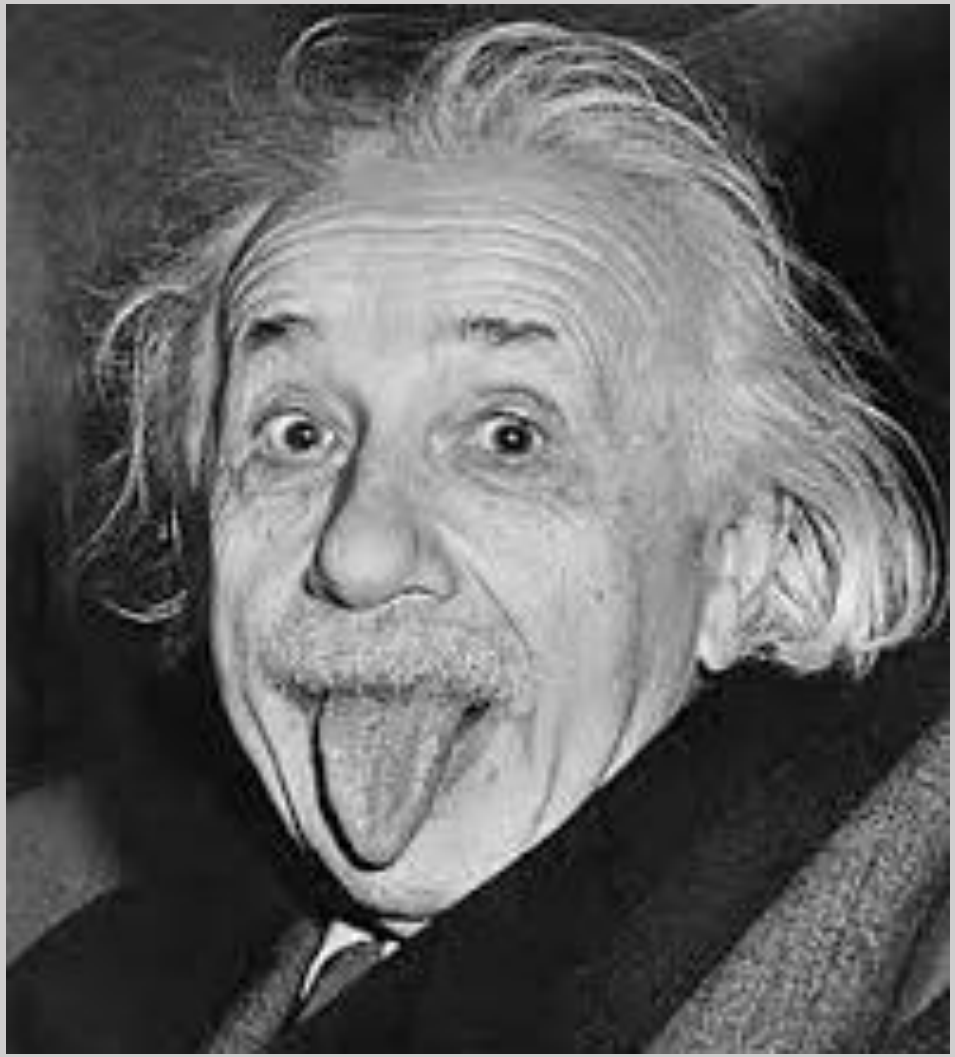

Argument from expert opinion


Argument from example


Argument from analogy


Argument from alternatives


Slippery slope argument



Argument from expert opinion

CQ1: How credible is E as an expert source?

CQ2: Is E an expert in the field that A is in?

CQ3: What did E assert that implies A?

CQ4: Is E personally reliable as a source?

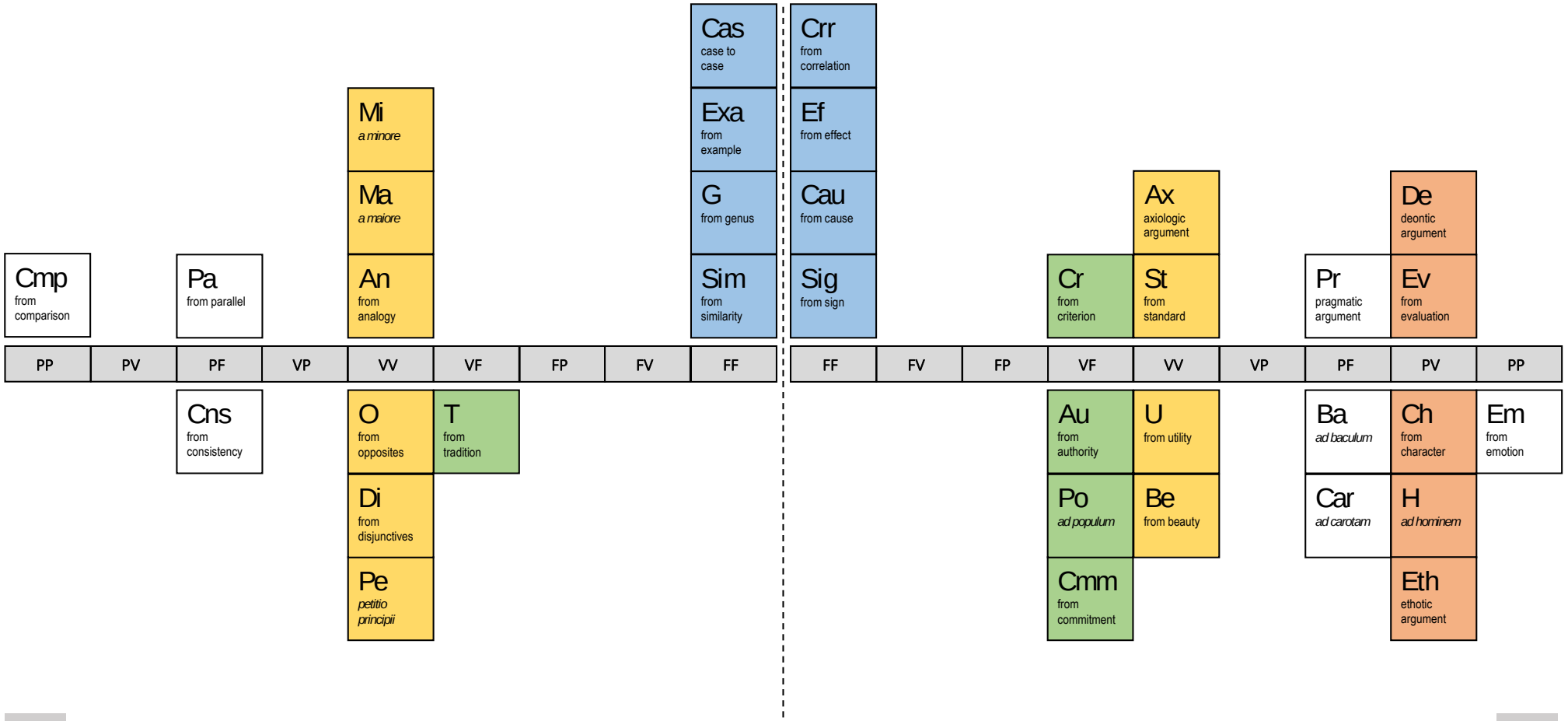
CQ5: Is A consistent with what other experts assert?

CQ6: Is E's assertion based on evidence?

PERIODIC TABLE OF ARGUMENTS

β subject 1

predicate α 1



2 γ subject

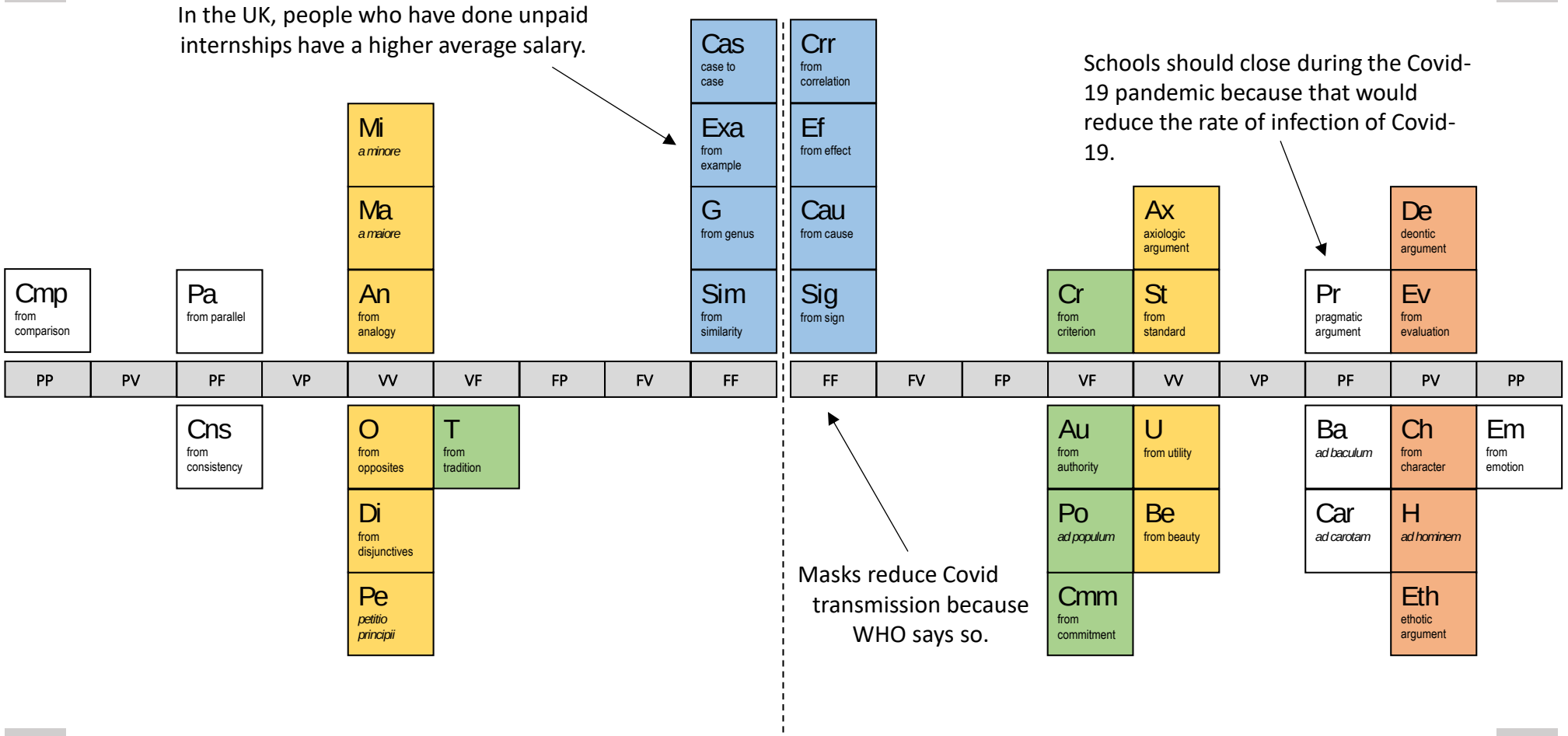
2 predicate δ

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PERIODIC TABLE OF ARGUMENTS

β subject 1

predicate α 1

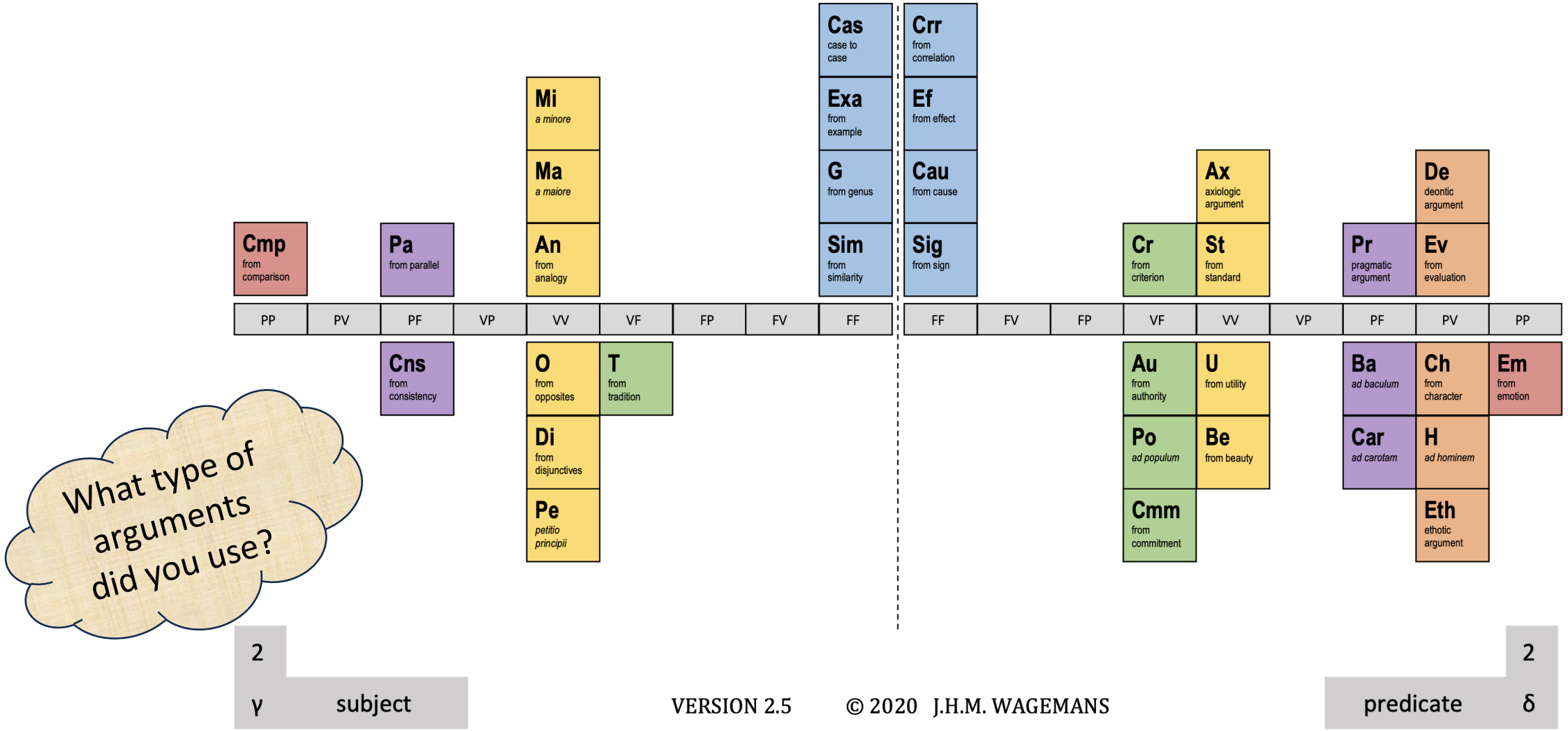


2 γ subject

2 predicate δ

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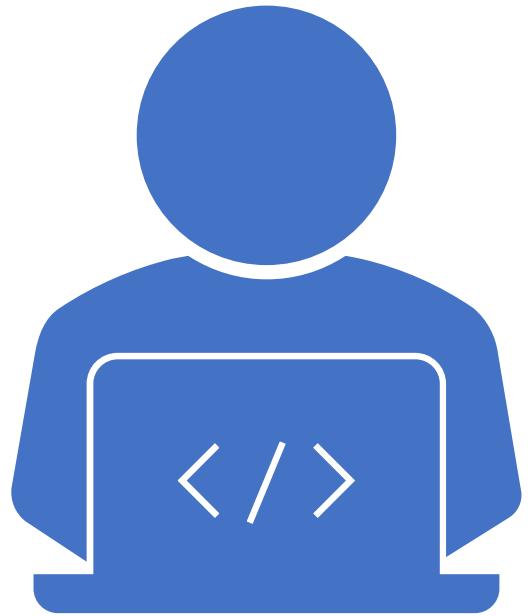
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For more info please visit www.periodic-table-of-arguments.org

Formalising Argumentation

What is Computational Argumentation?

- Formalisation of argumentation theory
- Can support human-computer connections and computer-computer connections
- Applications include:
  - providing reasoning and explaining decision-making
  - natural language processing and generation tasks



Abstract Argumentation

Disregards the internal structure of arguments and focusses on acceptability conditions that allow certain sets of arguments to co-exist in a rational manner based on a **given attack relationship between arguments**.

(P. M. Dung, 1995)

Should I go right or left?



Argument 1 (A1)

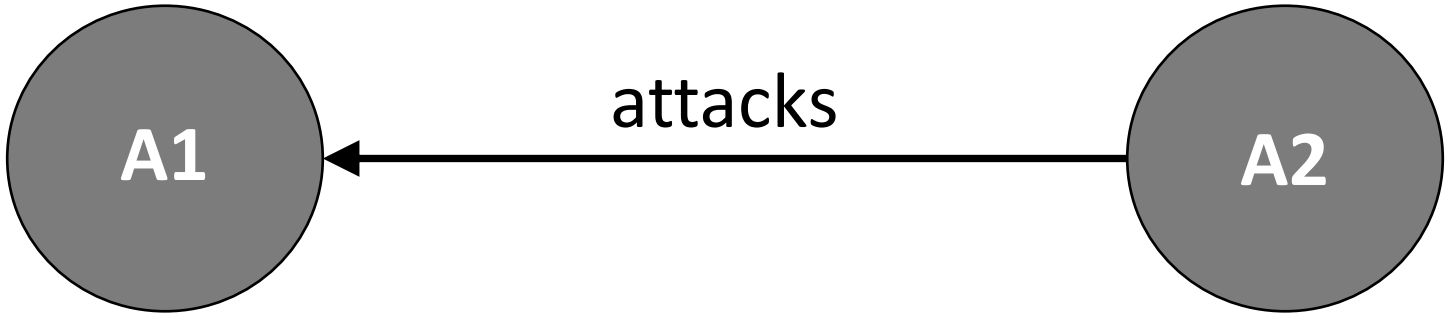
Going left is the fastest route,

therefore I should go left

Argument 2 (A2)

Today there is an obstacle to the left,

therefore I should go right



[1] Phan Minh Dung (1995). "On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming, and n–person games". *Artificial Intelligence*. **77** (2): 321–357.

Has social media been good for humanity?



A1: Social media has been good for humanity



A2: Social media has not been good for humanity



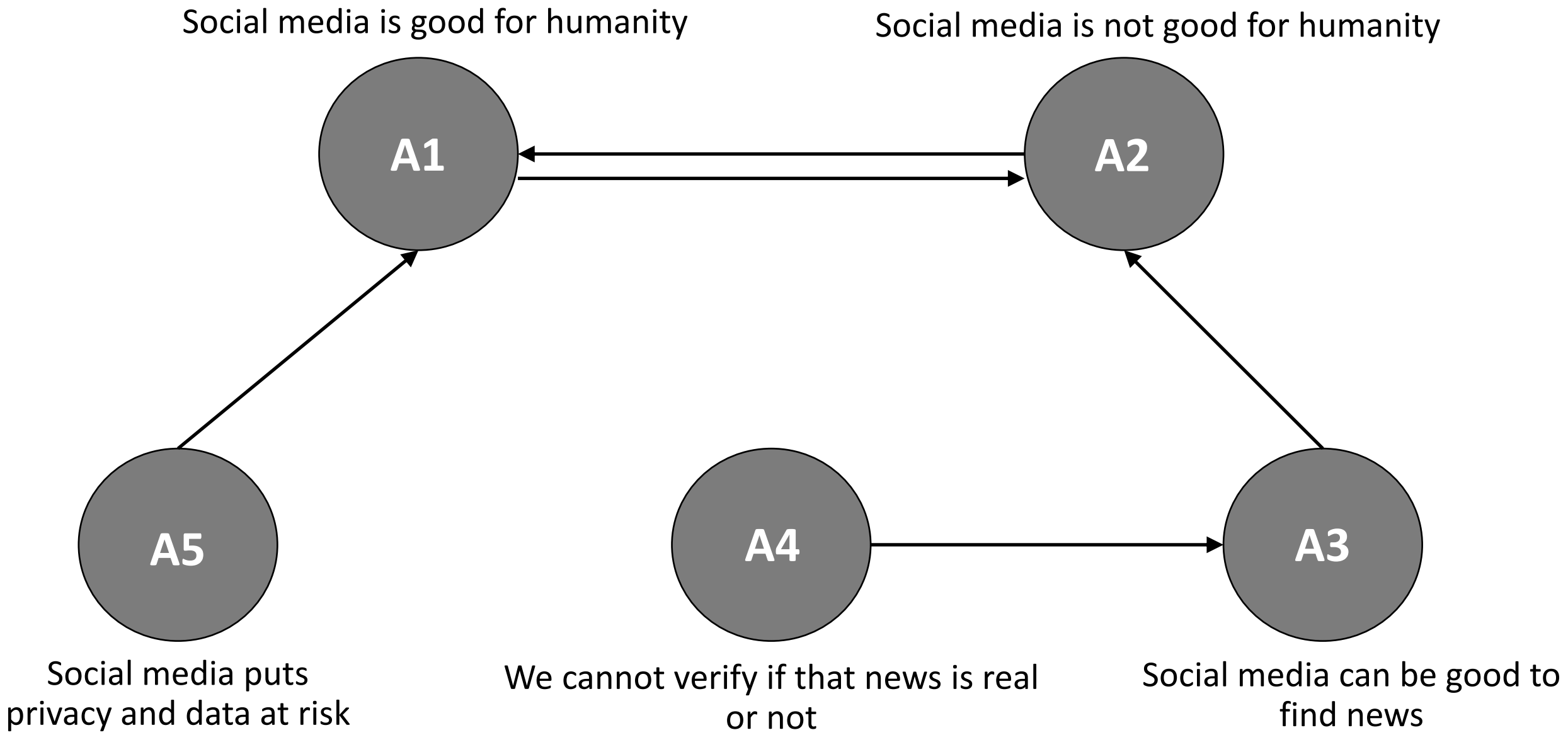
A3: Social media can be good to find news

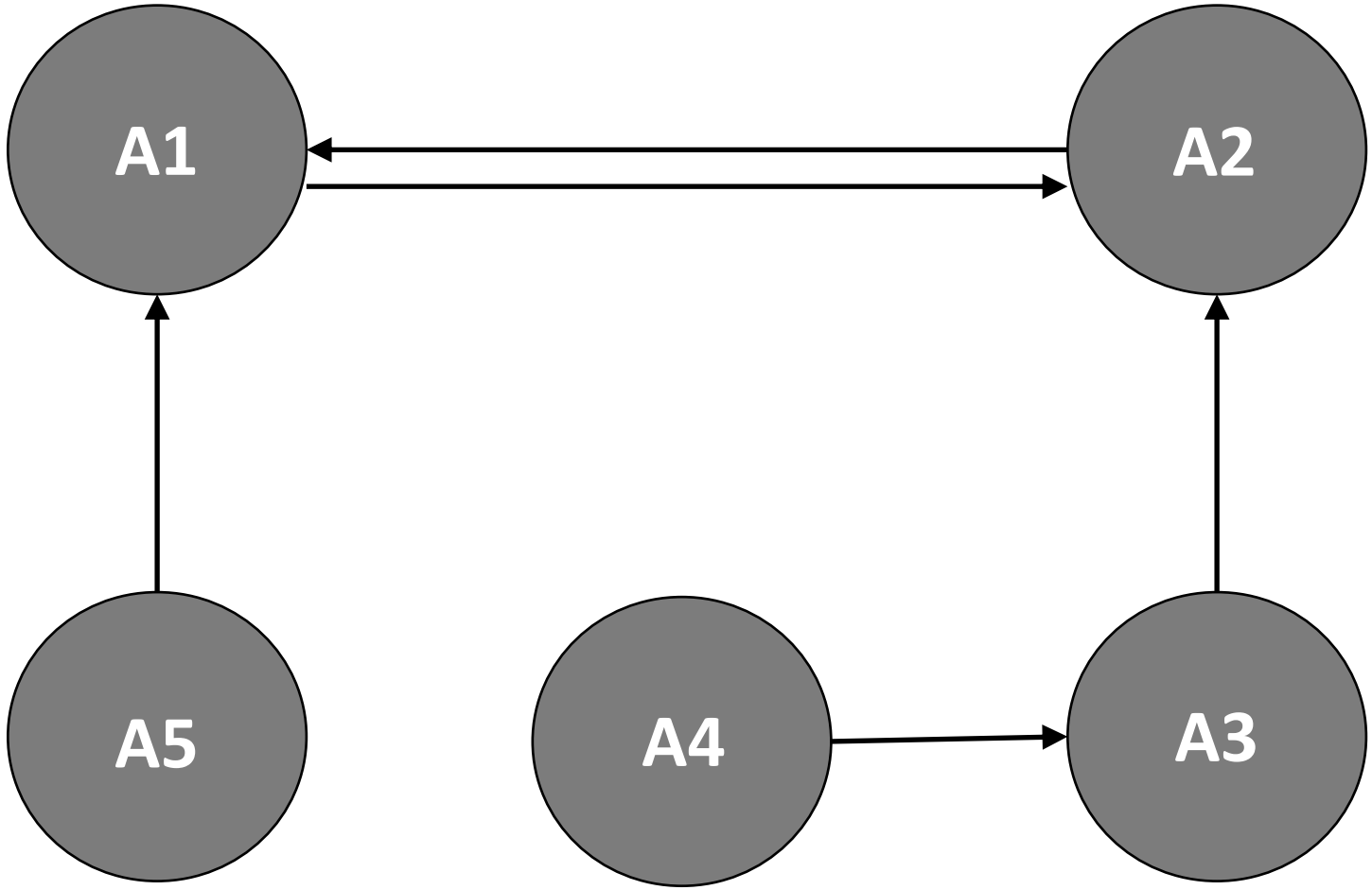


A4: We cannot verify if that news is real or not



A5: Social media puts privacy and data at risk


Social media is good for humanity
Social media is not good for humanity
A1
A2
A5
A4
A3
Social media puts privacy and data at risk
We cannot verify if that news is real or not
Social media can be good to find news


A1
A2
A5
A4
A3

Quest Time!

Create your own abstract argumentation graph

- Choose one of the debates you had before
- Extract the key arguments
- Think about how they attack each other
- Draw a graph of arguments and attacks

Abstract Argumentation

Disregards the internal structure of arguments and **focusses on acceptability conditions** that allow certain sets of arguments to co-exist in a rational manner based on a given attack relationship between arguments.

(P. M. Dung, 1995)



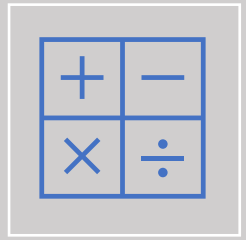
Label-based

Labellings of arguments in the graphs that have specific properties



Extension-based

Subsets of arguments in the graphs that have specific properties



Equations

Solution to a defined set of equations representing the interactions between arguments

Label-based semantics

IN if all its attackers are OUT (or no attackers)

OUT if it has an attacker that is IN

UNDEC if not all its attackers are OUT/at least one of its attackers is UNDEC and it does not have an attacker that is in

IN if all its attackers are out

OUT if it has an attacker that is in

UNDEC if not all its attackers are out
and it does not have an attacker that is in

Argument 1 (A1)

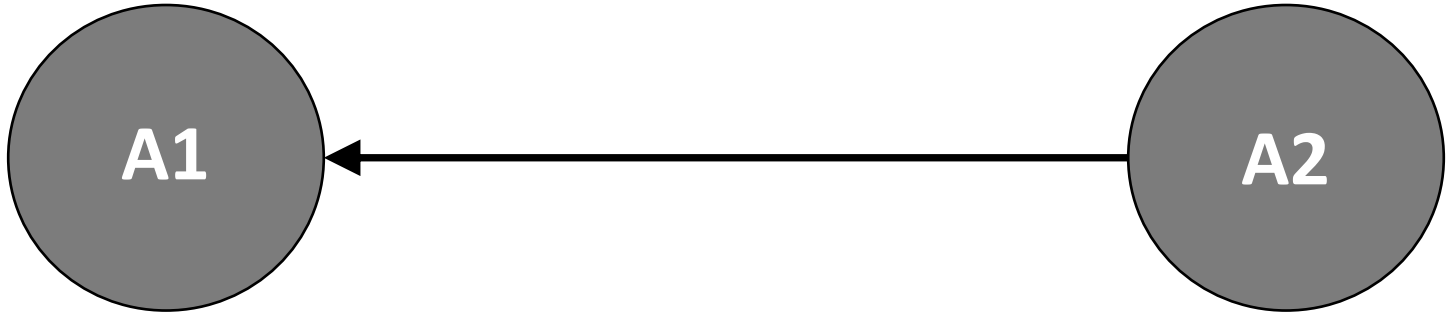
Going left is the fastest route,

therefore I should go left

Argument 2 (A2)

Today there is an obstacle to the left,

therefore I should go right



IN if all its attackers are out

OUT if it has an attacker that is in

UNDEC if not all its attackers are out
and it does not have an attacker that is in

Argument 1 (A1)

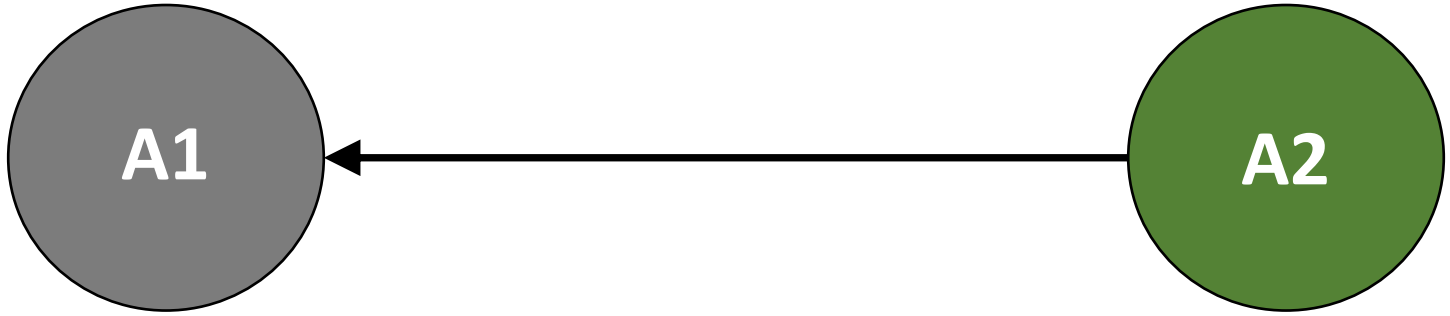
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IN if all its attackers are out

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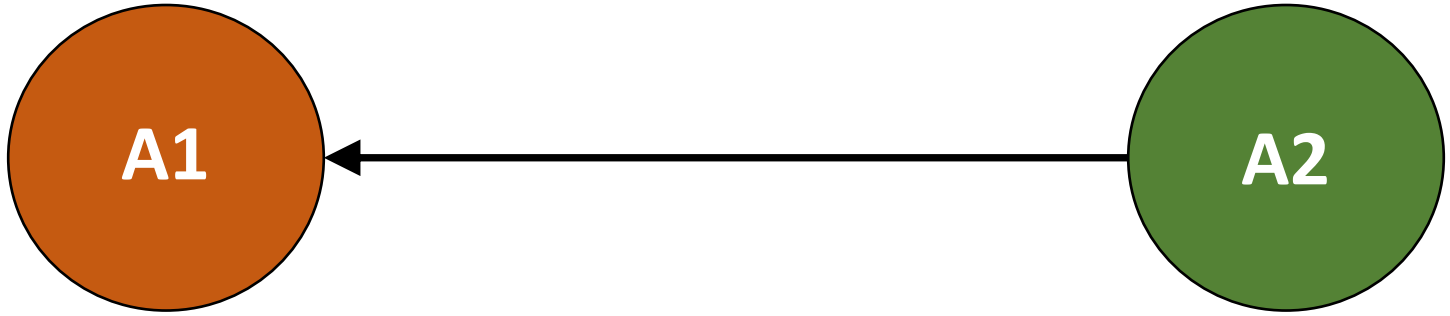
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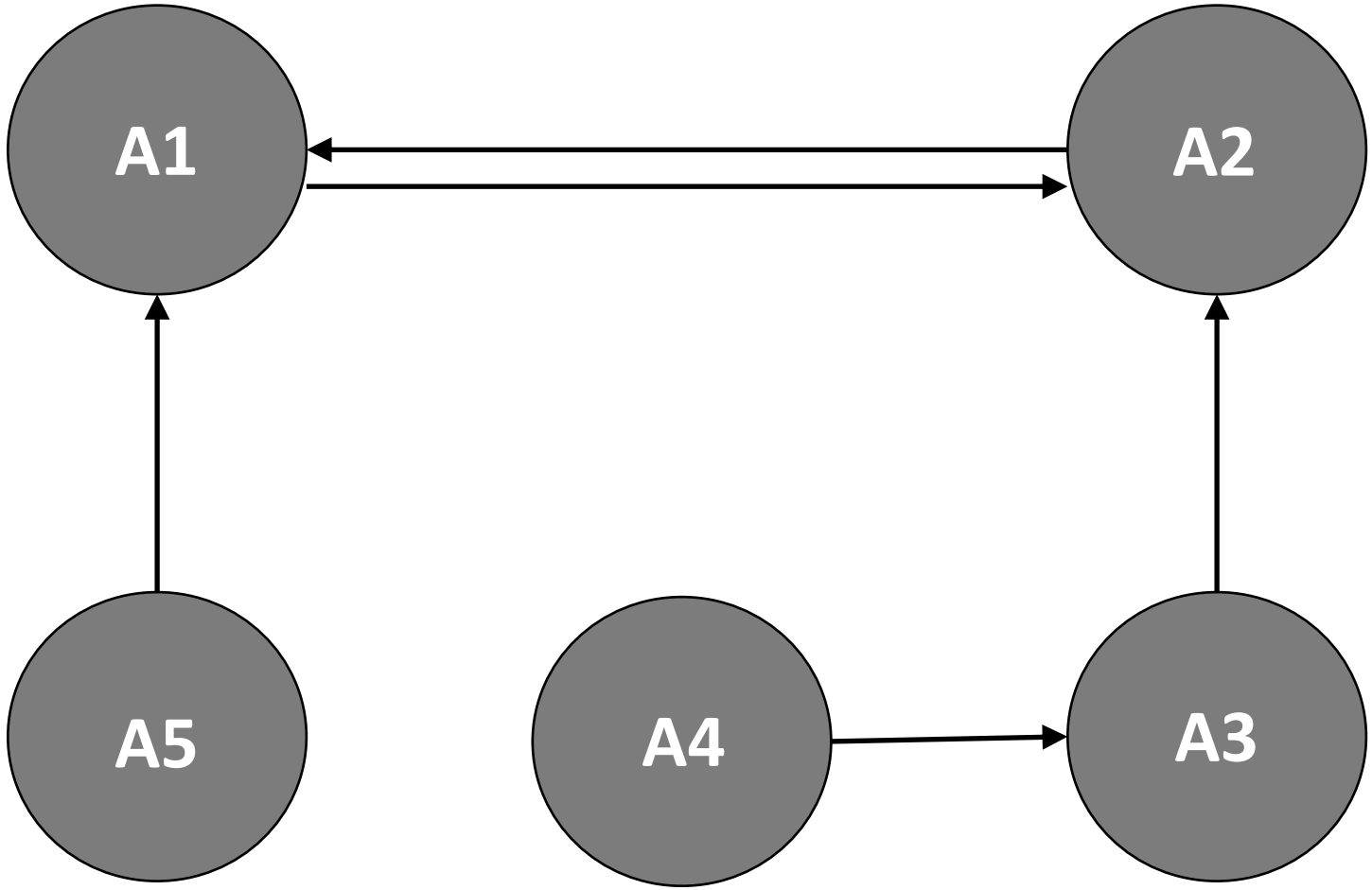
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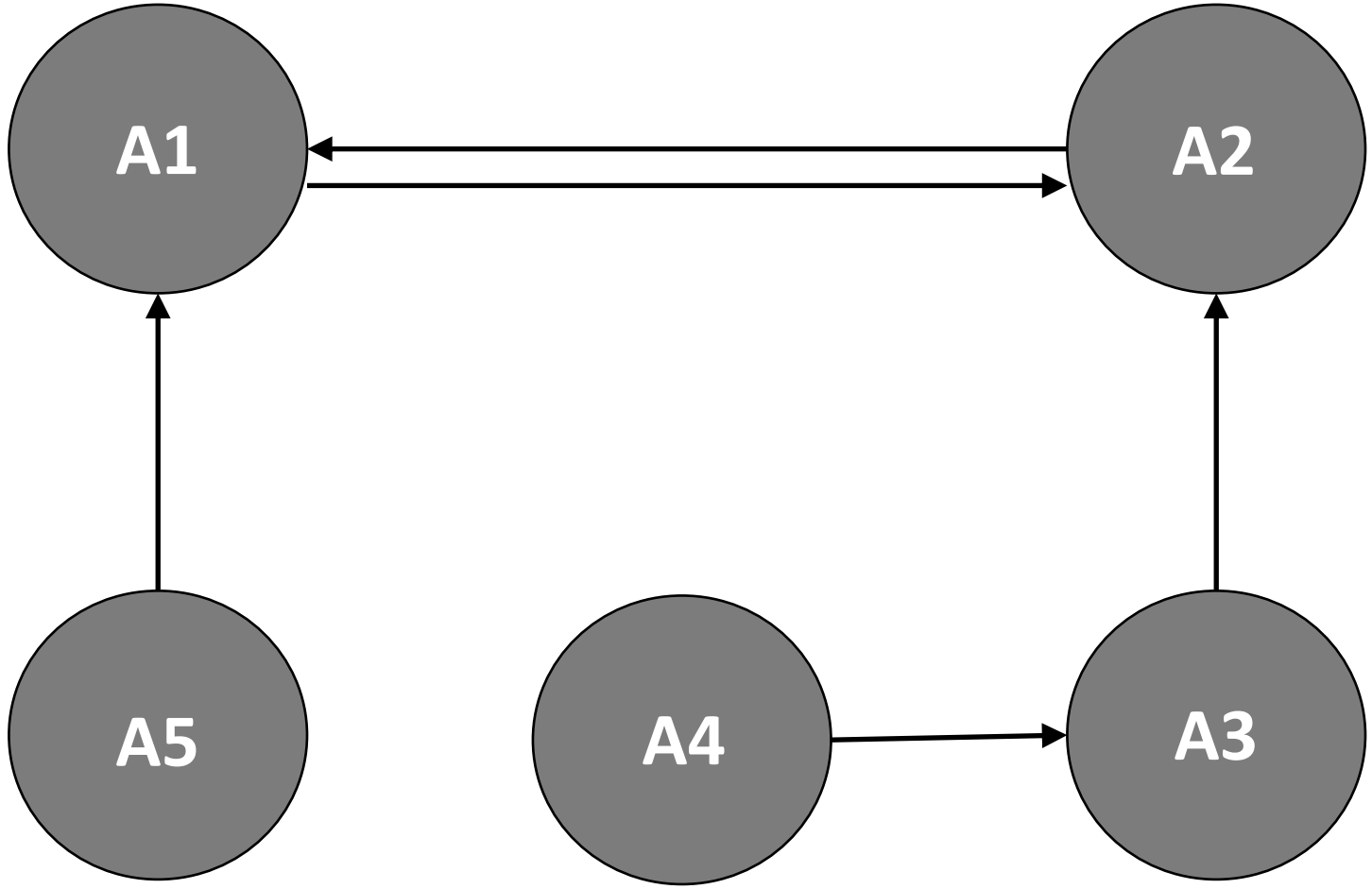



A1
A2
A5
A4
A3

IN if all its attackers are out

OUT if it has an attacker that is in

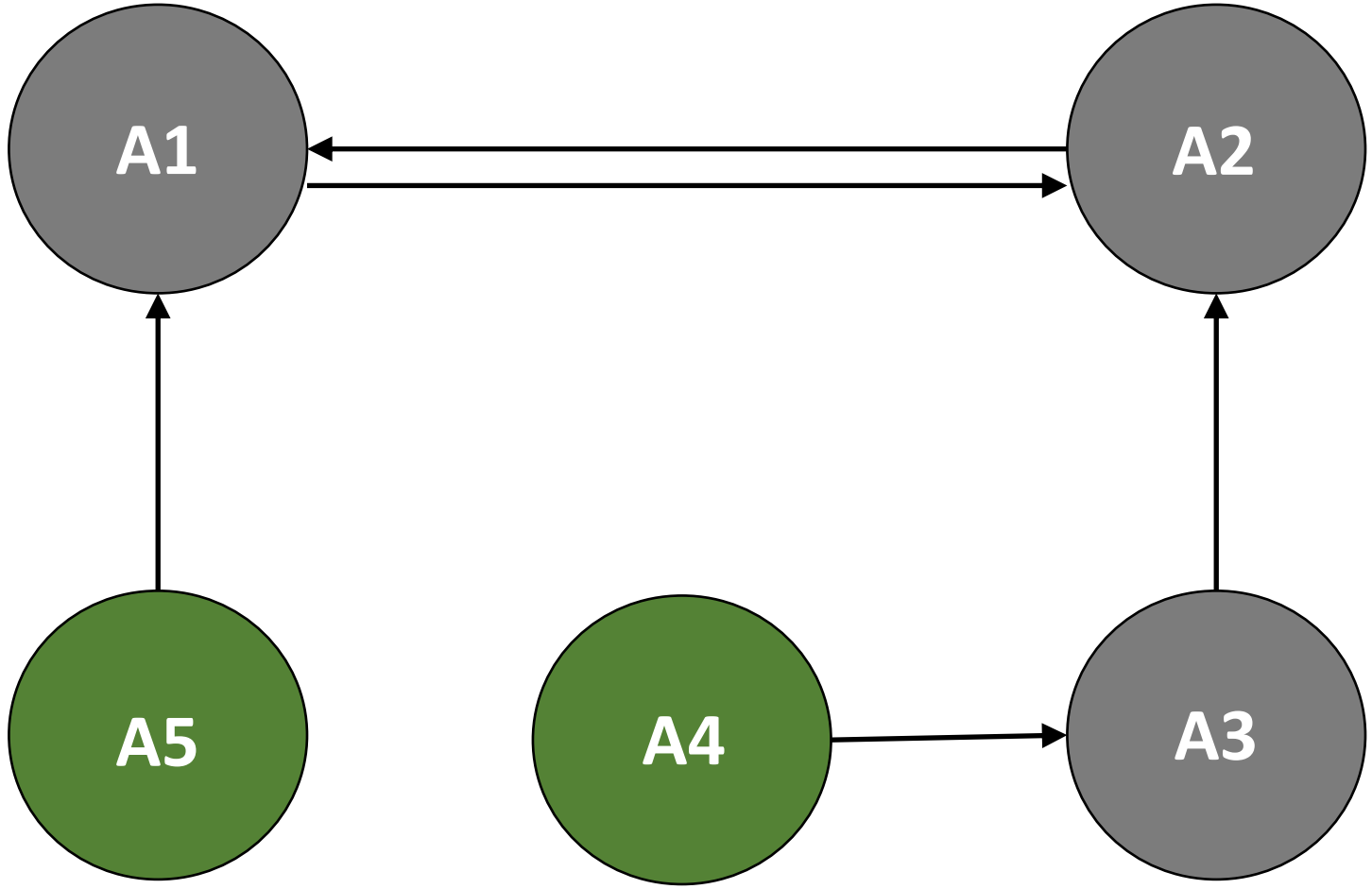
UNDEC if not all its attackers are out
and it does not have an attacker that is in



IN if all its attackers are out

OUT if it has an attacker that is in

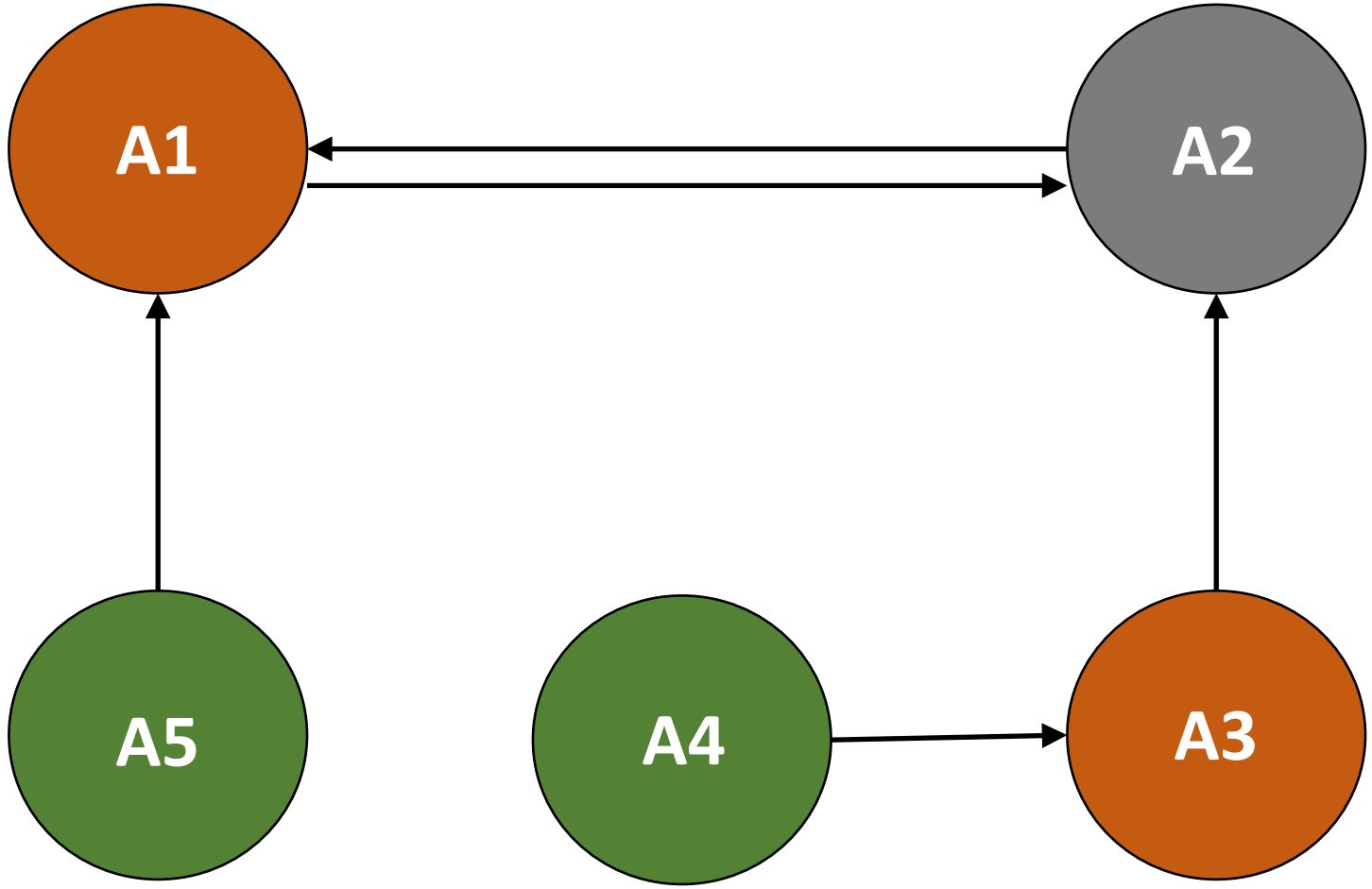
UNDEC if not all its attackers are out
and it does not have an attacker that is in



IN if all its attackers are out

OUT if it has an attacker that is in

UNDEC if not all its attackers are out
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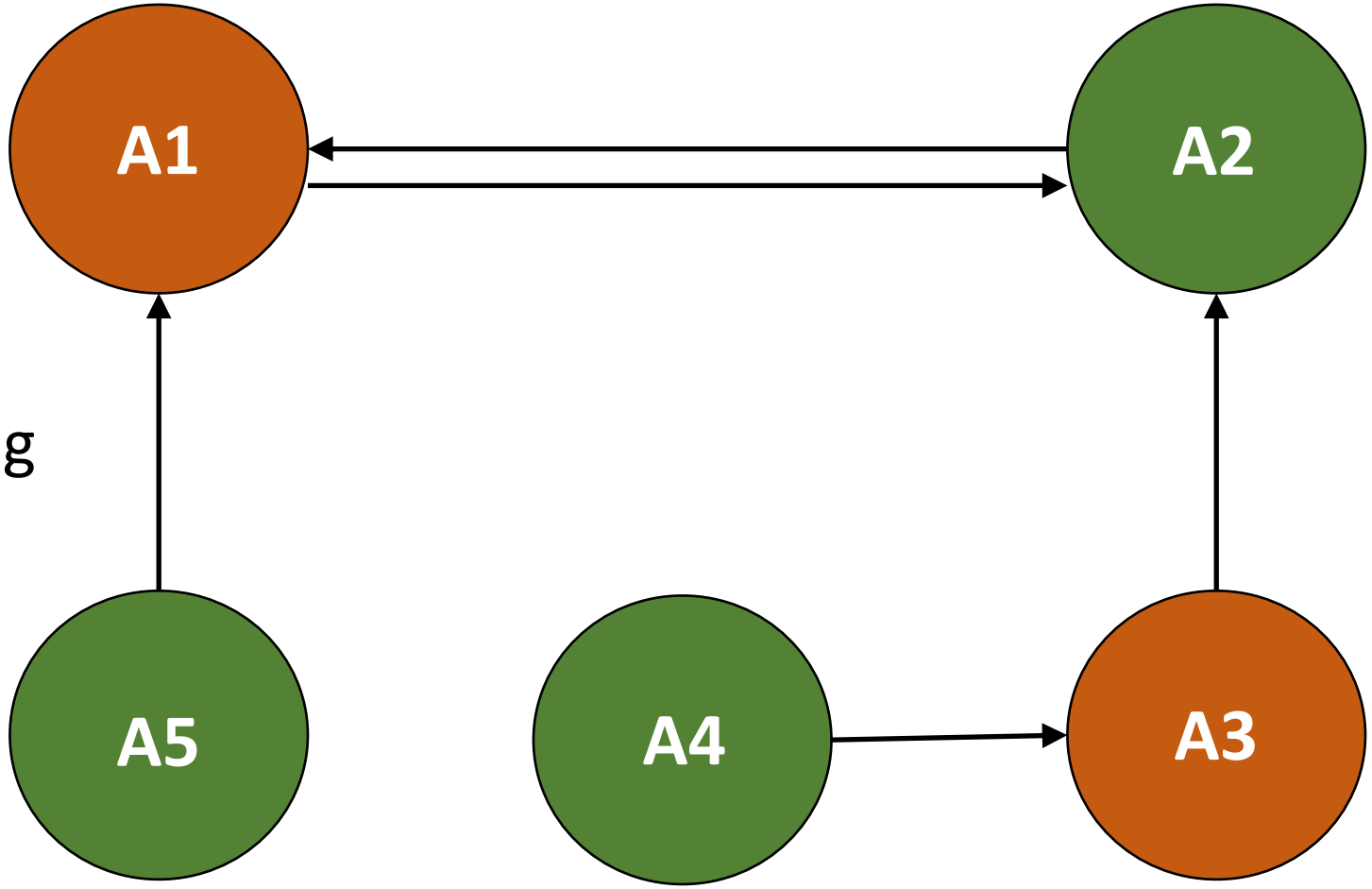


IN if all its attackers are out

OUT if it has an attacker that is in

UNDEC if not all its attackers are out
and it does not have an attacker that is in

We call this a complete labelling



Has social media been good for humanity?



A1: Social media has been good for humanity



A2: Social media has not been good for humanity



A3: Social media can be good to find news



A4: I read on social media that we cannot verify whether news on social media is real or fake

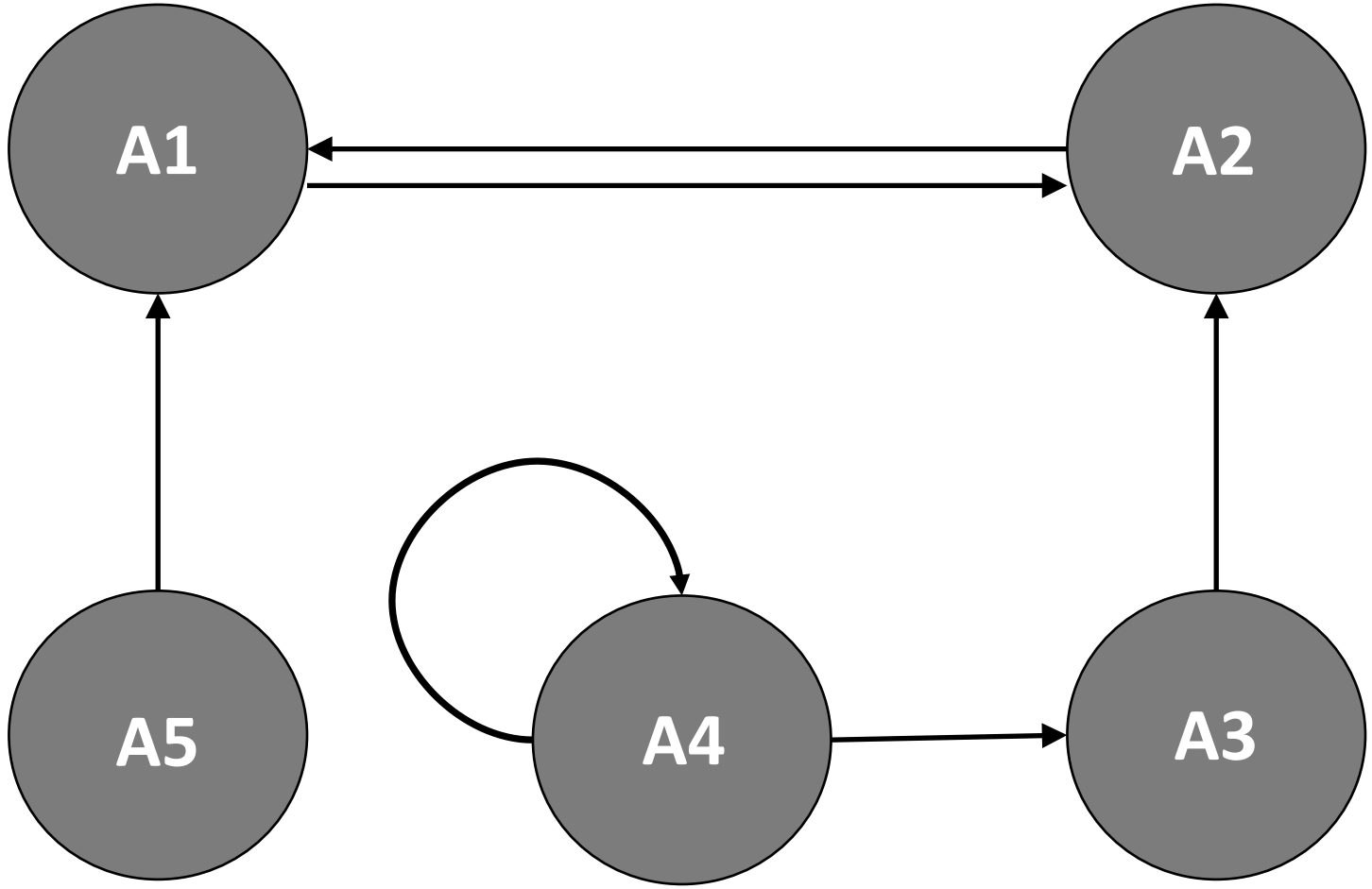


A5: Social media puts privacy and data at risk

IN if all its attackers are out

OUT if it has an attacker that is in

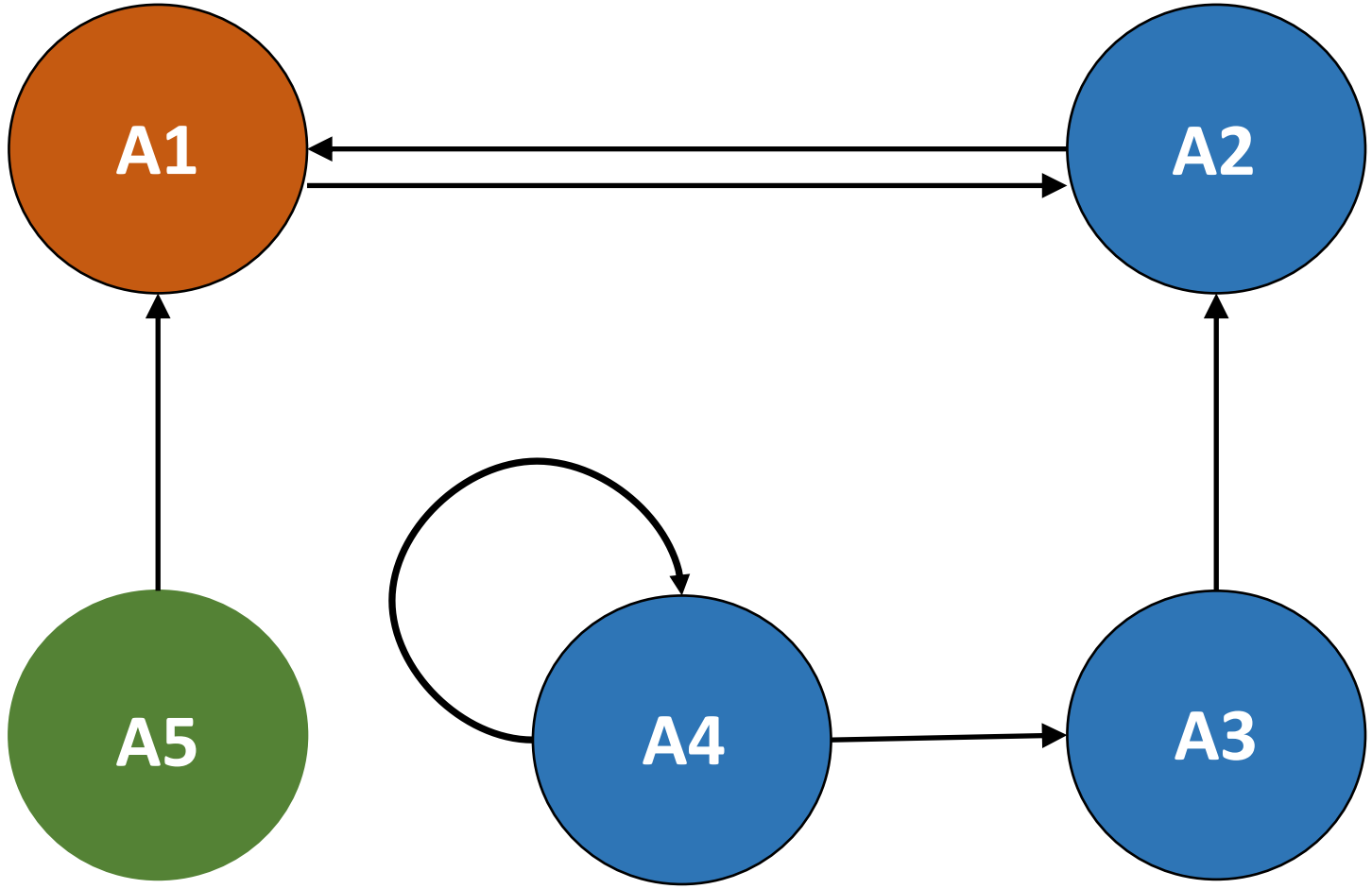
UNDEC if not all its attackers are out
and it does not have an attacker that is in



IN if all its attackers are out

OUT if it has an attacker that is in

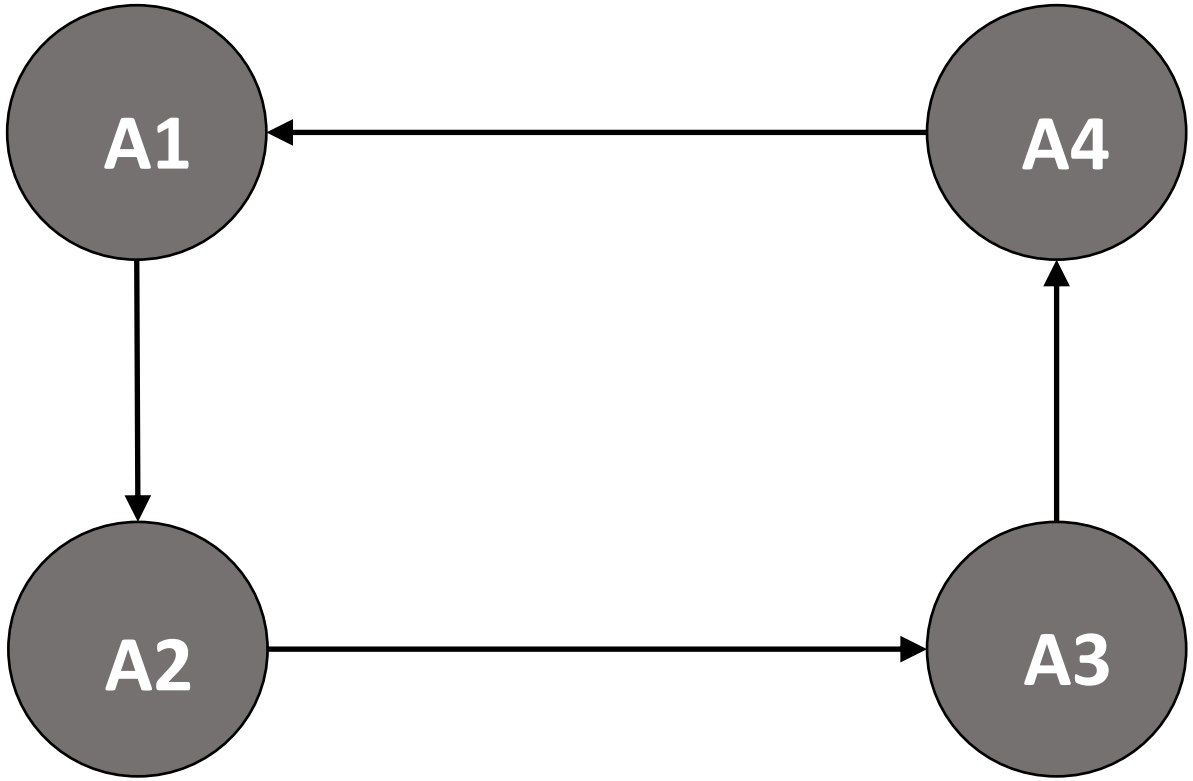
UNDEC if not all its attackers are out
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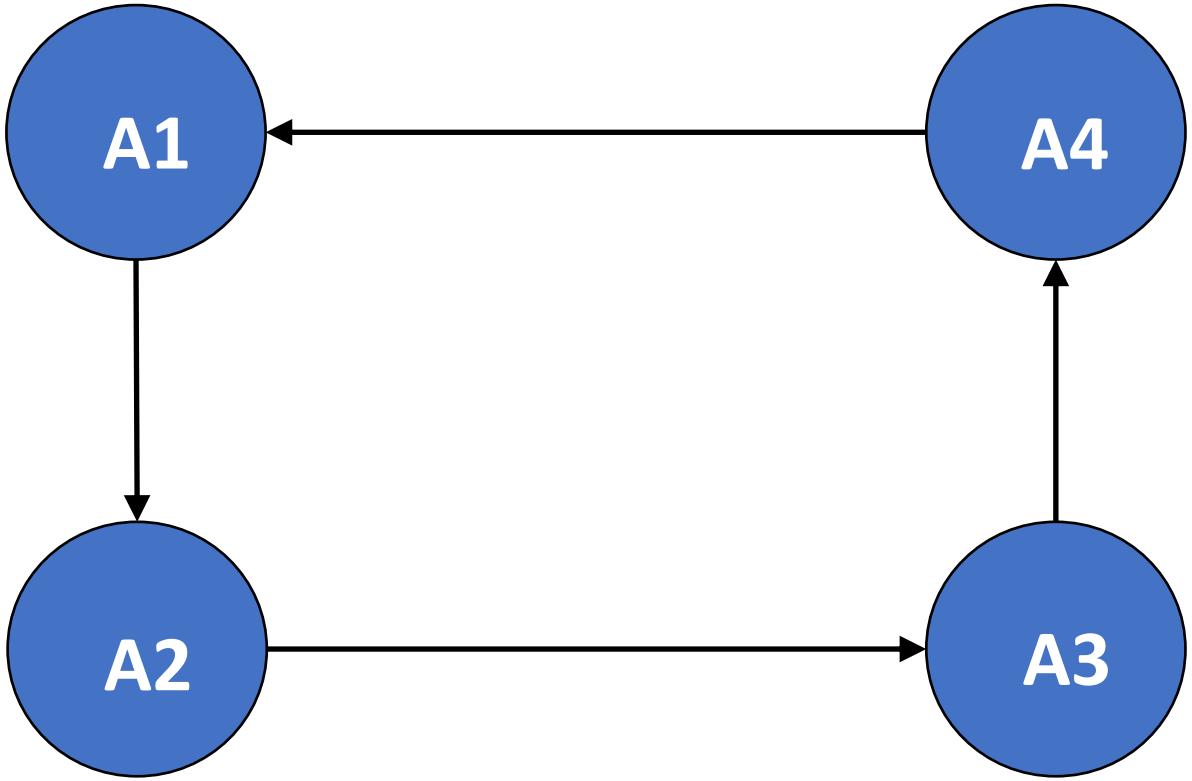


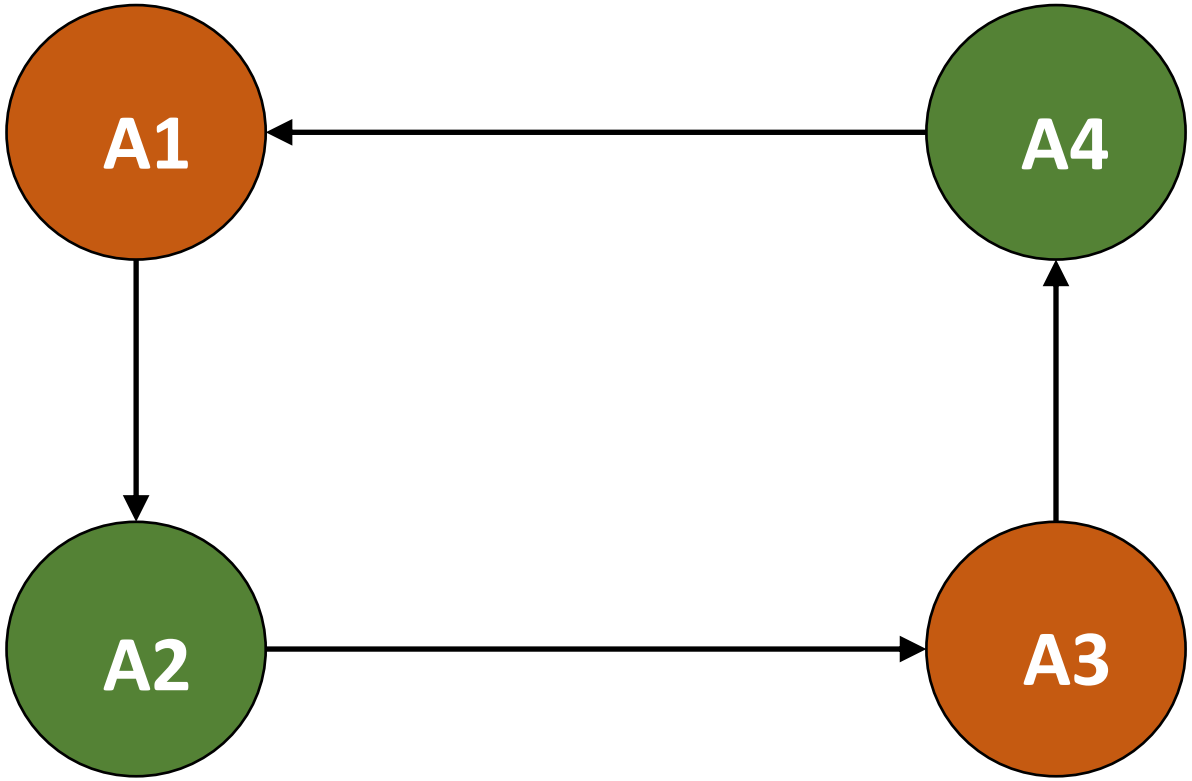
IN if all its attackers are out

OUT if it has an attacker that is in

UNDEC if not all its attackers are out
and it does not have an attacker that is in



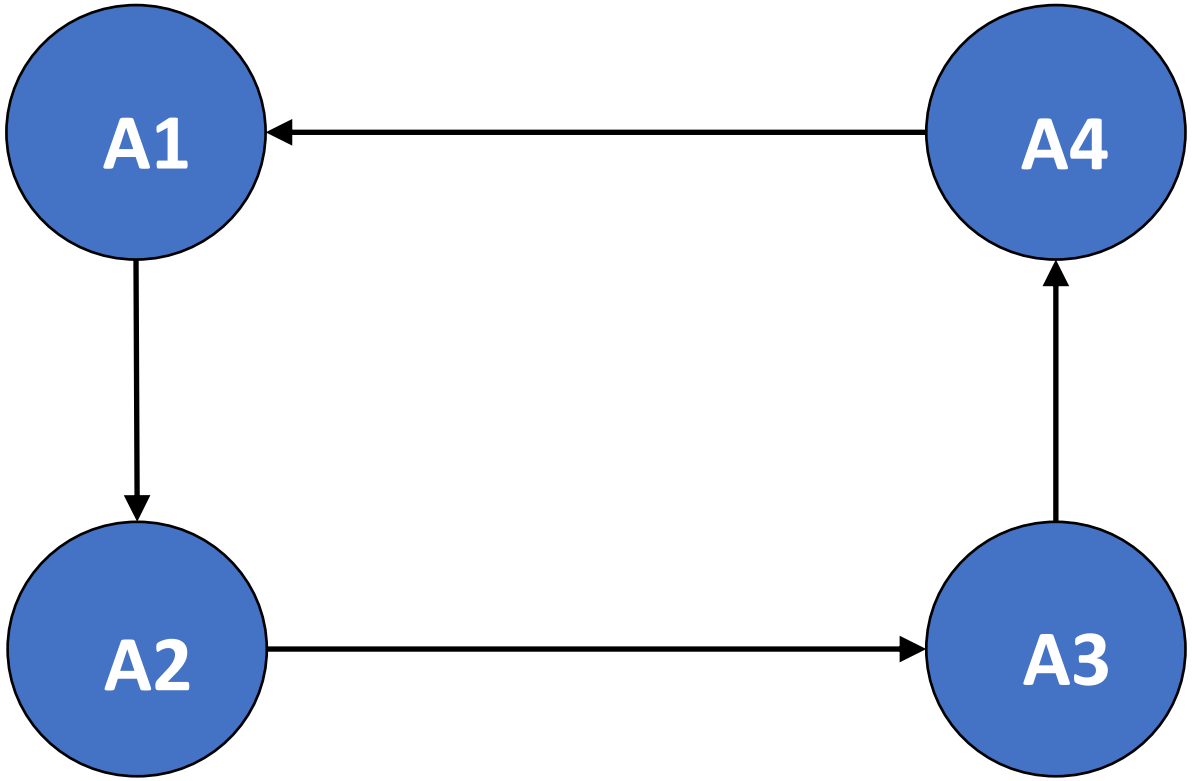

A1
A4
A2
A3


A1
A4
A2
A3

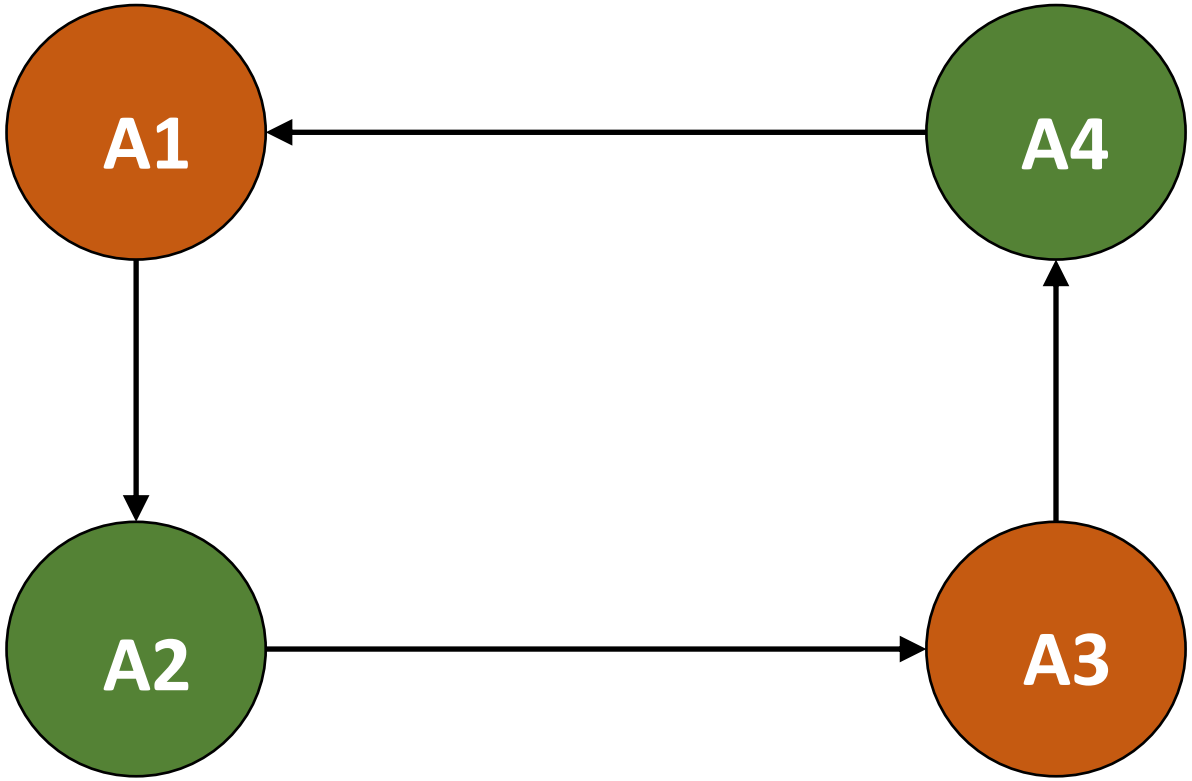
Labellings

- Grounded labelling – minimise the arguments that are IN
- Preferred labelling – maximise the arguments that are IN
- Stable labelling – no UNDEC arguments
- Semi-stable labelling – minimise the arguments that are UNDEC

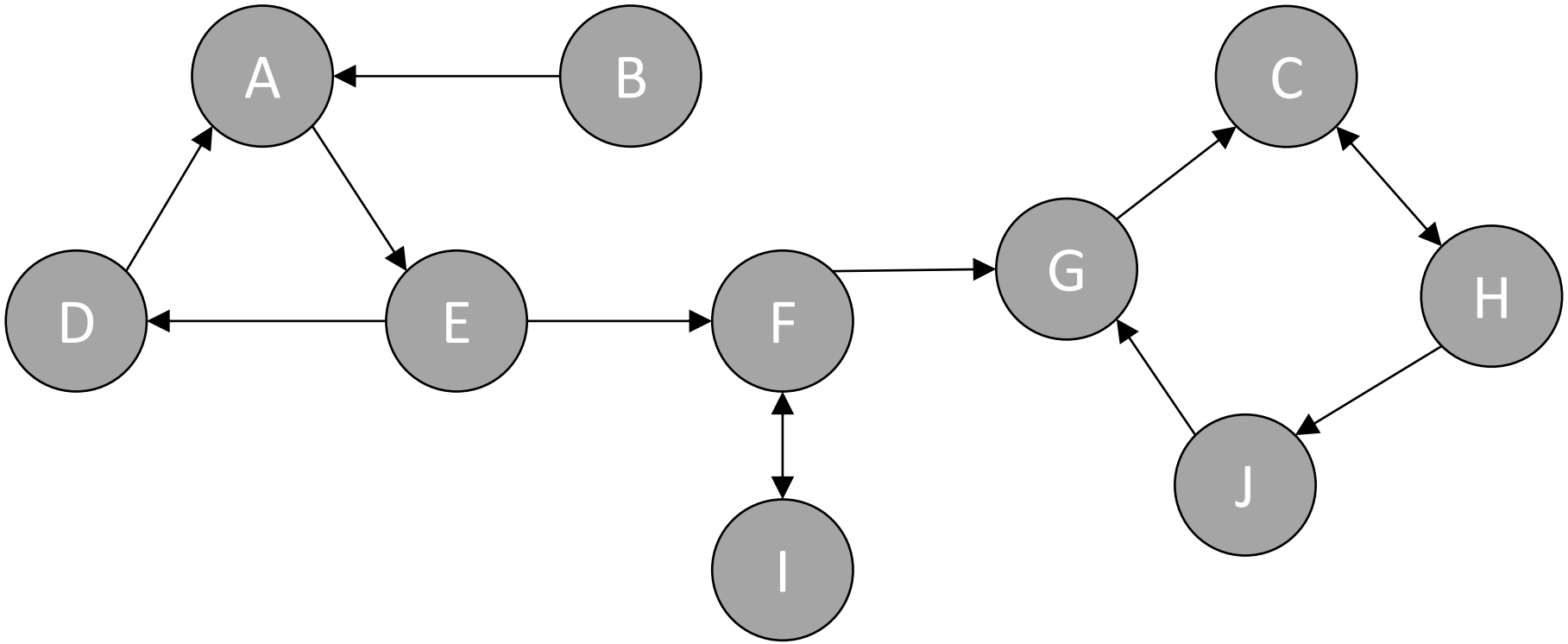
Grounded labelling



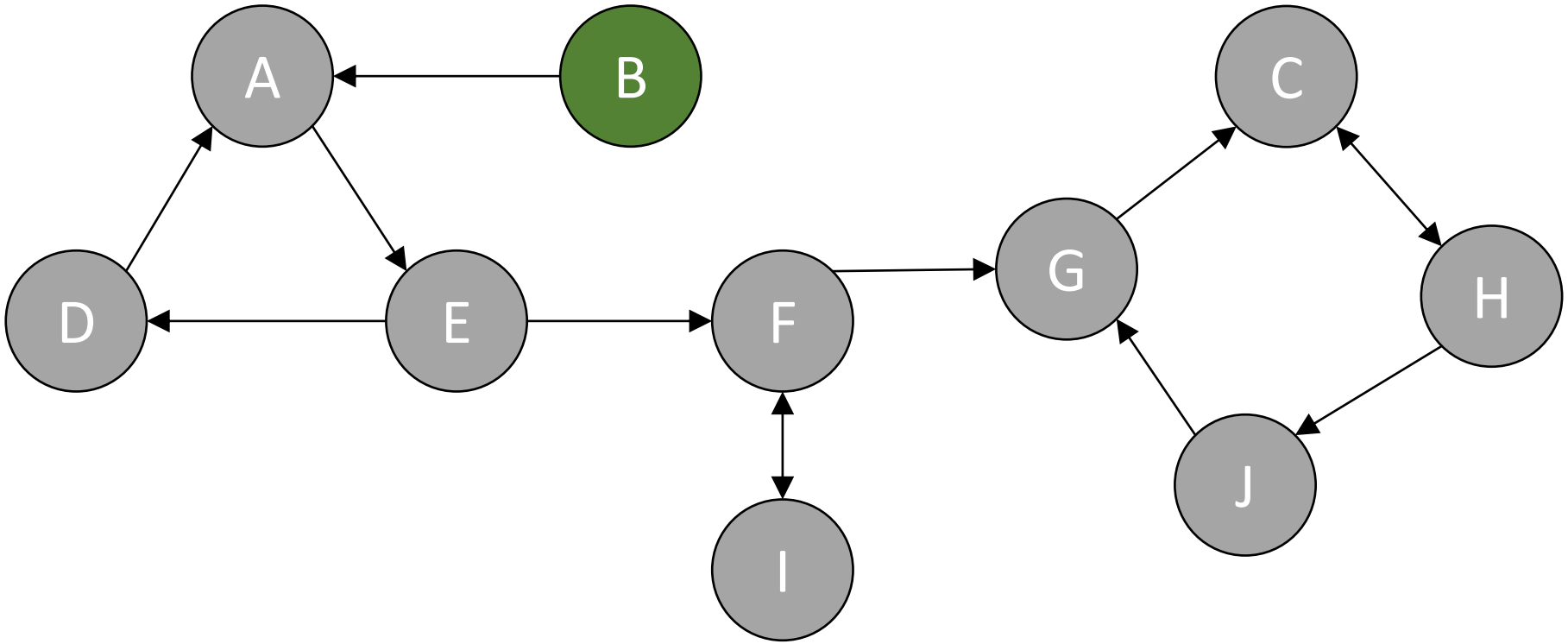
Preferred labelling
Stable labelling
Semi-stable labelling



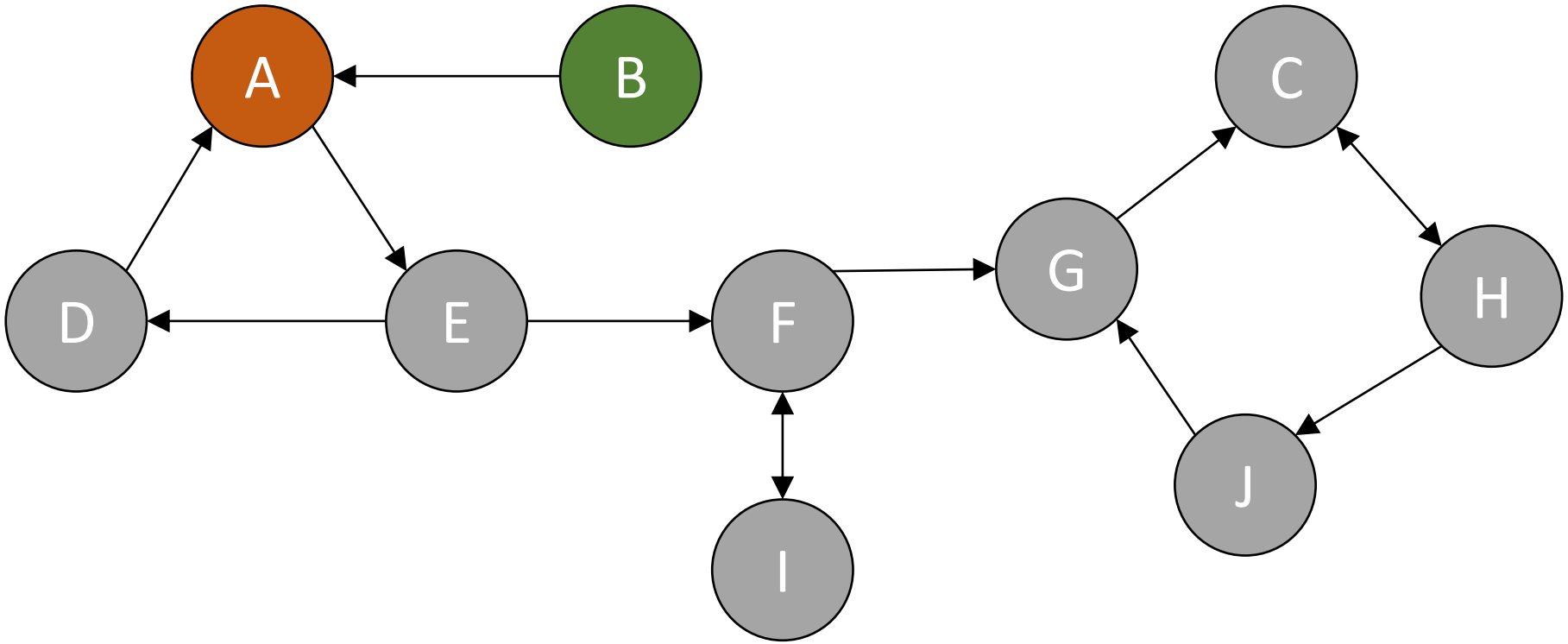
Which are the grounded labellings?



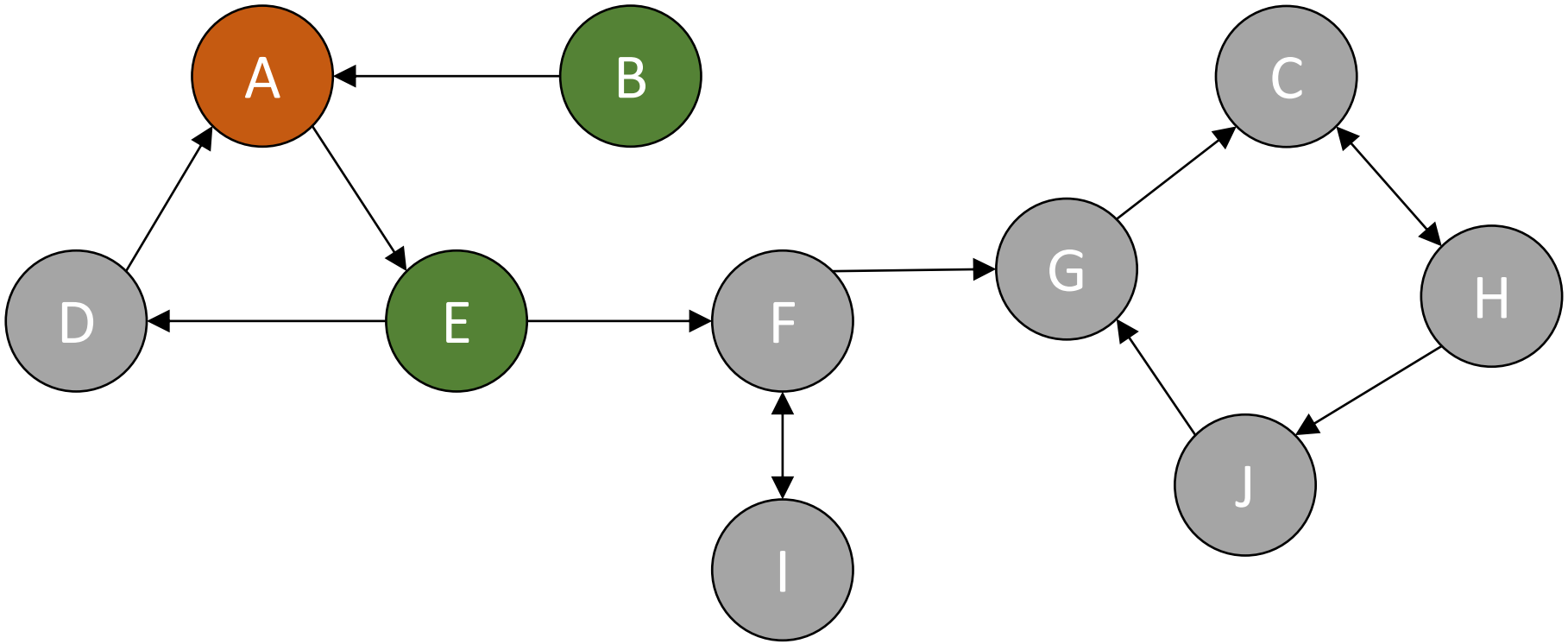
Which are the grounded labellings?



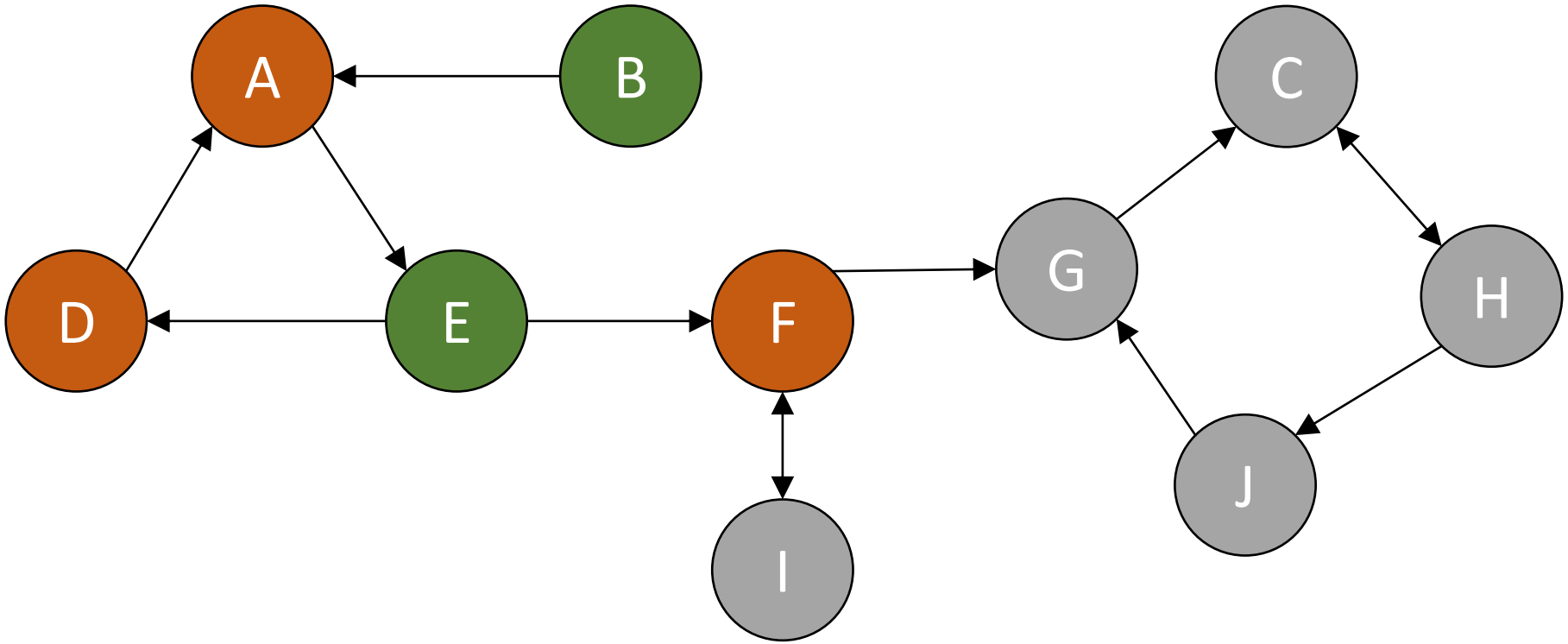
Which are the grounded labellings?



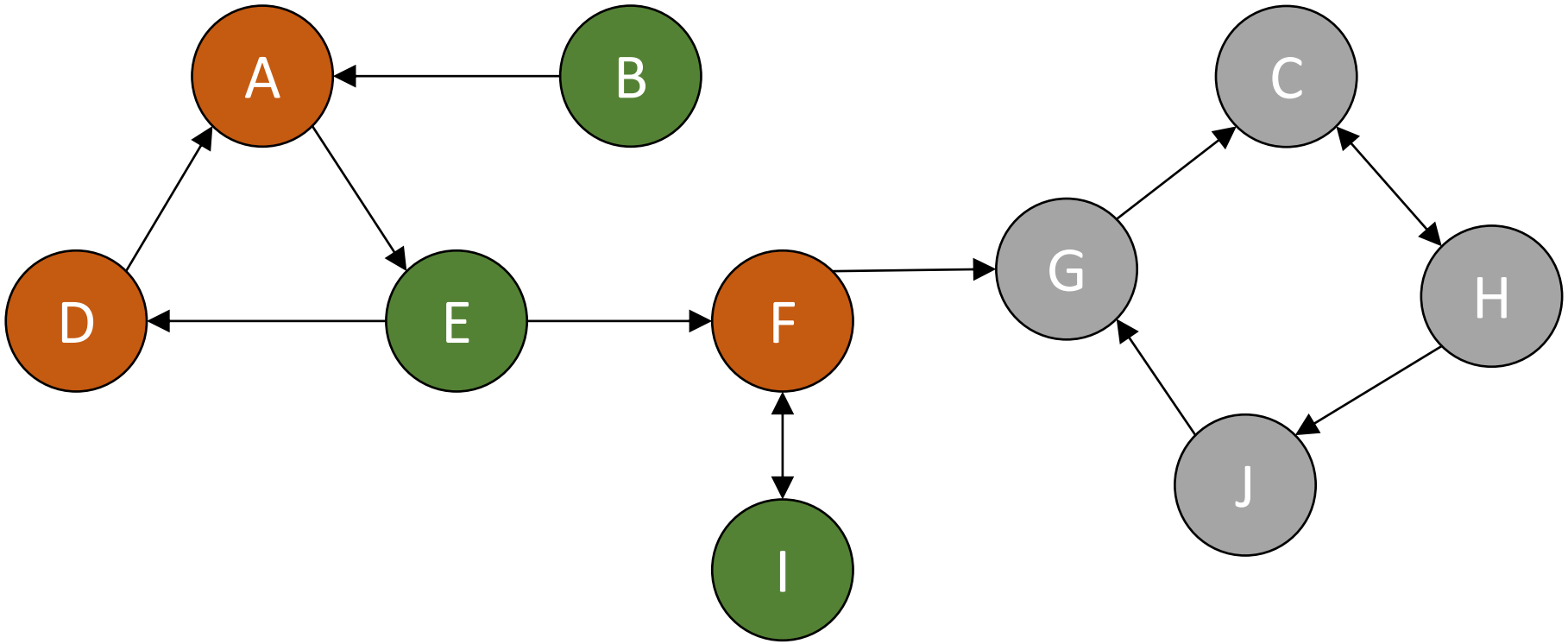
Which are the grounded labellings?



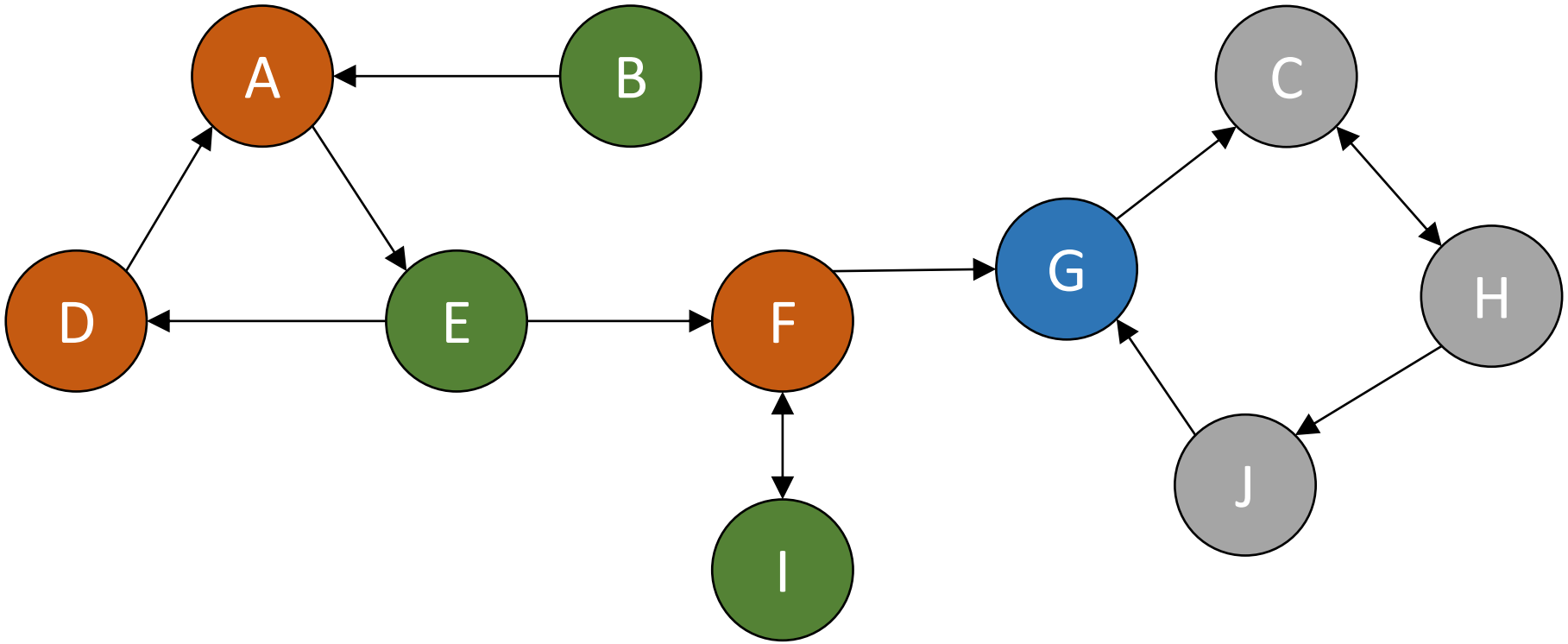
Which are the grounded labellings?



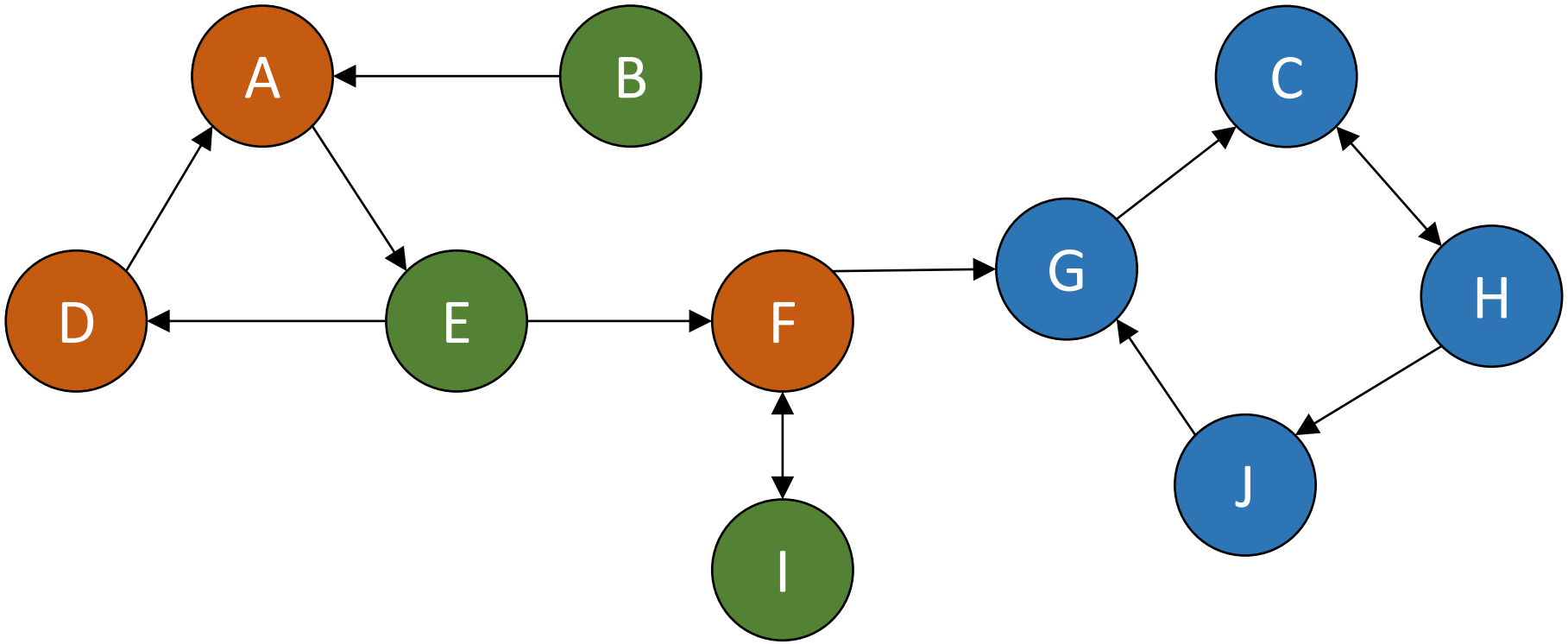
Which are the grounded labellings?



Which are the grounded labellings?



Which are the grounded labellings?



What are the labellings of the graphs you created for your debate?

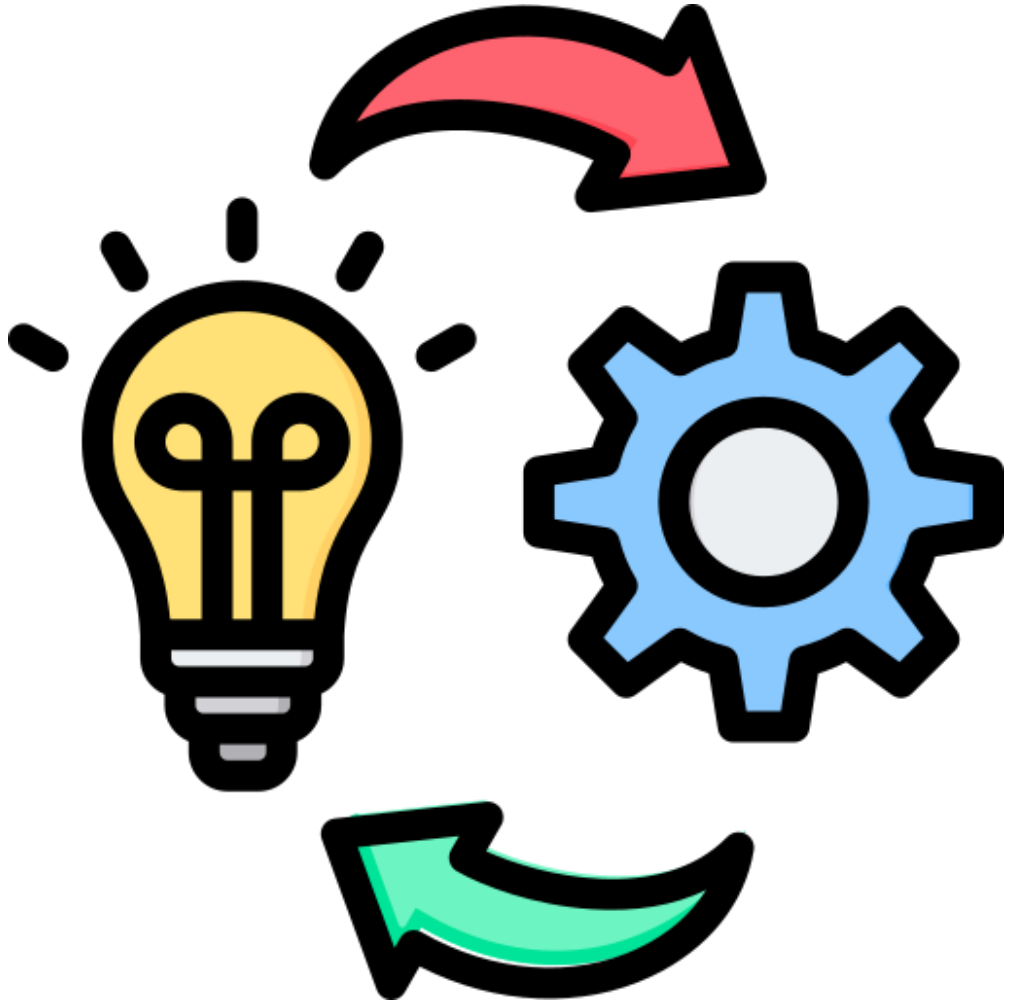
- Grounded labelling – minimise the arguments that are IN
- Preferred labelling – maximise the arguments that are IN
- Stable labelling – no UNDEC arguments
- Semi-stable labelling – minimise the arguments that are UNDEC



Break

Implementation time!

http://argteach.herokuapp.com



Variations of abstract argumentation frameworks

Bipolar argumentation frameworks (BAFs)



Adds support relations to abstract argumentation frameworks



Semantics defined differently

An argument is accepted only if it is directly defended or supported by arguments that are themselves already accepted in a grounded manner.

Has social media been good for humanity?



A1: Social media has been good for humanity



A2: Social media has not been good for humanity



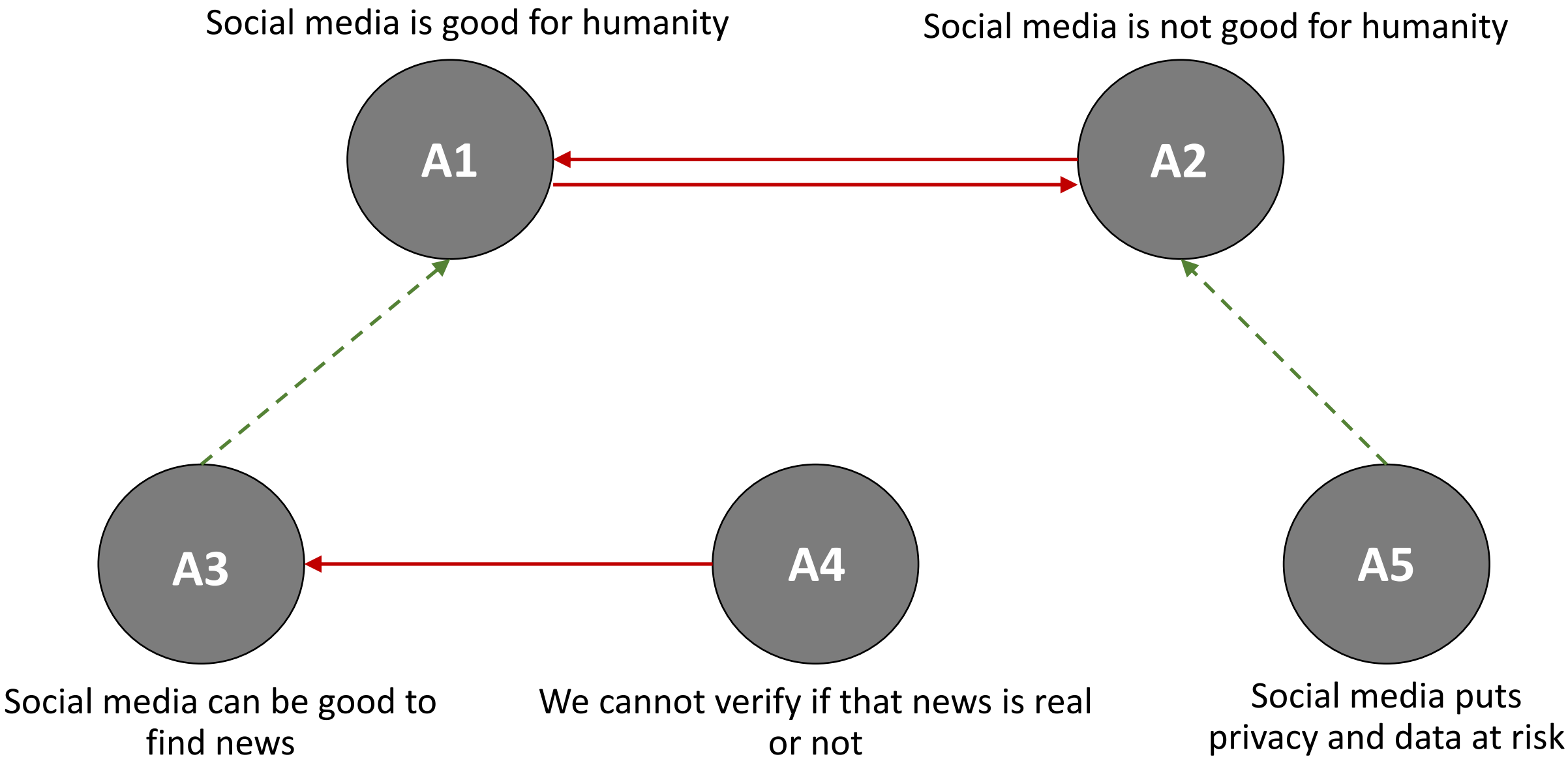
A3: Social media can be good to find news

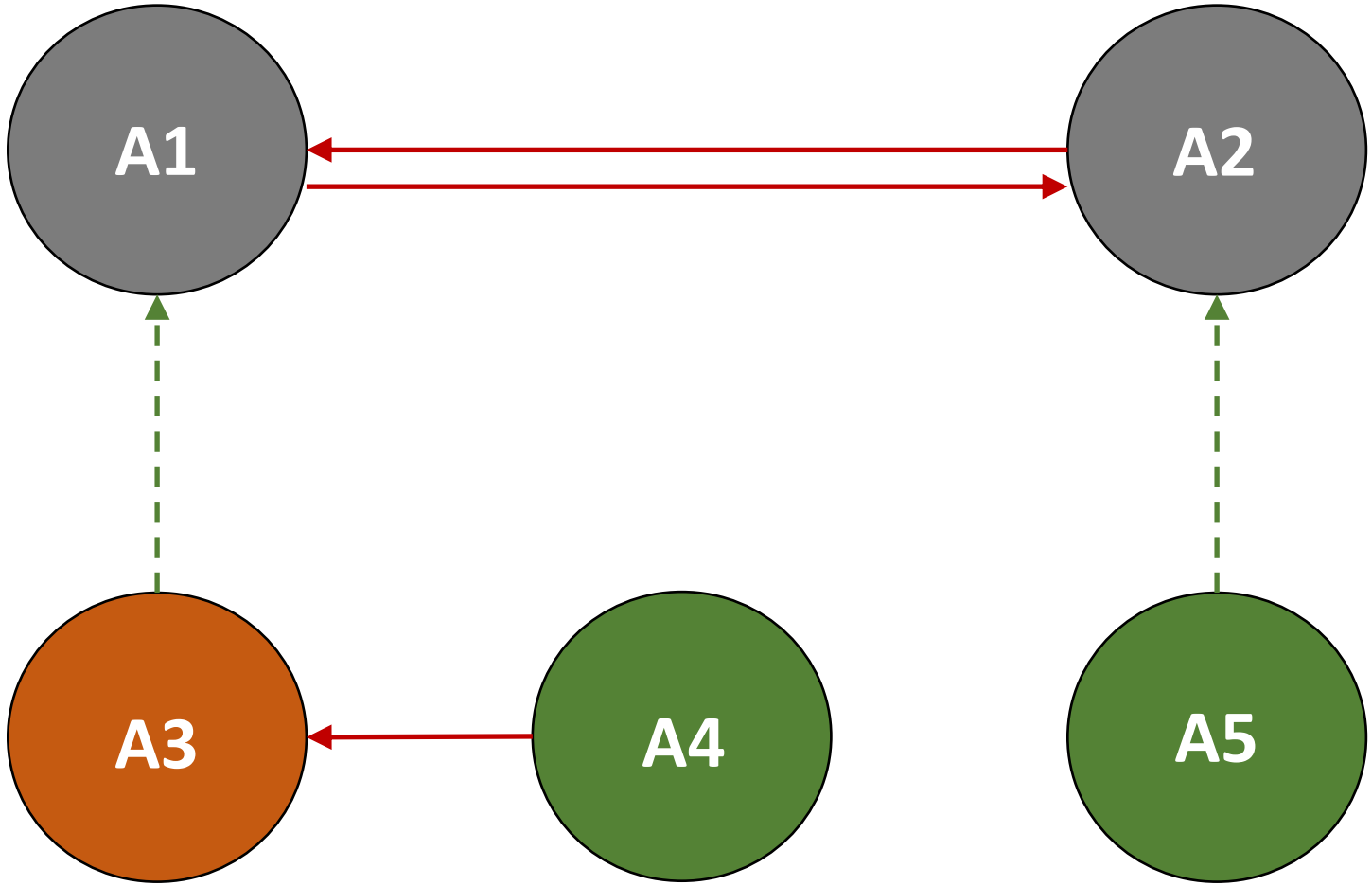


A4: We cannot verify if that news is real or not

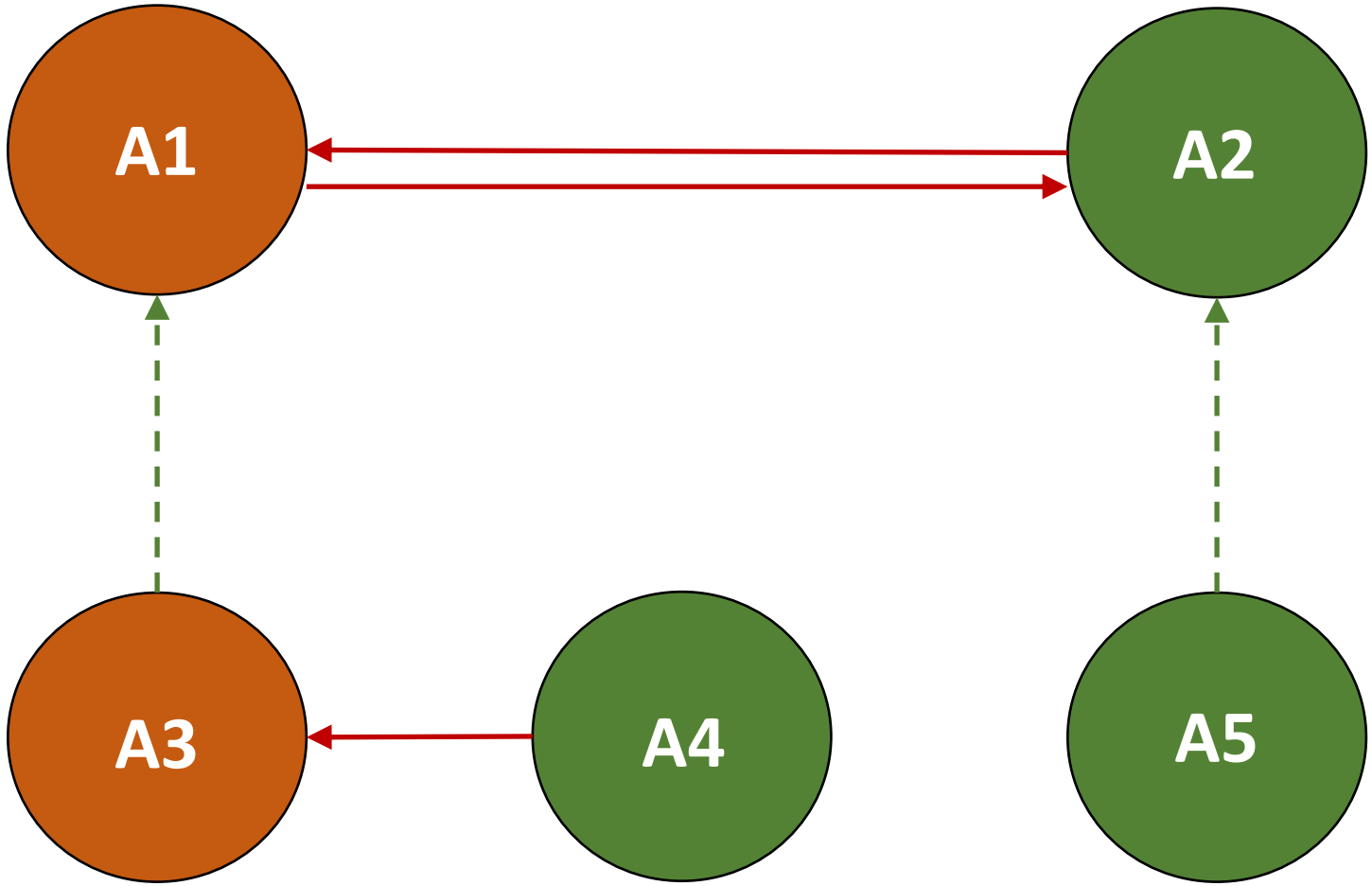


A5: Social media puts privacy and data at risk

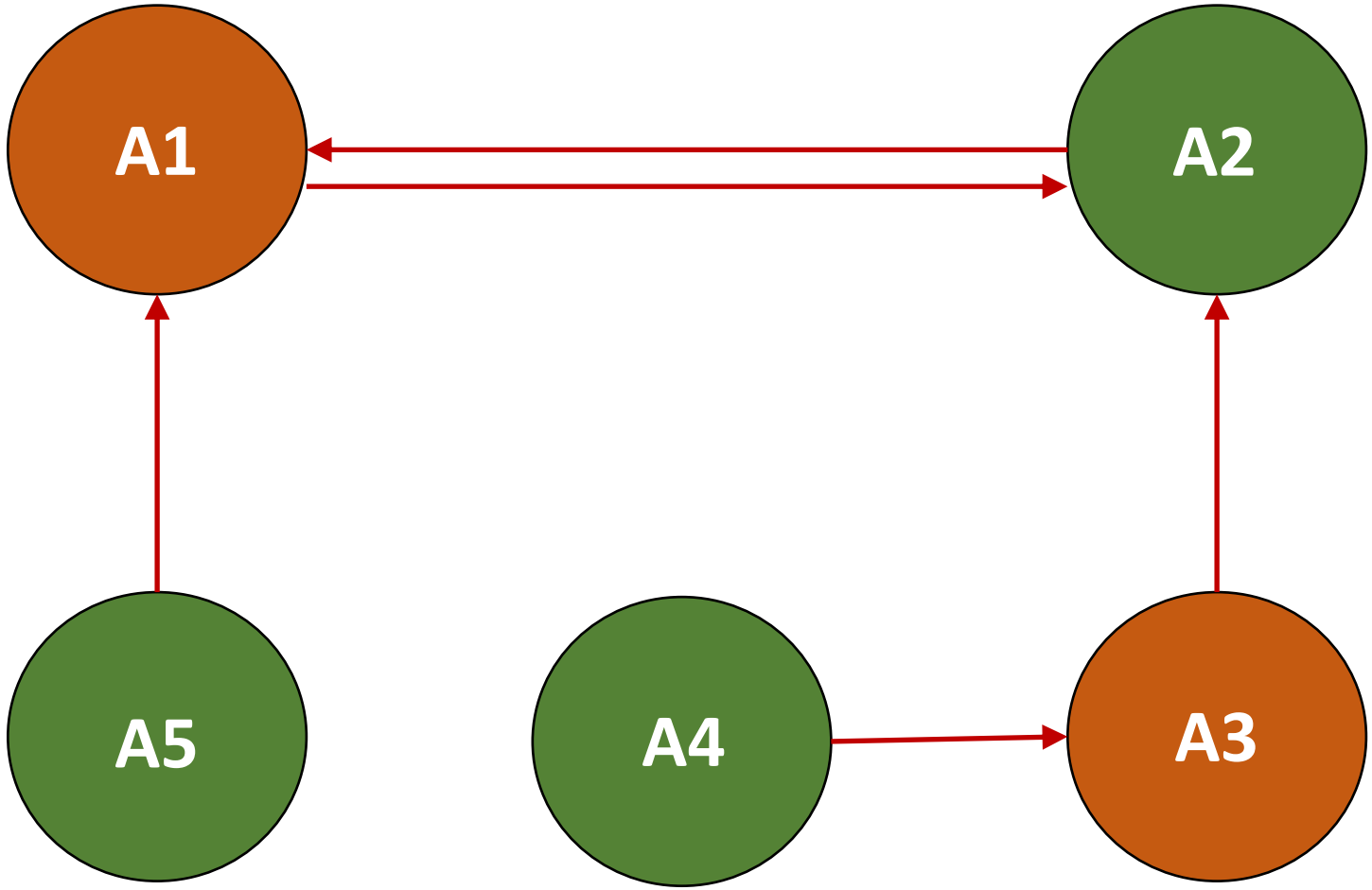

Social media is good for humanity
Social media is not good for humanity
A1
A2
A3
A4
A5
Social media can be good to find news
We cannot verify if that news is real or not
Social media puts privacy and data at risk


A1
A2
A3
A4
A5

Graph with attack and support relations



Graph with only attack relations



Weighted argumentation frameworks (WAFs)



Adds numerical values to the abstract argumentation graph



Intrinsic weights assigned to arguments/attacks/supports representing their initial strength



Higher weights indicate stronger arguments/attacks/supports and therefore have more influence on the final acceptability calculated



Semantics used to calculate final weights of arguments based on the weights of incoming arguments/attacks/supports

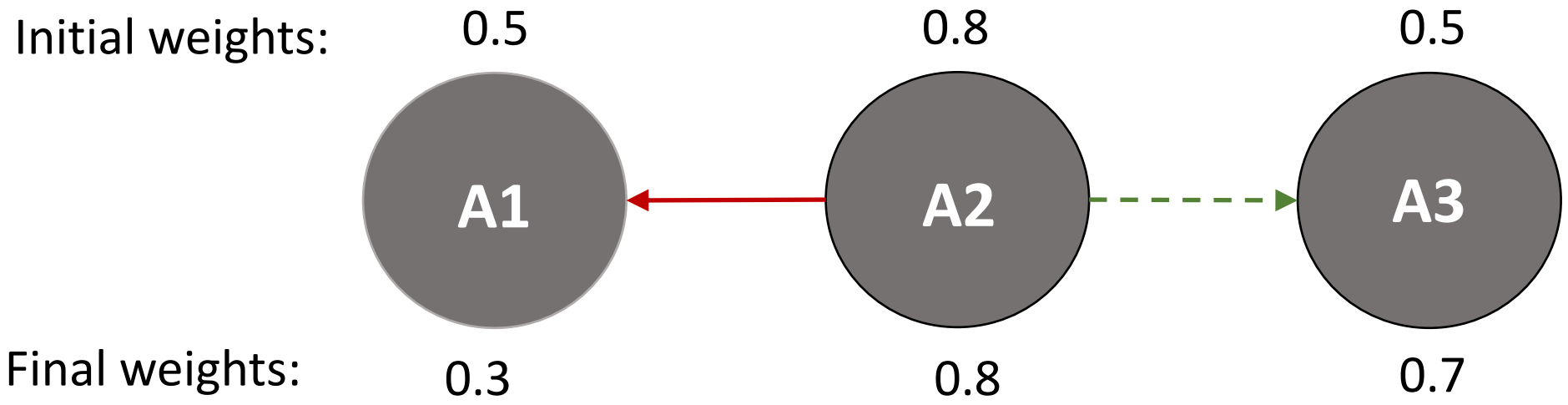
WAF semantics

Attackers force the strength value towards 0 proportionally to their strength

Supporters force the strength value towards 1 proportionally to their strength

Strength values are attracted by their initial weight [0,1]

Using the Quadratic Energy Model Semantics (Potyka 2019)



Quest Time!

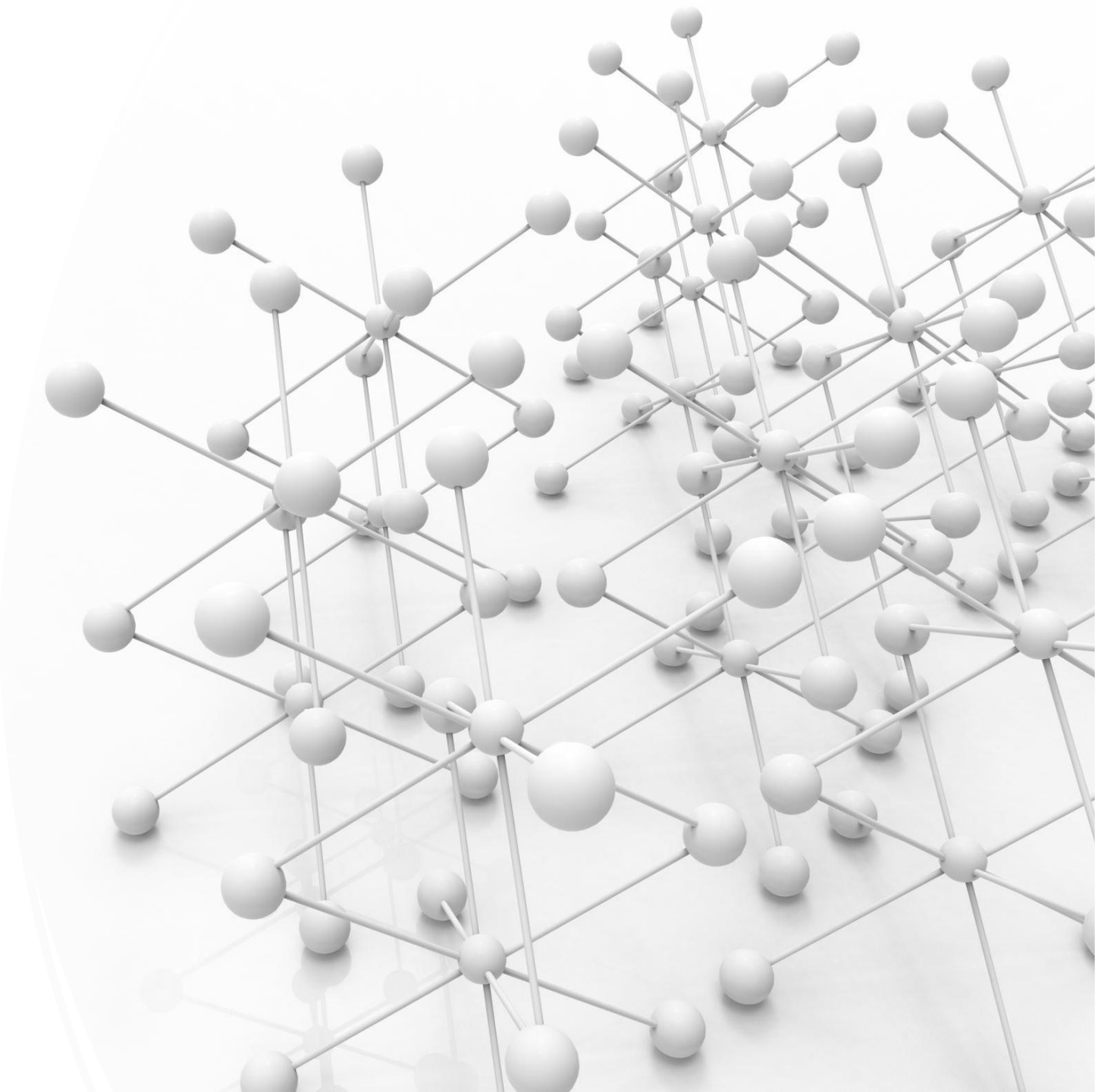
Look back at your debate graphs

- Would it be easier to create them with support relations as well?
- Make one variation of the graph that includes a support relation.
- Would it be helpful to have any other relationship represented in your graph?

Structured argumentation

Building on abstract argumentation, structured argumentation additionally details:

- the *internal* structure of arguments
- the explicit relationships among them





Structured argumentation models premises, conclusions, and inferential rules

p: Left is the fastest route.

c: I will go left.

r: I want to take the fastest route.

Strict and defeasible rules

Strict Rules

If A and B then **always** C



All tuba players are mortal.
I am a tuba player.
 I am mortal.

Defeasible rules

If A and B then **usually** C



After complaints, I don’t play the tuba.
Neighbours complained.
 I don’t play the tuba…

Unless I have a concert.

Statements, axioms, ordinary premises

Statements

Facts, beliefs, derivations, …



Axioms (necessary premises)

Facts



We are at the Multimedia Center.

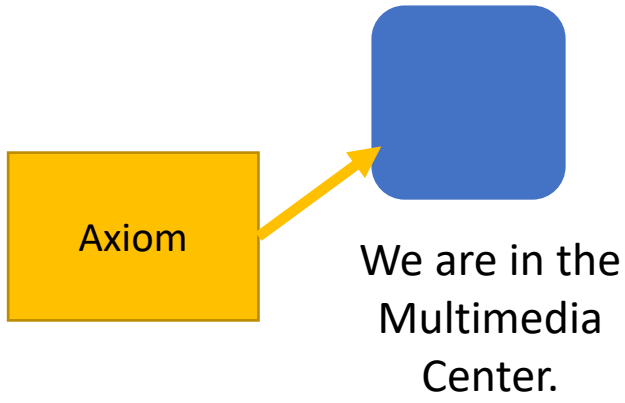
Ordinary premises

Things you believe

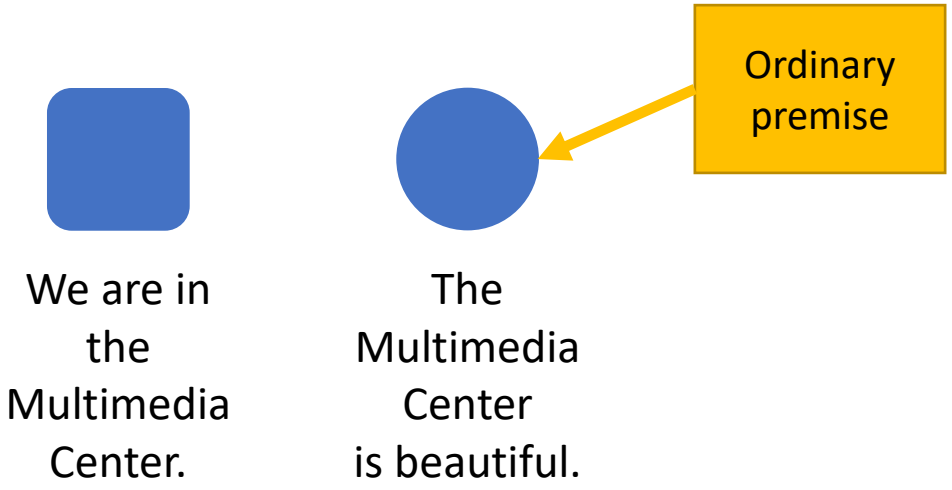


The Multimedia Center is beautiful.

Constructing arguments

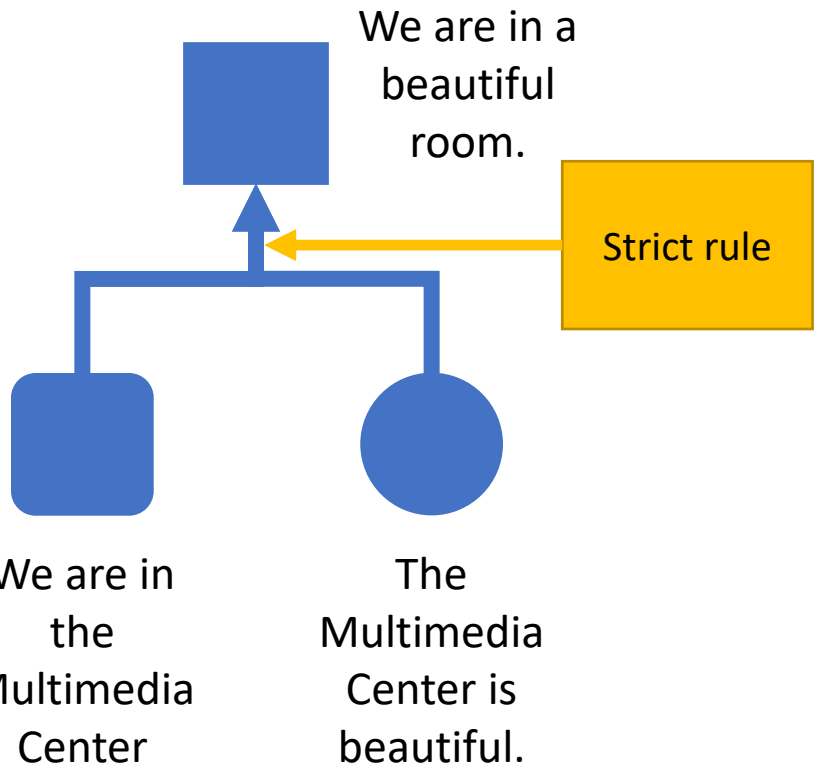


Constructing arguments

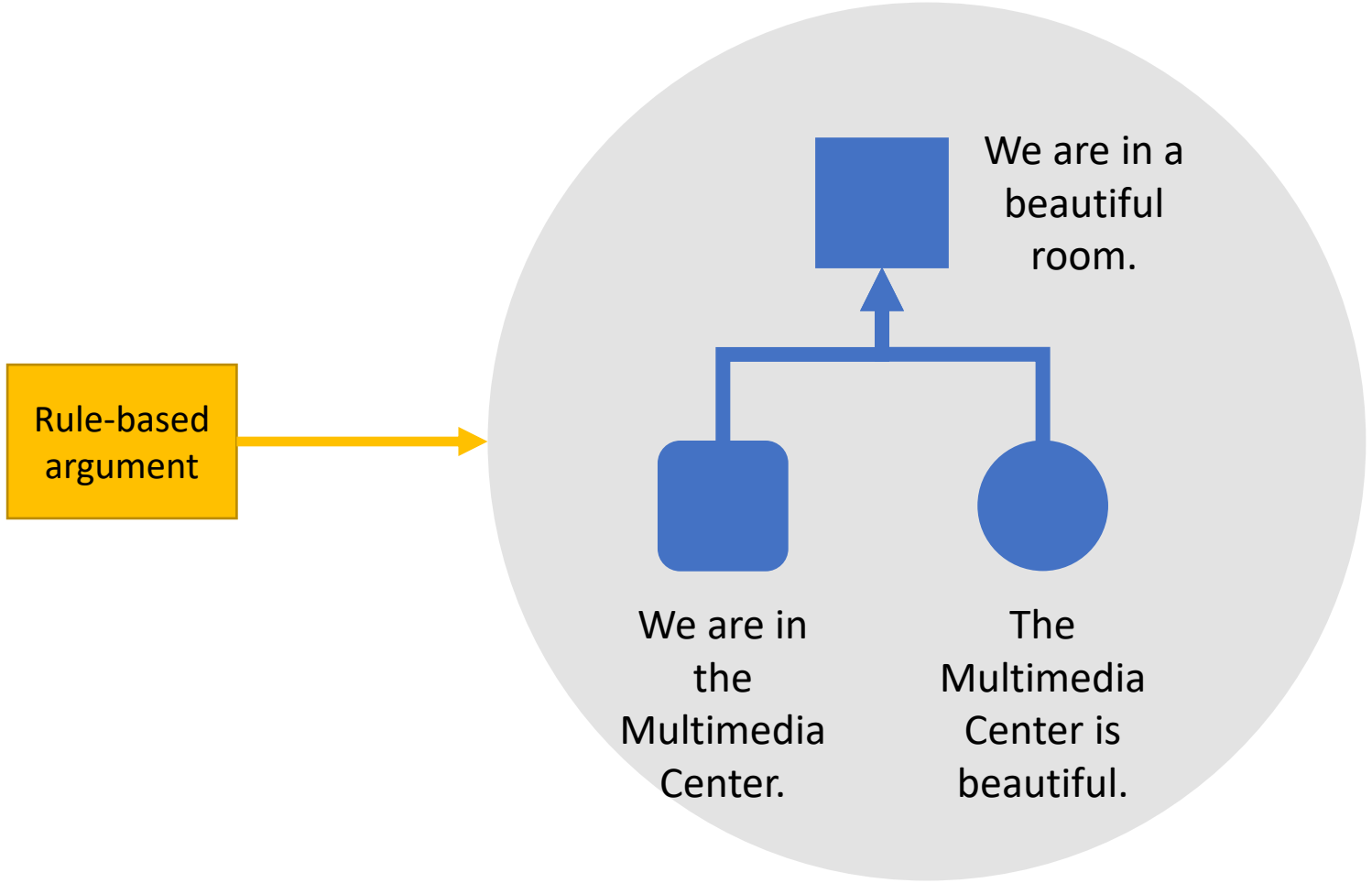


Constructing arguments

If we are in the Multimedia Center.
and the Multimedia Center is beautiful,
then we are in a beautiful room.



Constructing arguments



Defeasible Rules

Structured argumentation (ASPIC$^+$): An argument consists of a conclusion deductively and/or defeasibly inferred from some premises.

e.g.

(conclusion)	b	Take an umbrella with you.
(inference rule)	$a \Rightarrow b$	If it's raining outside, then take an umbrella with you.
(premise)	a	It's raining outside.

Argument from Expert Opinion

ASPIC+ / Rd

From a scheme to a structured argument

DEFEASIBLE RULE d_{ex}

expert(e, d) , asserts(e, p) , within(p, d) ⇒ p

Ordinary premises (Kp)

A1 · EXPERTISE

expert(Lee, met)

Dr Lee is an expert in meteorology.

A2 · ASSERTION

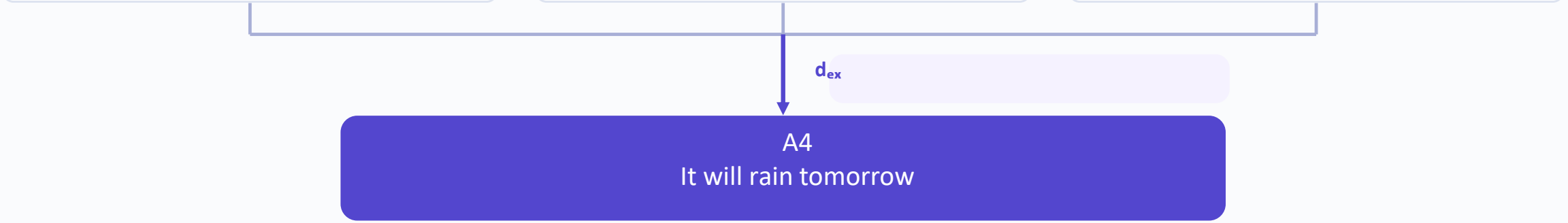
asserts(Lee, rain)

Dr Lee says it will rain tomorrow.

A3 · DOMAIN

within(rain, met)

Rain forecasts fall within meteorology.



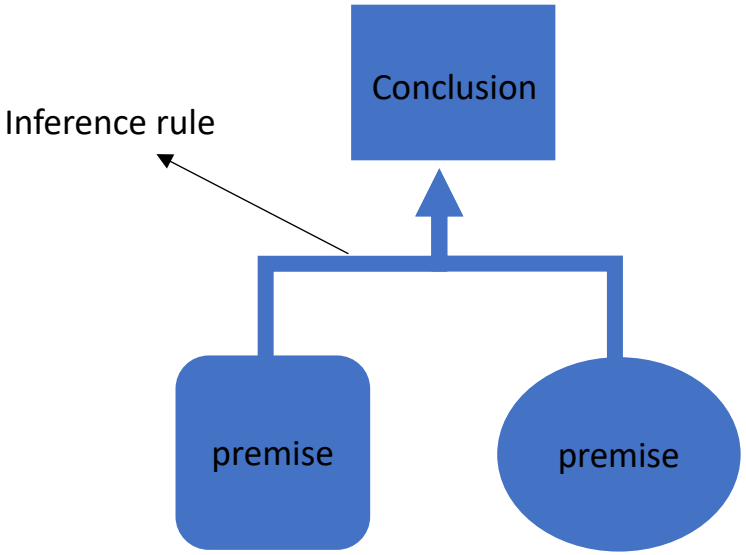
A4 = [A1, A2, A3 ⇒ rain]

⇒ PRESUMPTIVE Unlike a strict rule (→), this inference can be defeated.

CRITICAL QUESTIONS

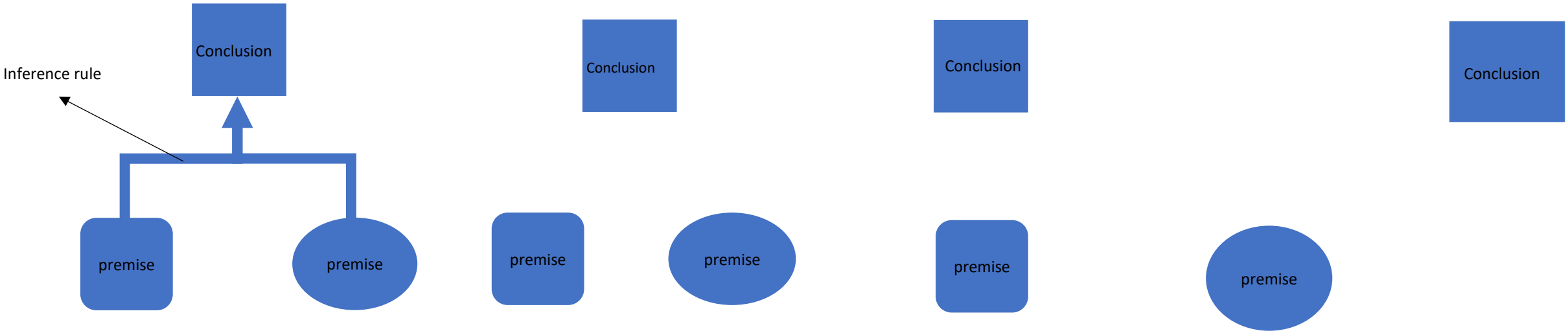
Is Lee credible? Do other experts disagree?

ASPIC+ notation: Modgil & Prakken (2014).

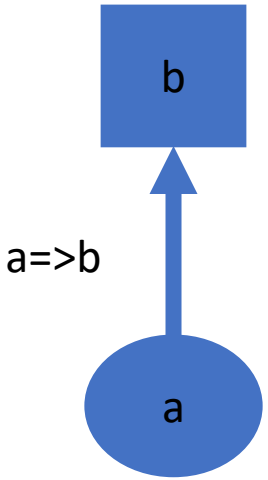


Complete Argument

Enthymemes:
arguments with incomplete logical structure



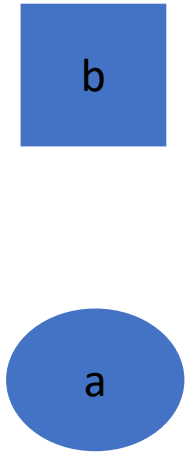
Enthymemes:
arguments with incomplete logical structure



It’s raining outside. (a)

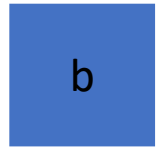
Take an umbrella with you. (b)

If it’s raining outside, then take an umbrella with you. (a=>b)

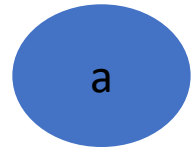


It’s raining outside.

Take an umbrella with you.



Take an umbrella with you.



It’s raining outside.

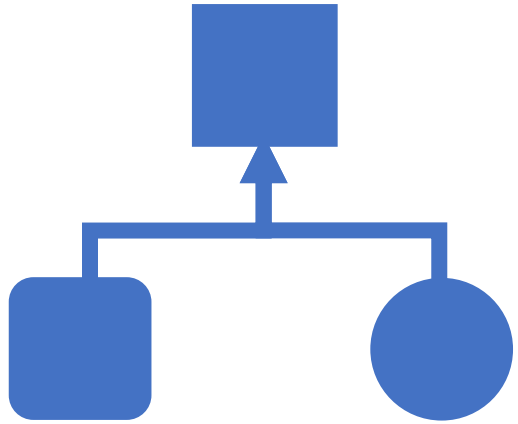
Enthymemes



Backward extension

Enthymemes

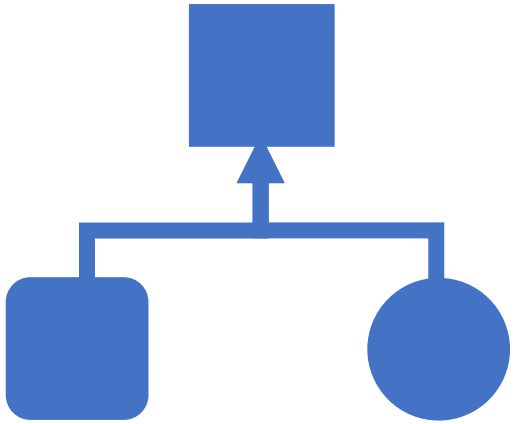
Backward extension

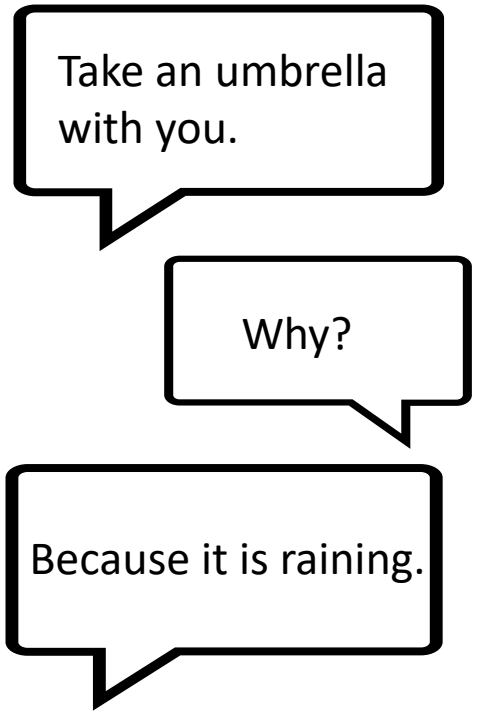


Enthymemes

Backward extension

Locutions: *why*, *because*

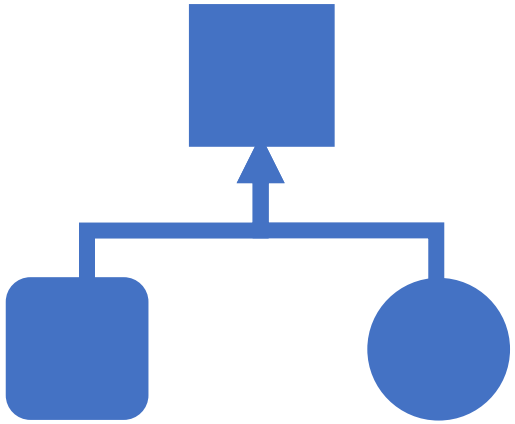


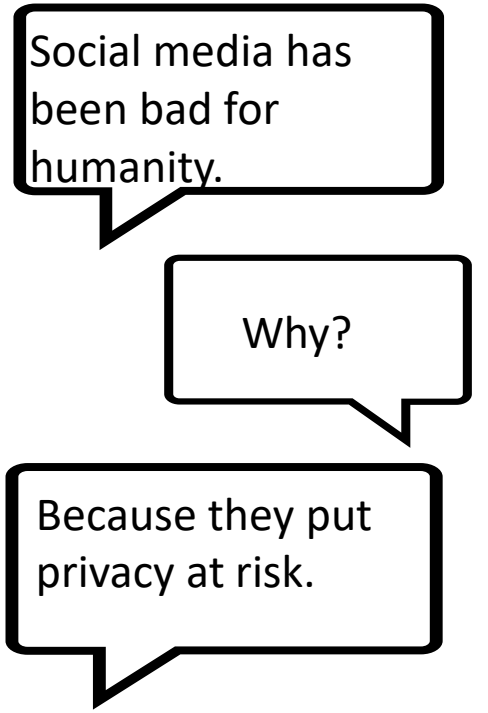


Enthymemes

Backward extension

Locutions: *why*, *because*





Enthymemes



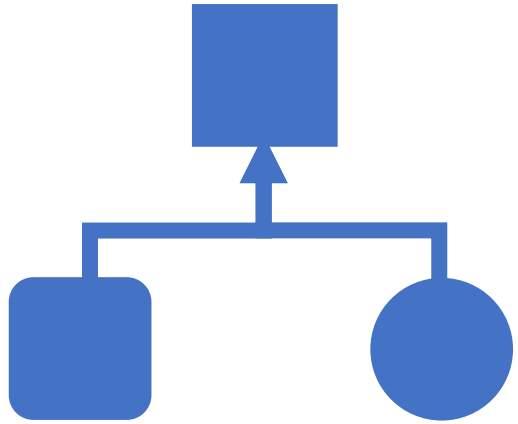
Enthymemes

Forward extension



Enthymemes

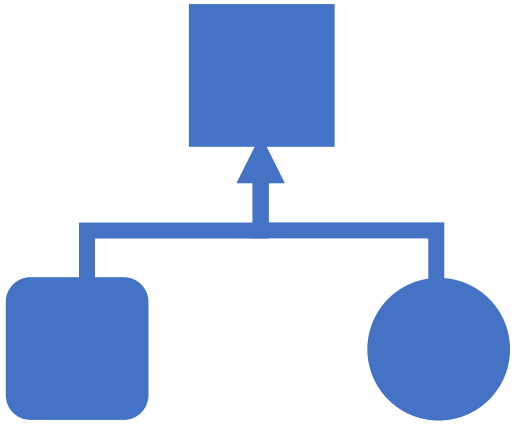
Forward extension

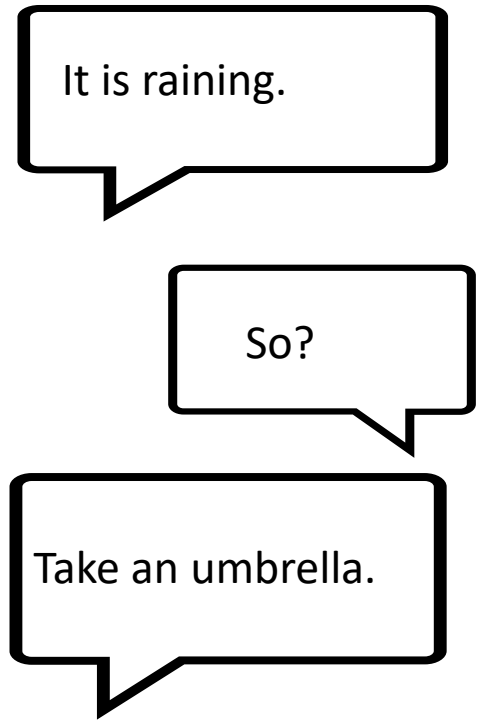


Enthymemes

Forward extension

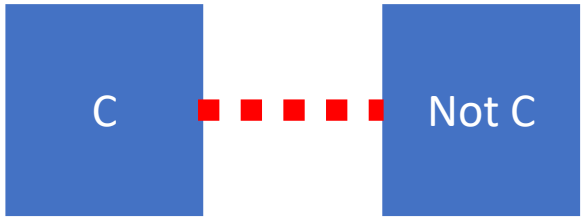
Locutions: *and-so*, *hence*



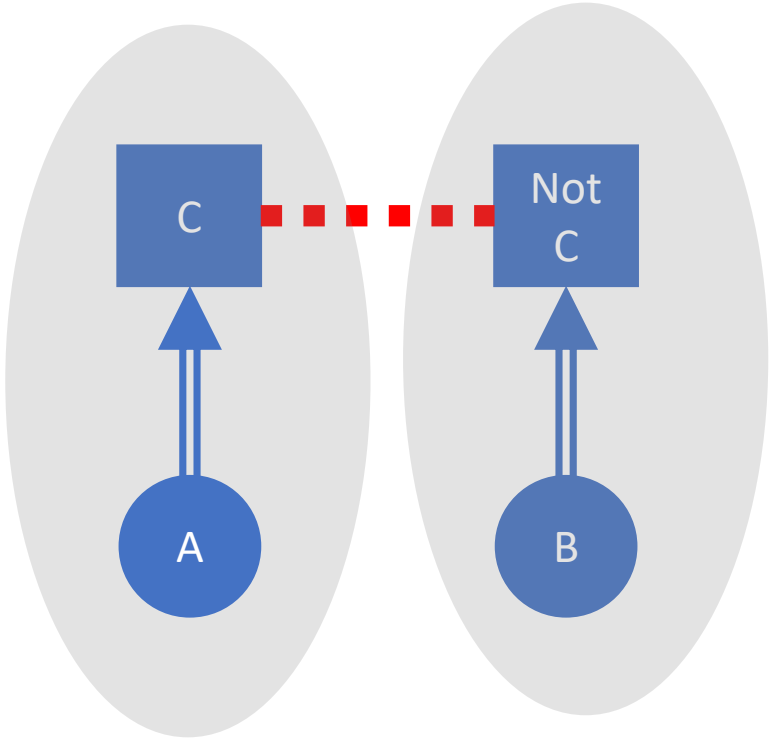


Attacks

C: The pencil is green.



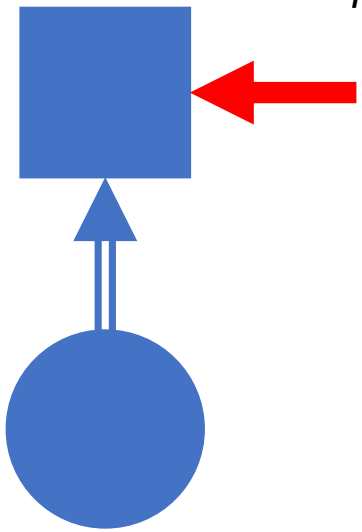
Attacking Arguments

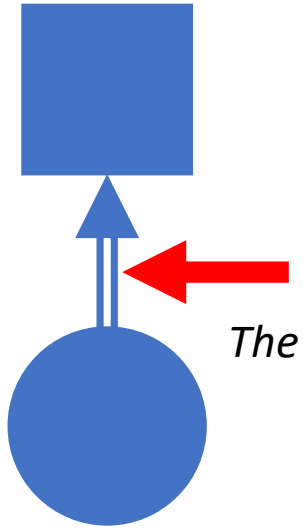


A: Elfia says she sees that the pencil is green.
B: Maddie sees that the pencil is not green.
r: If I see the pencil color X, then it is color X.

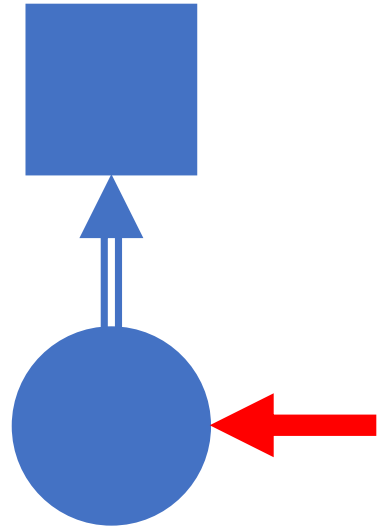
Points of attack

The pencil is red.



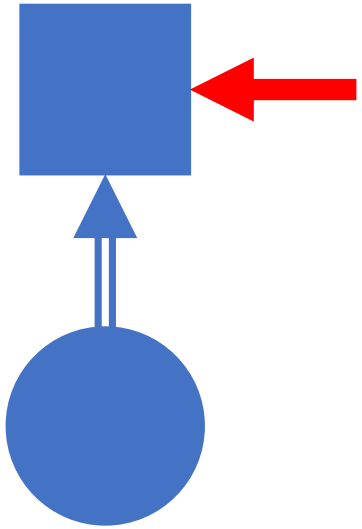


The pencil is lit with a green light.

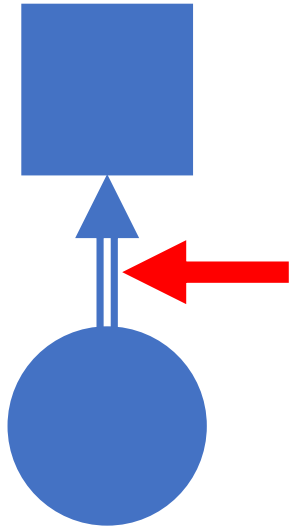


You don’t see the pencil green (eg, you are lying).

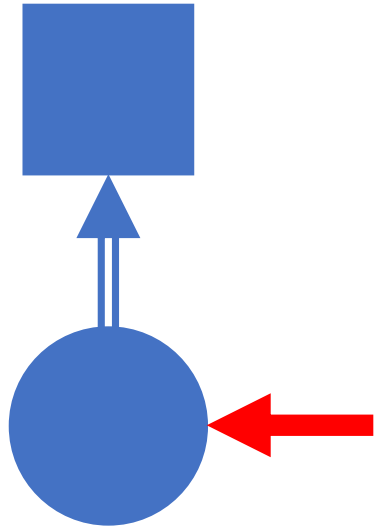
Points of attack



Rebuttal

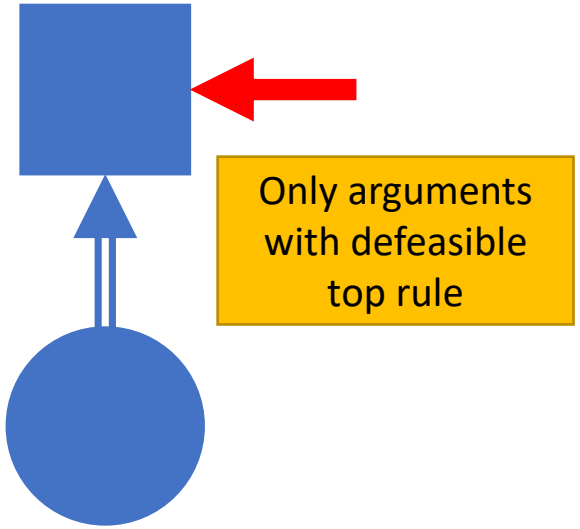


Undercutting



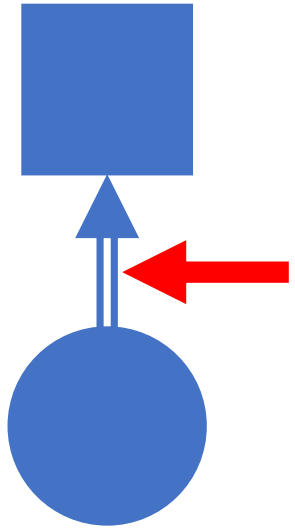
Undermining

Points of attack

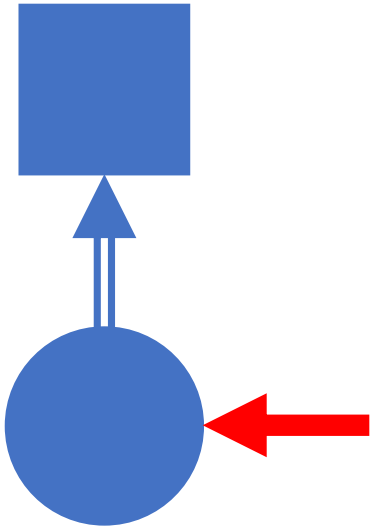


Only arguments with defeasible top rule

Rebuttal

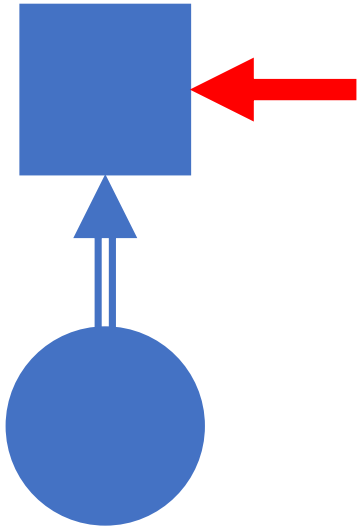


Undercutting

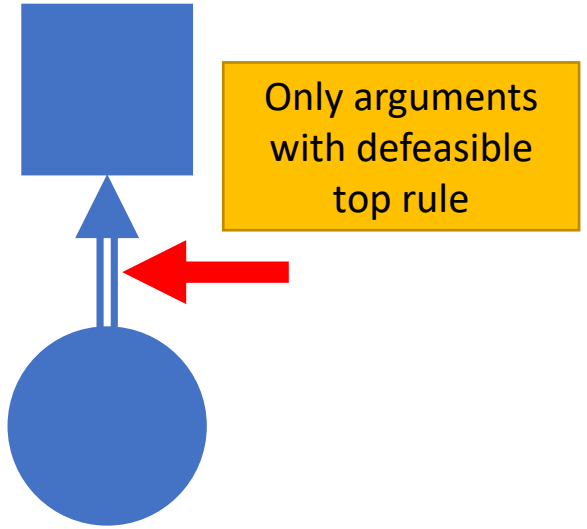


Undermining

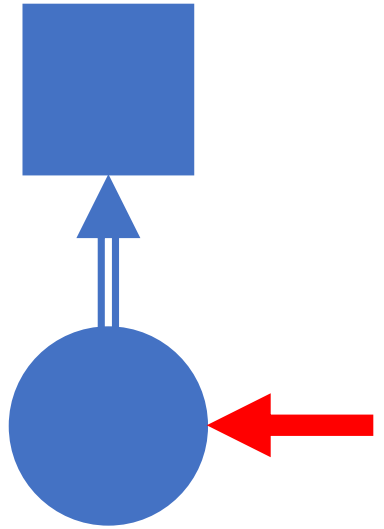
Points of attack



Rebuttal

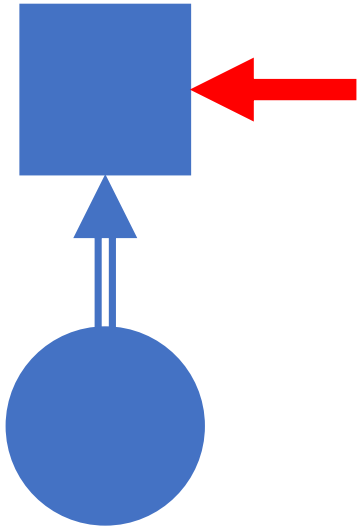


Undercutting

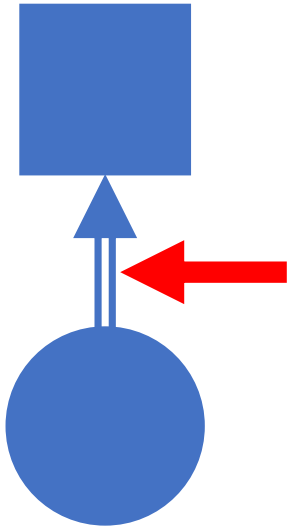


Undermining

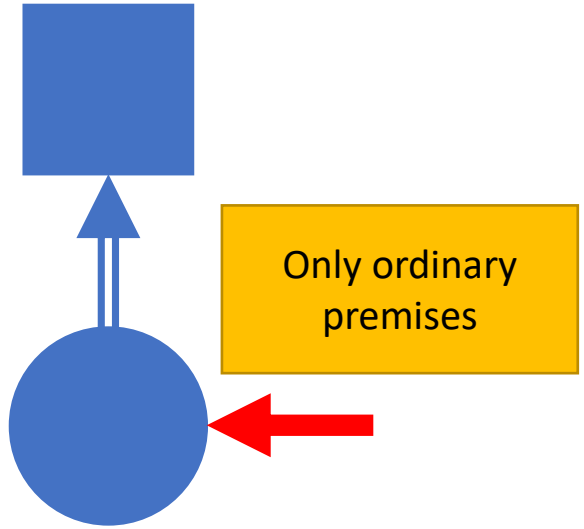
Points of attack



Rebuttal

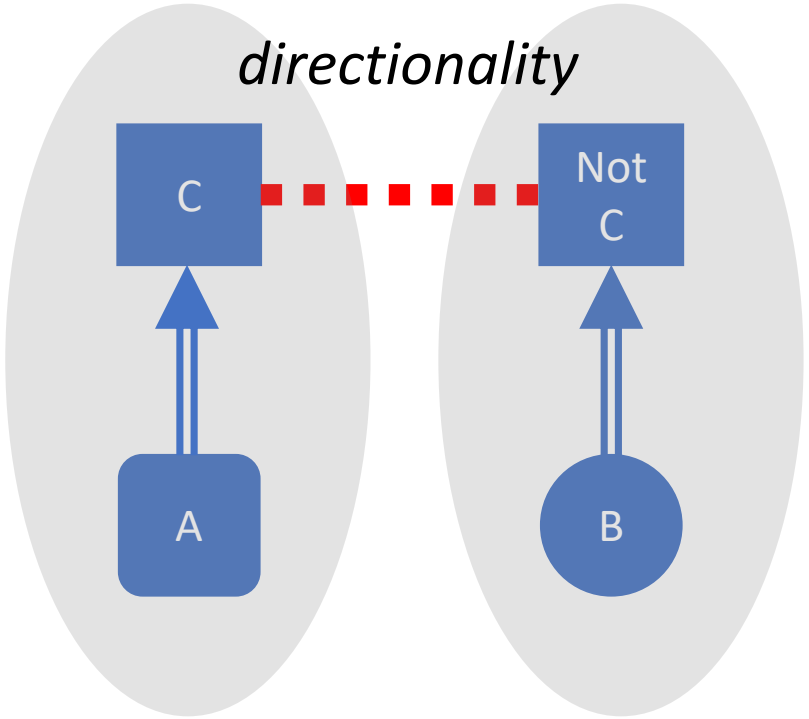


Undercutting



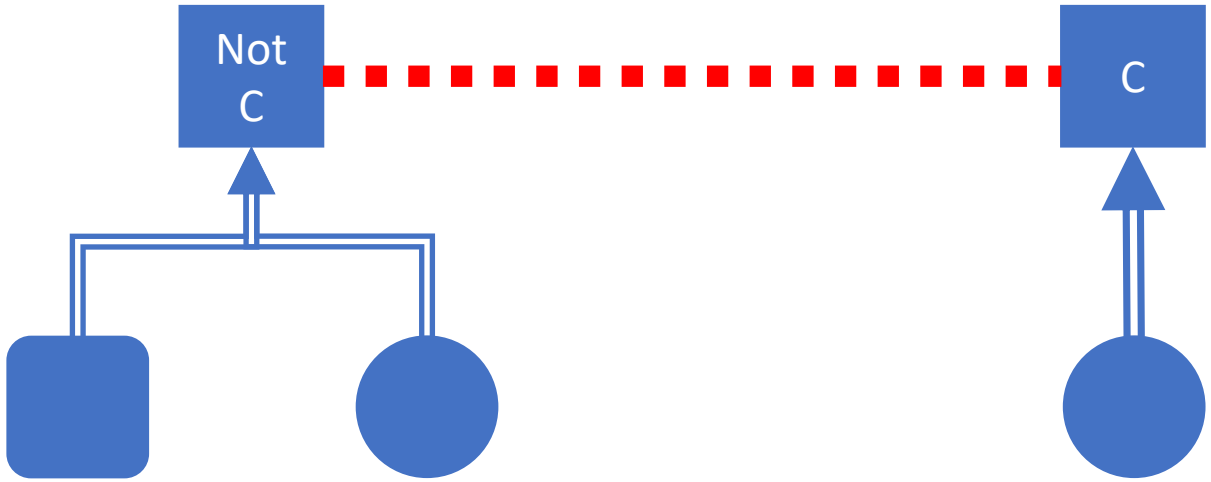
Undermining

Attacking Arguments



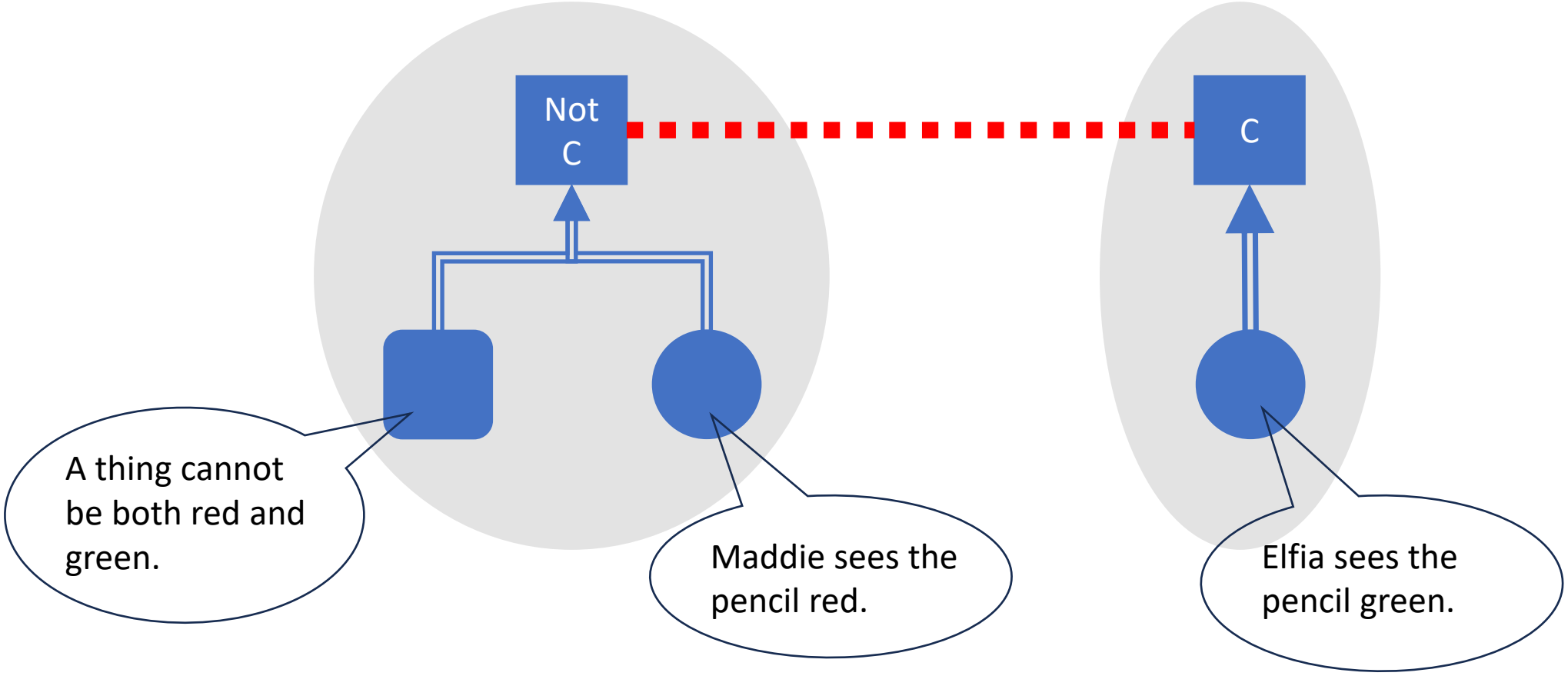
How do we resolve?

Rebuttal attack (symmetric)

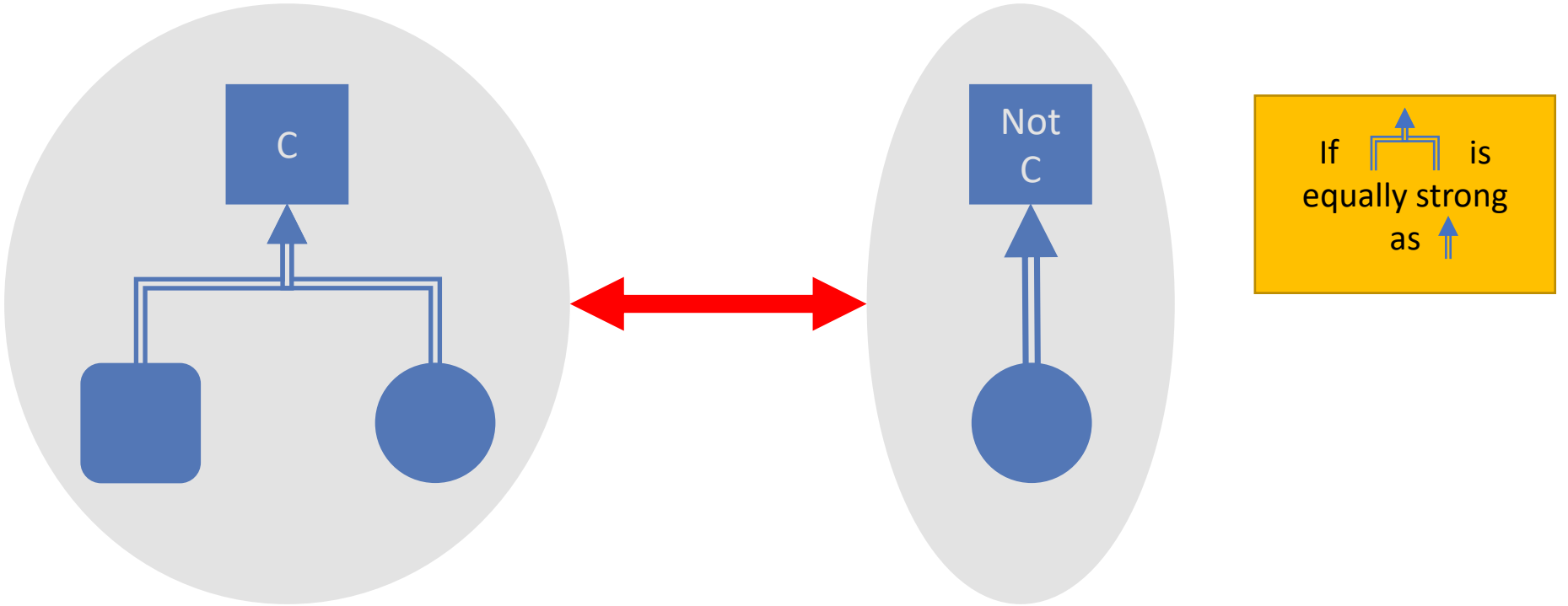


Rebuttal attack (symmetric)

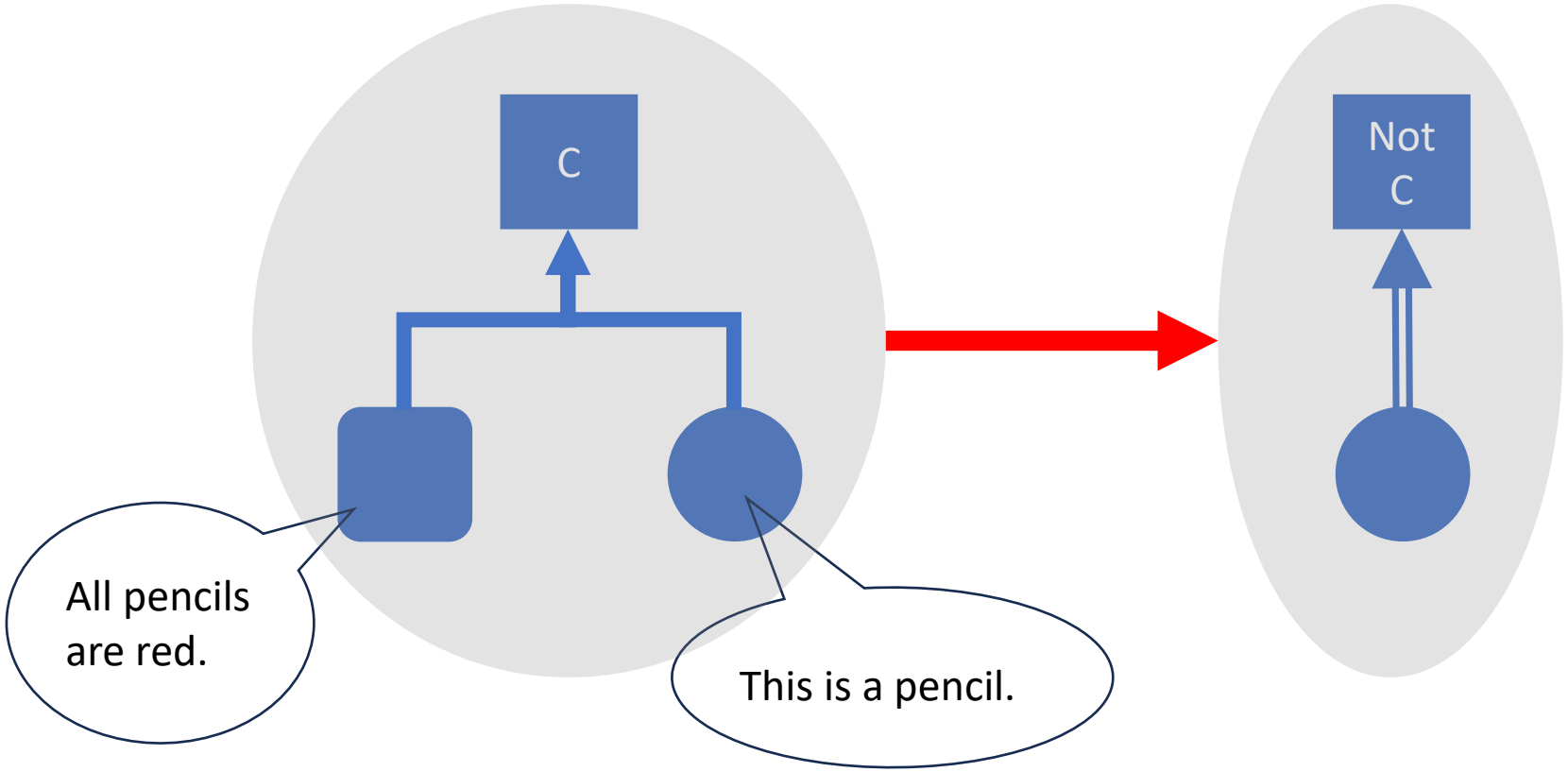
C: The pencil is green.



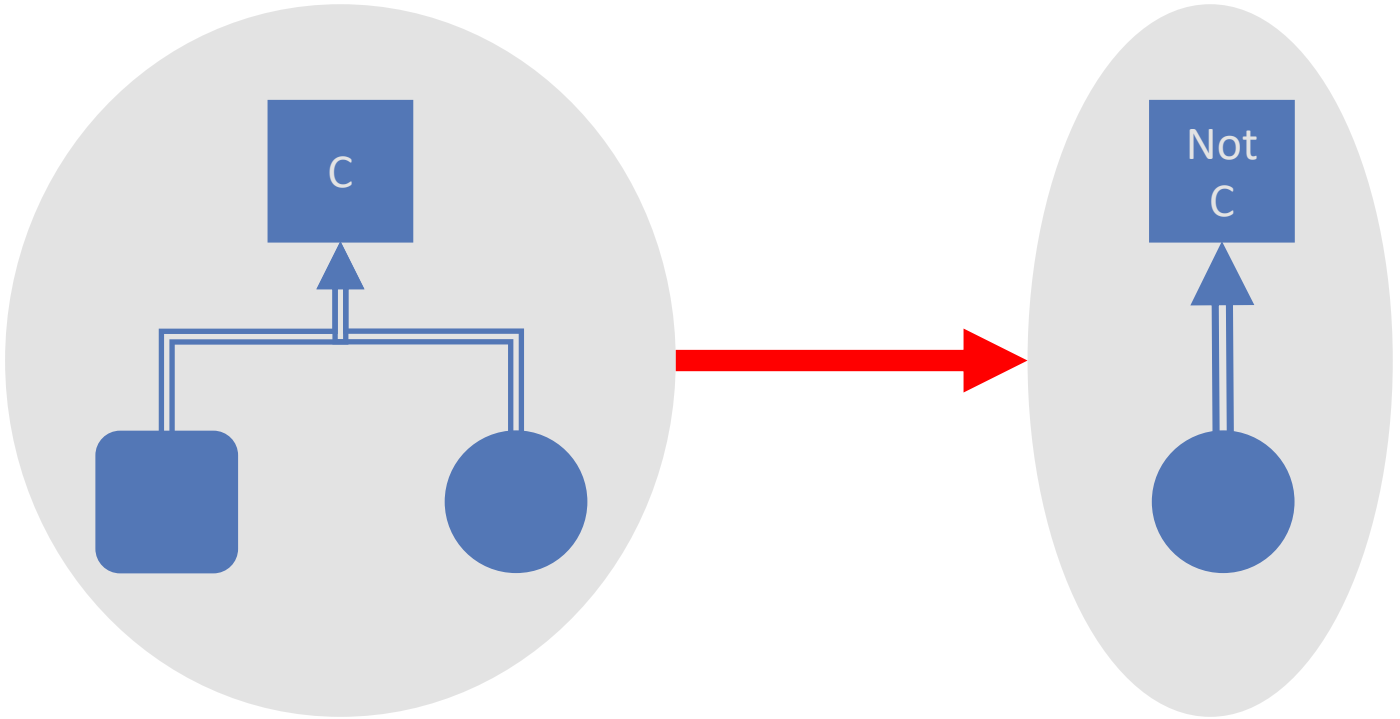
Rebuttal attack (symmetric)



Rebuttal attack (asymmetric)



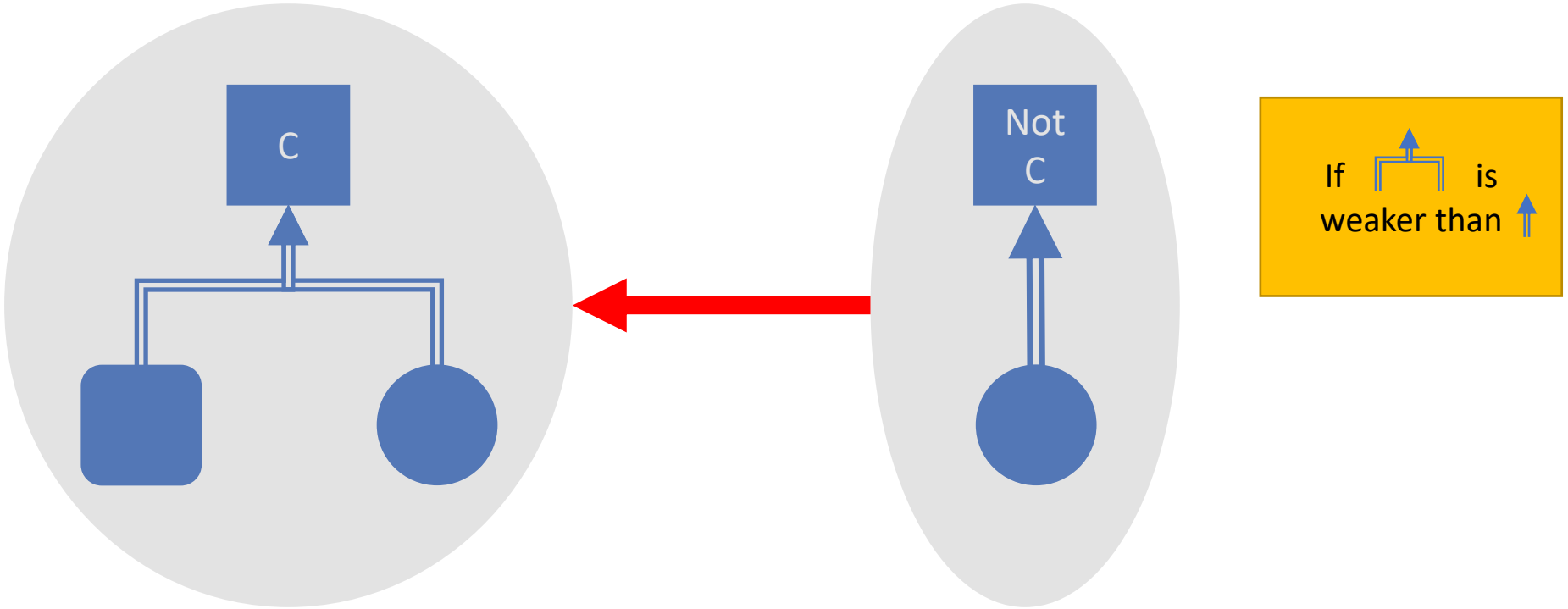
Rebuttal attack (asymmetric)



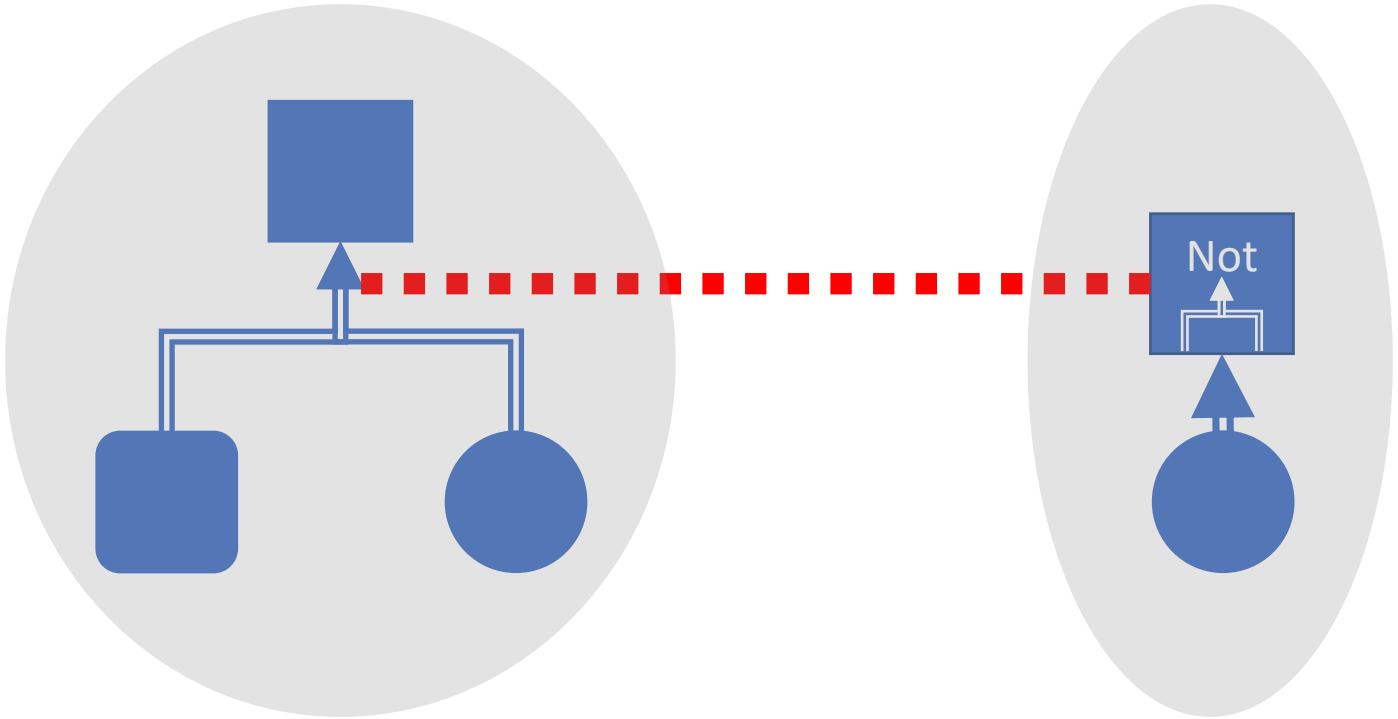
If  is stronger than 

Eg, argument from expert opinion > argument from example

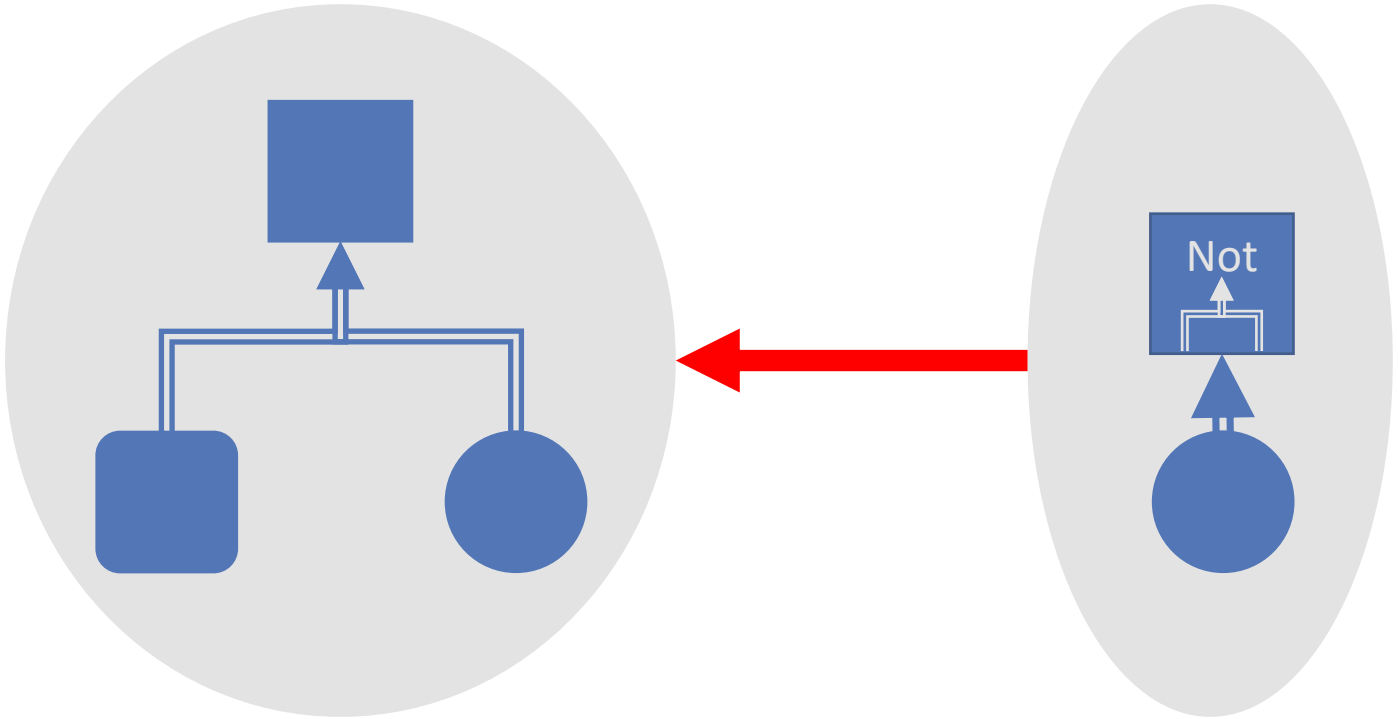
Rebuttal attack (asymmetric)



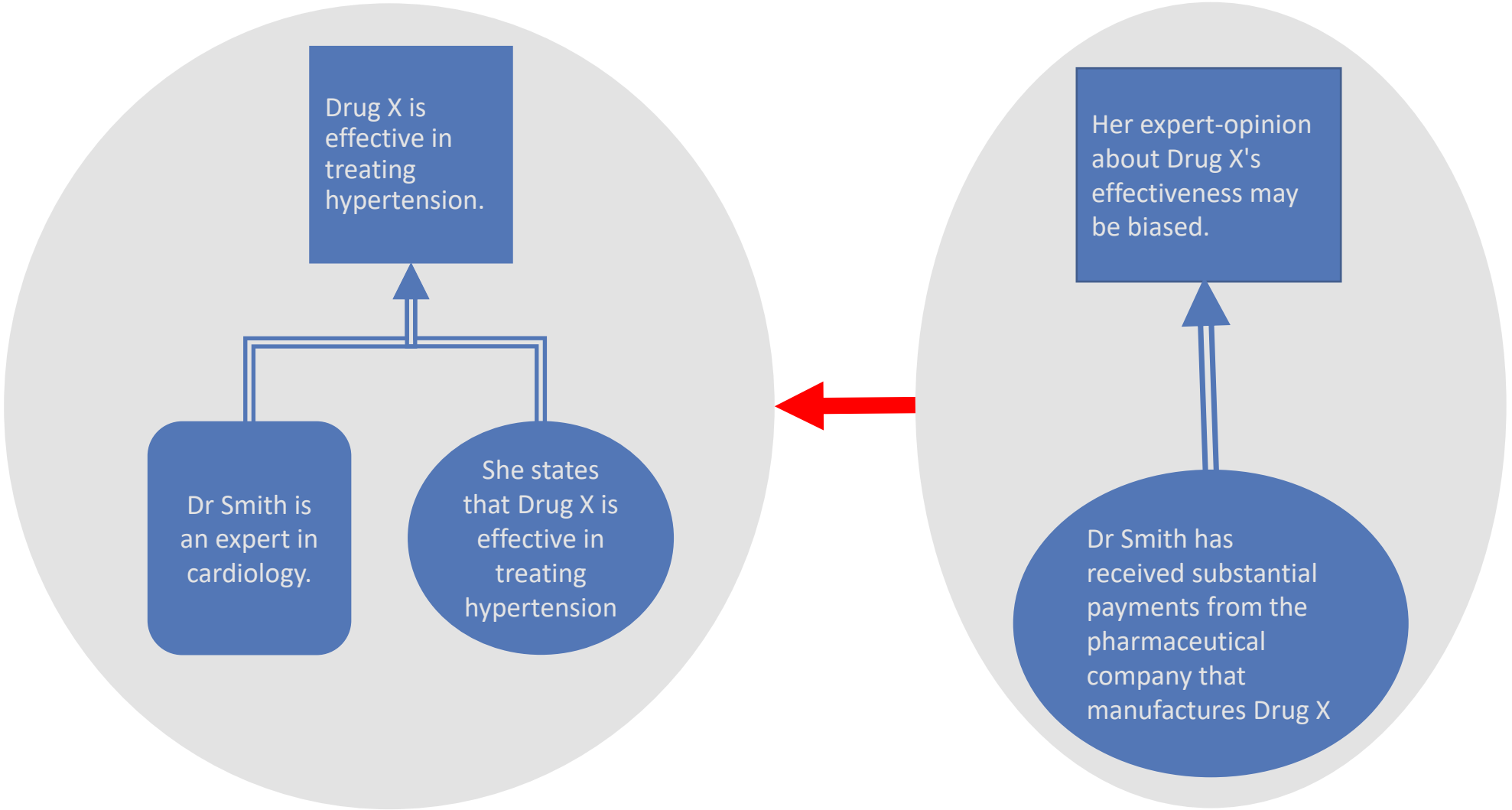
Undercutting attack (always asymmetric)



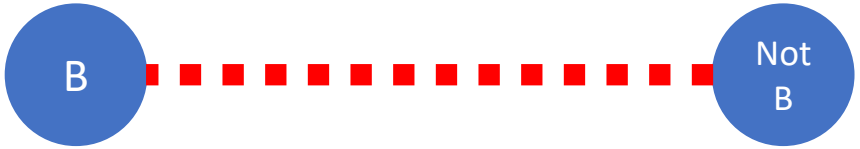
Undercutting attack (always asymmetric)



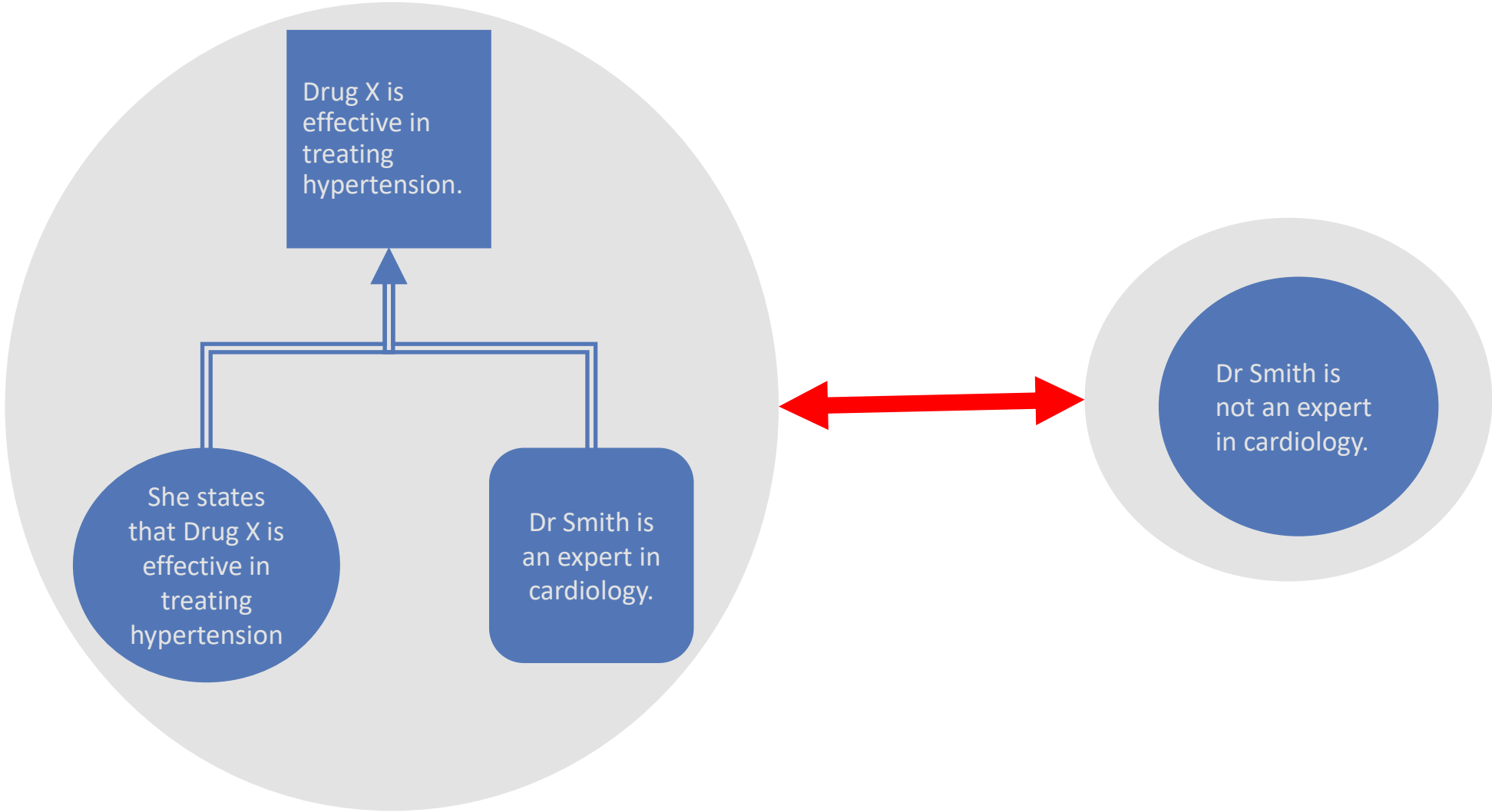
Undercutting attack (always asymmetric)



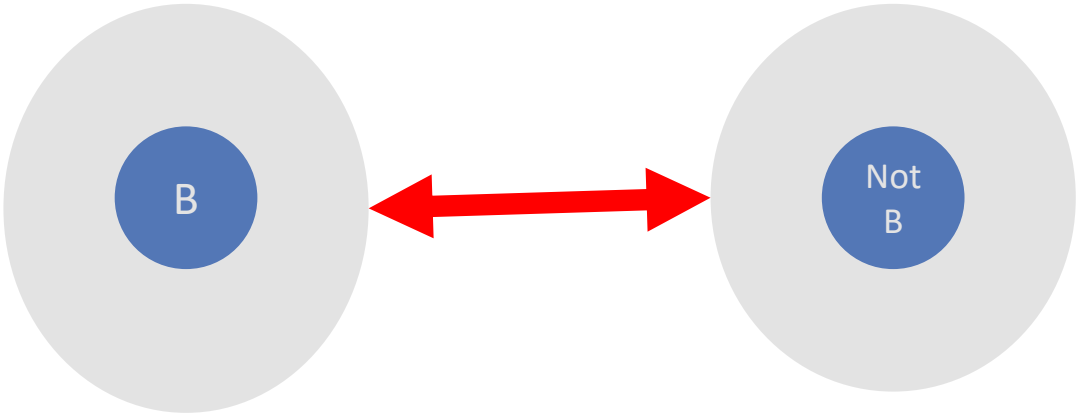
Undermining attack (symmetric)



Undermining attack (symmetric)

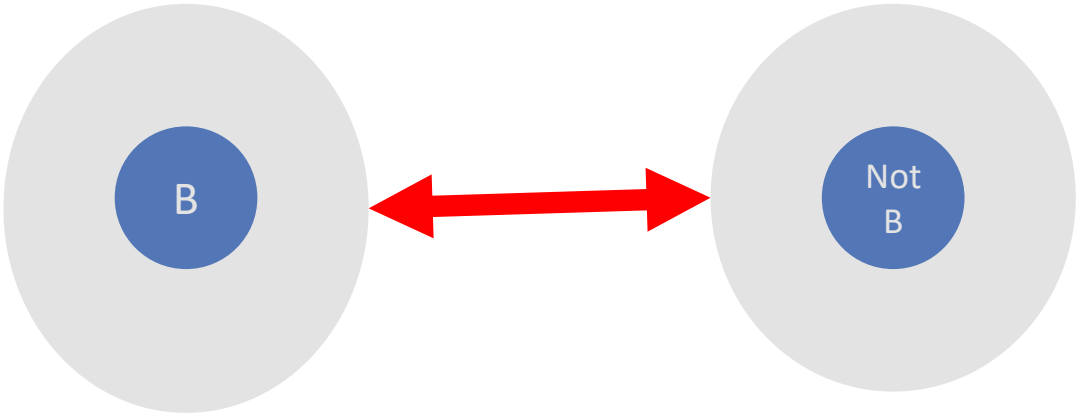


Undermining attack (symmetric)



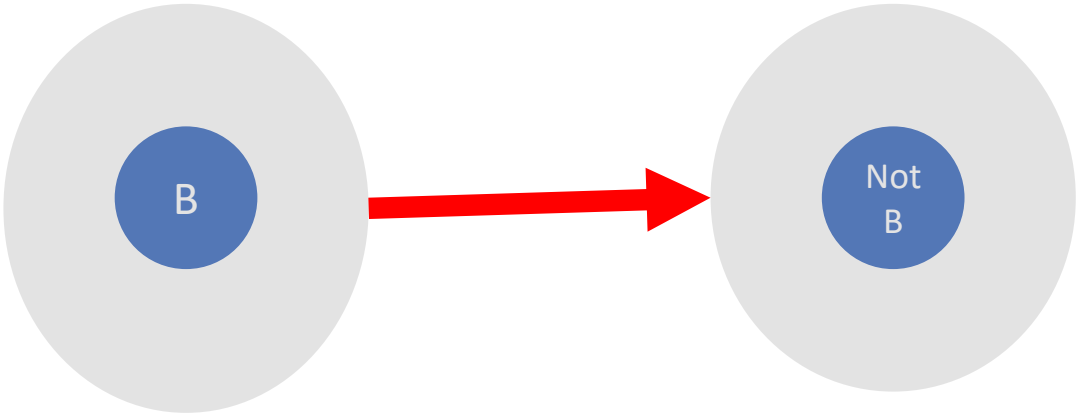
Undermining attack (symmetric)

If “B” is equally strong as “Not B”



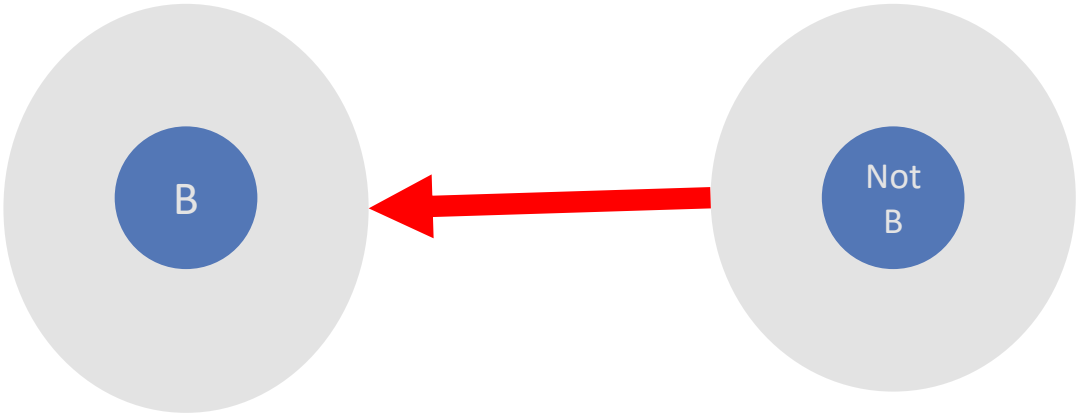
Undermining attack (asymmetric)

If “B” stronger than “Not B”

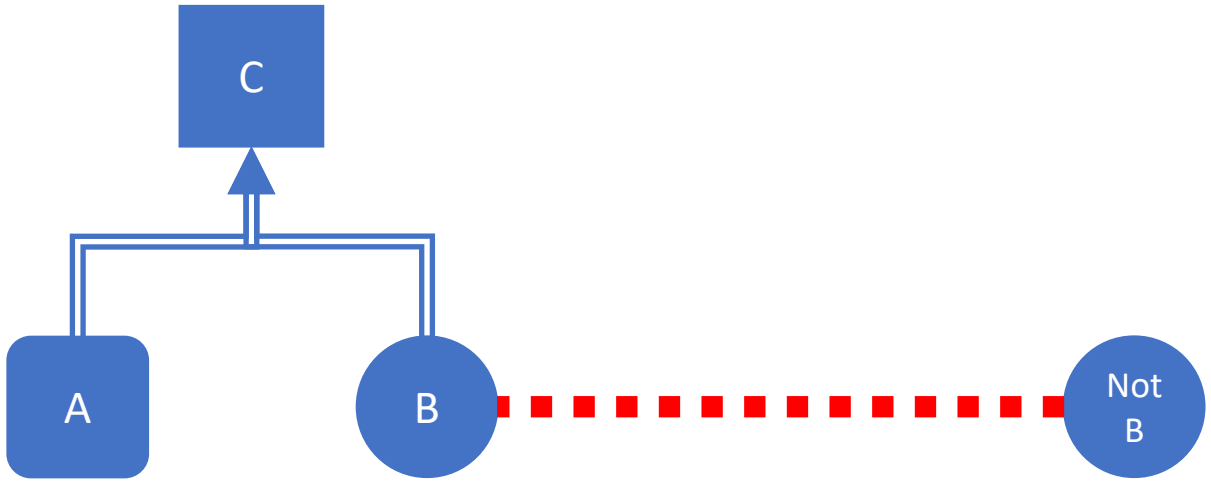


Undermining attack (asymmetric)

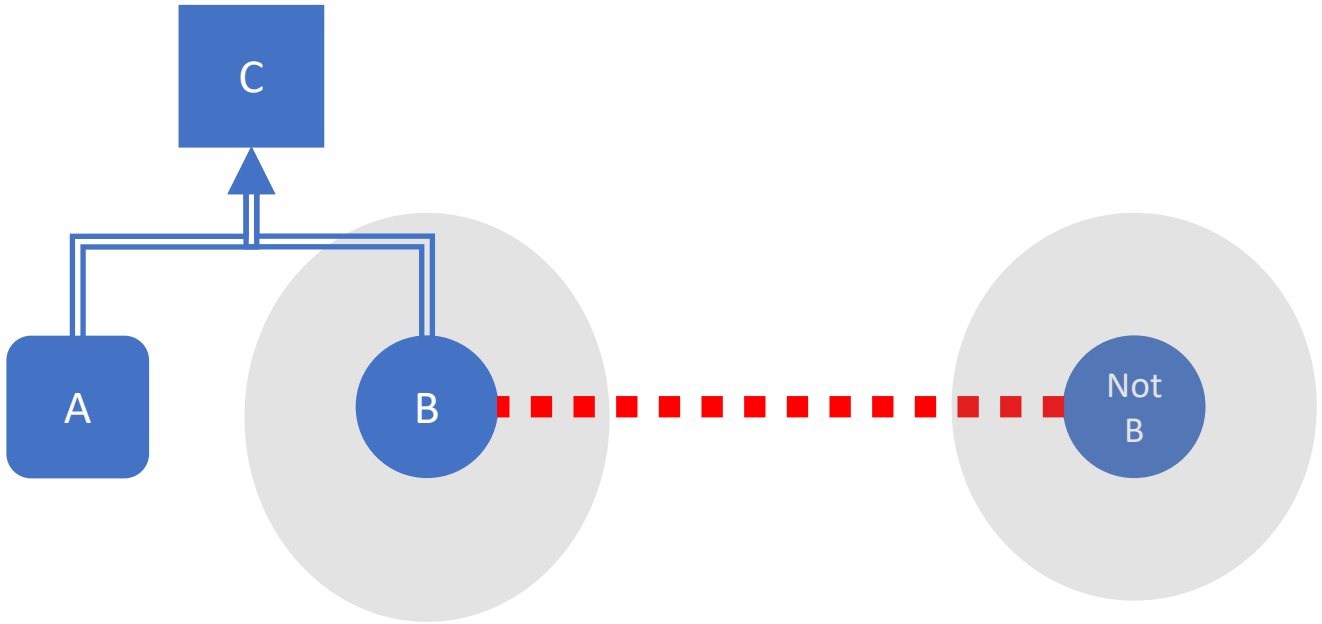
If “B” weaker than “Not B”



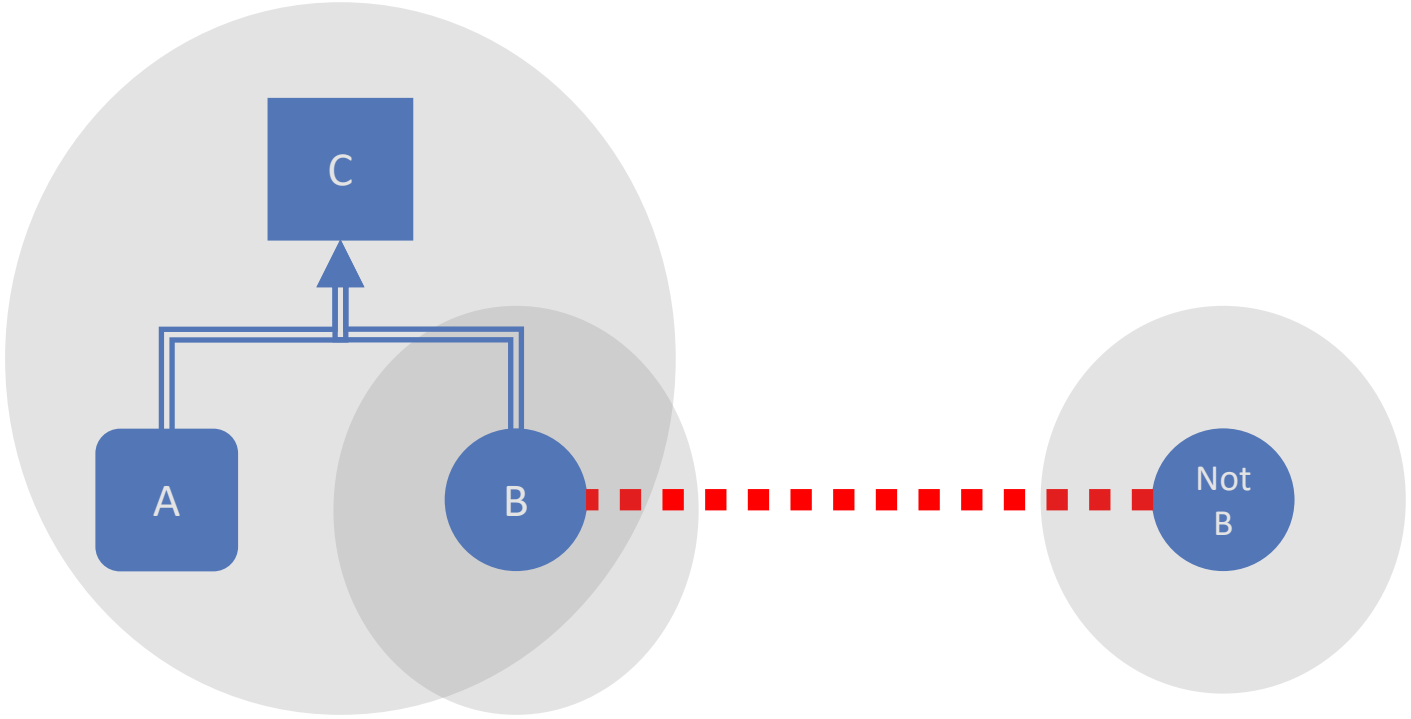
Indirect attacks



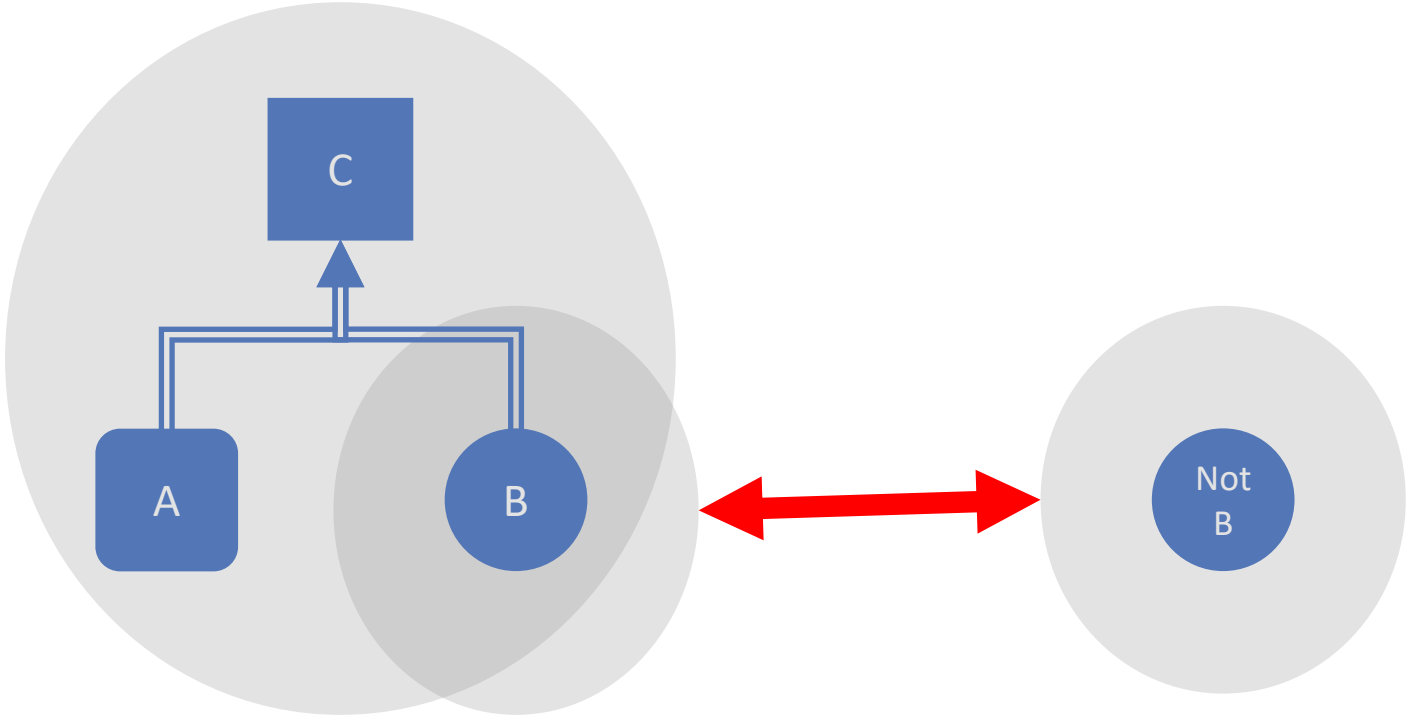
Indirect attacks



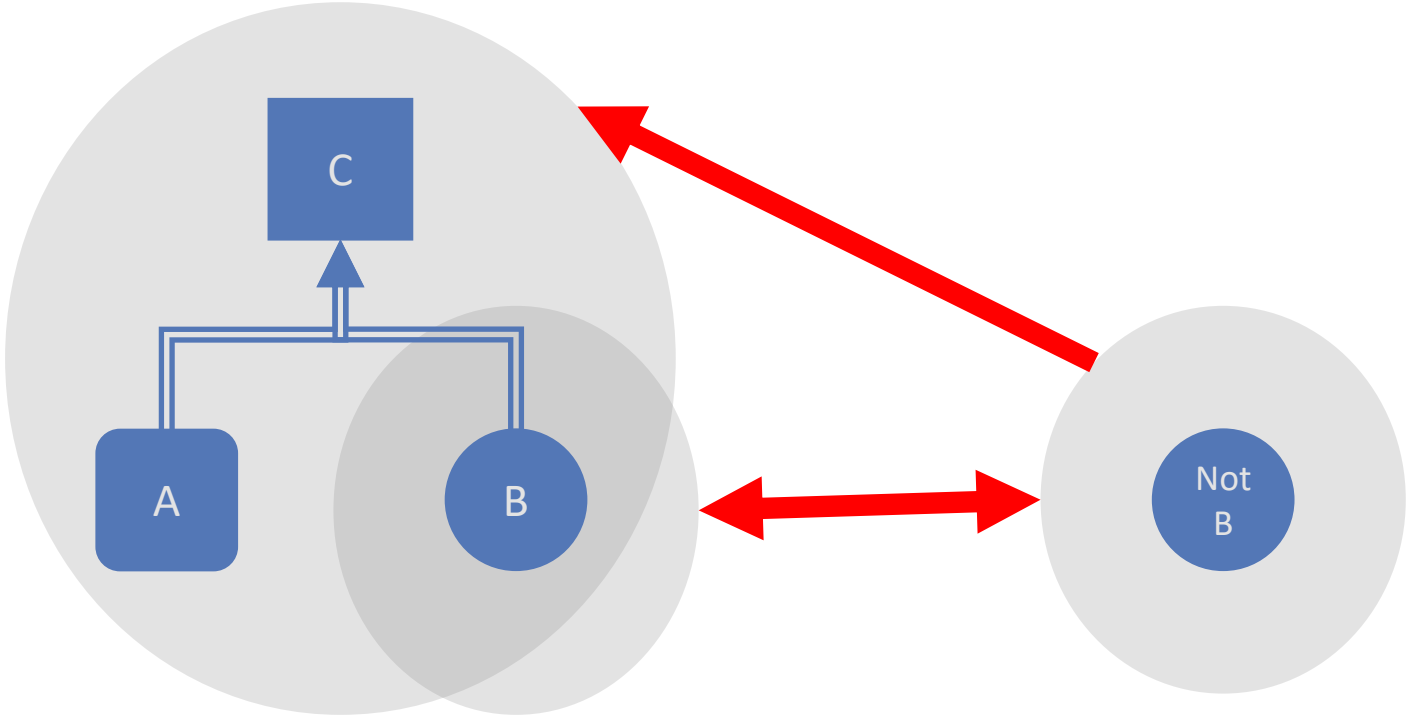
Indirect attacks



Indirect attacks



Indirect attacks



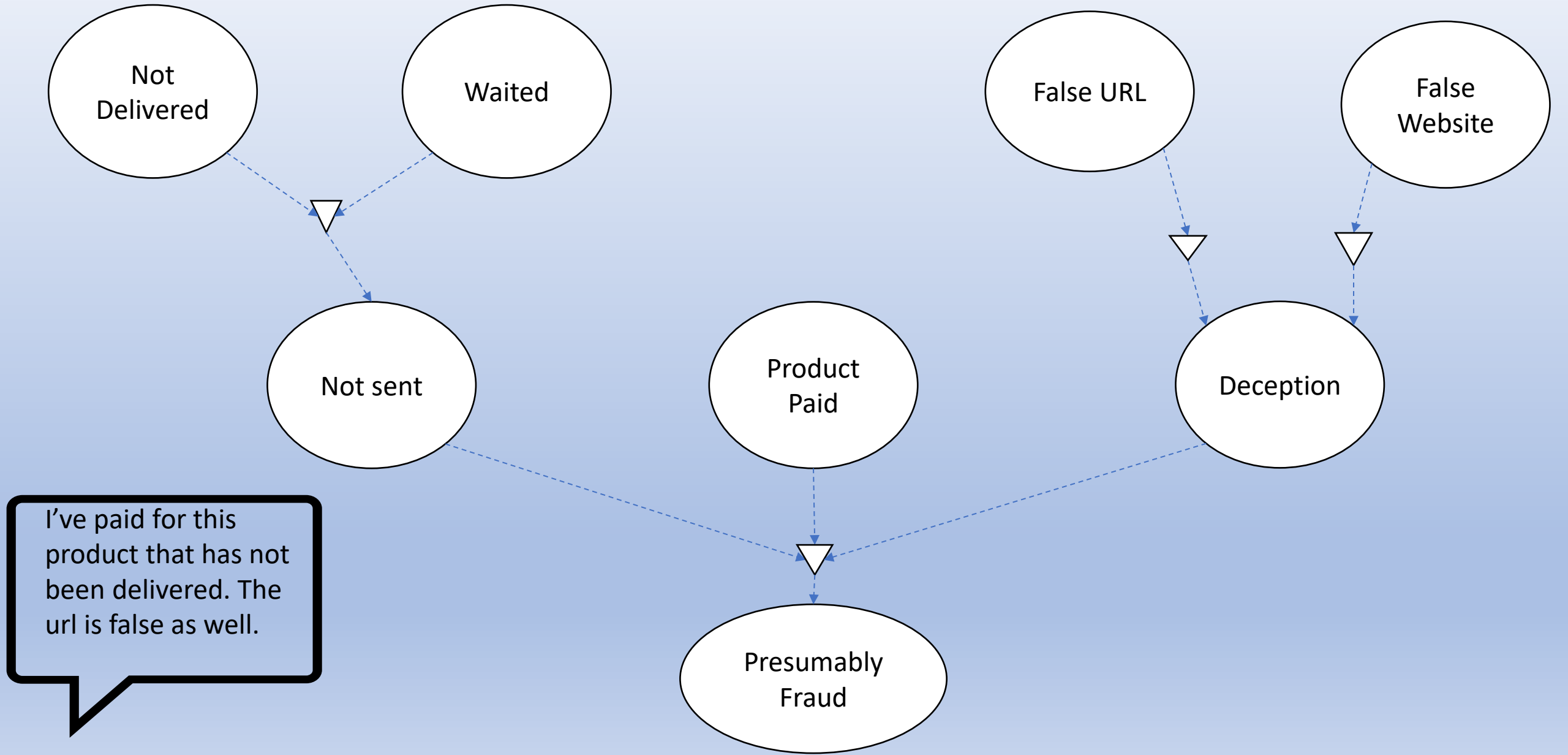
Quest Time!

Go back to your arguments and discuss within your groups:

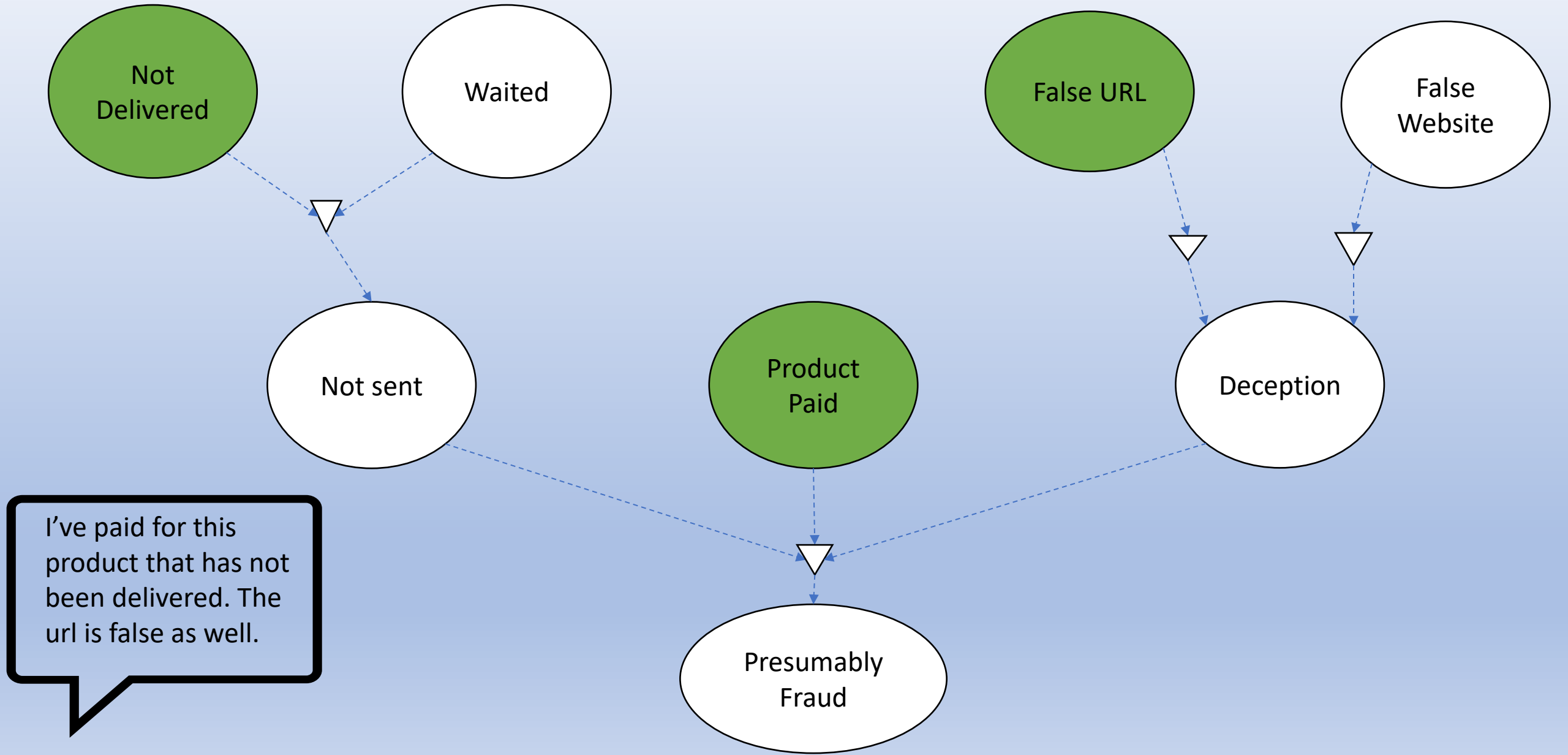
- Can you represent them as premise, rule, conclusion?
- Did you use enthymemes?
  - How did you find out the missing elements?
- How (where) do they attack each other (which attack types)?
  - Rebuttal (attacking conclusions)
  - Undercutting (attacking rules)
  - Undermining (attacking premises)

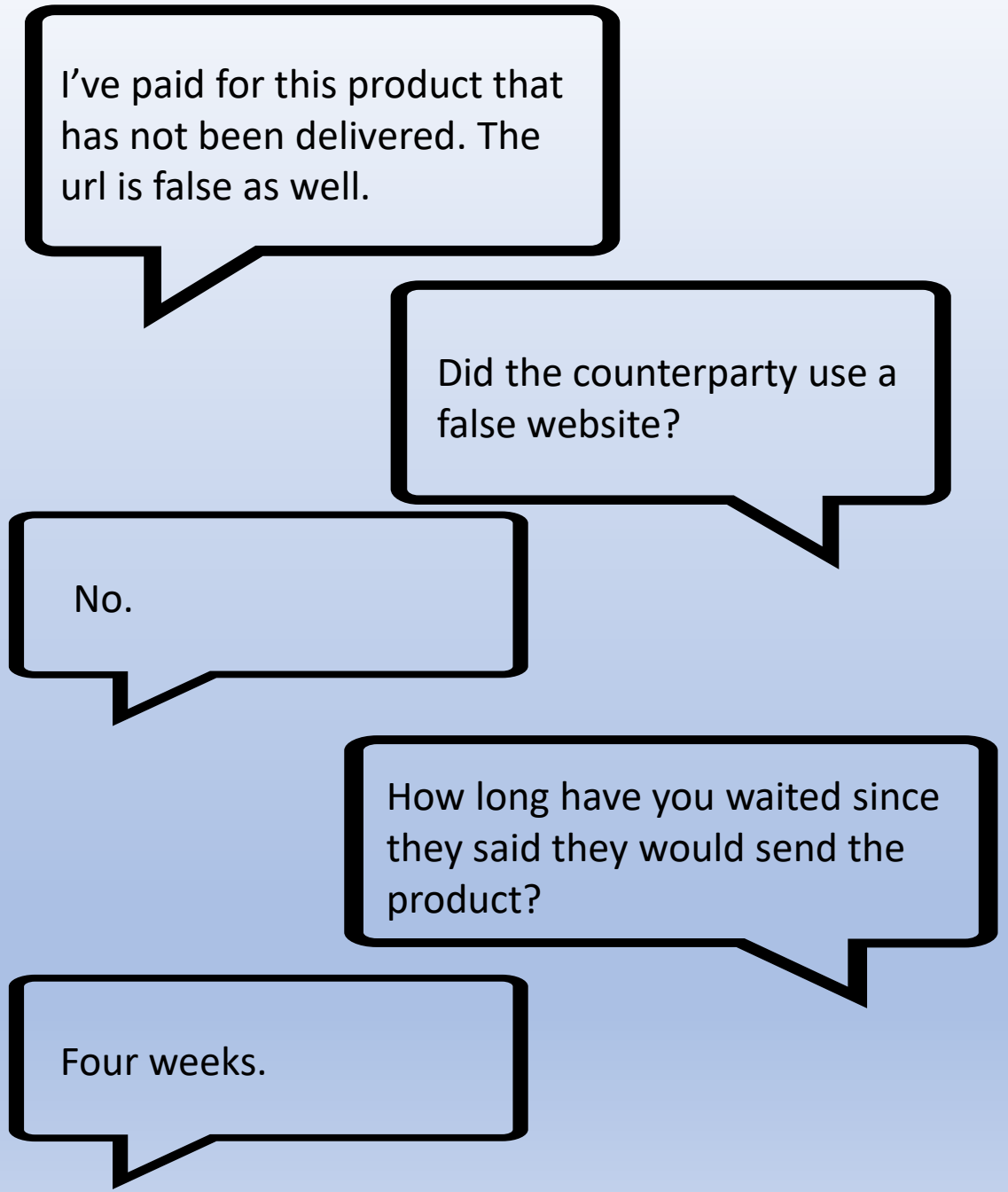

FROM THEORY TO PRACTICE:
ARGUMENTATION IN ACTION
POLICE
POLICE
POLICE

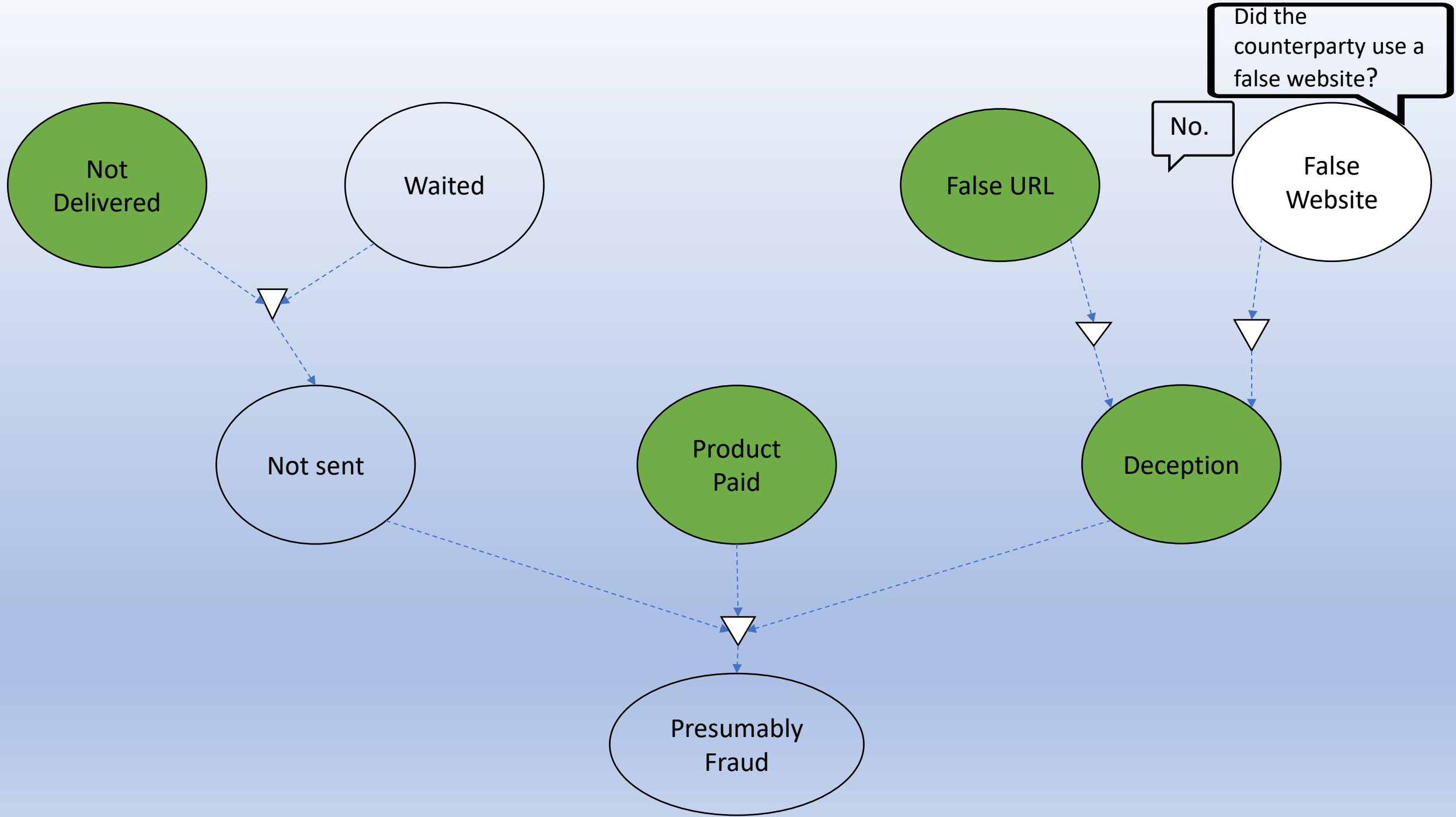
Application: Dutch Police

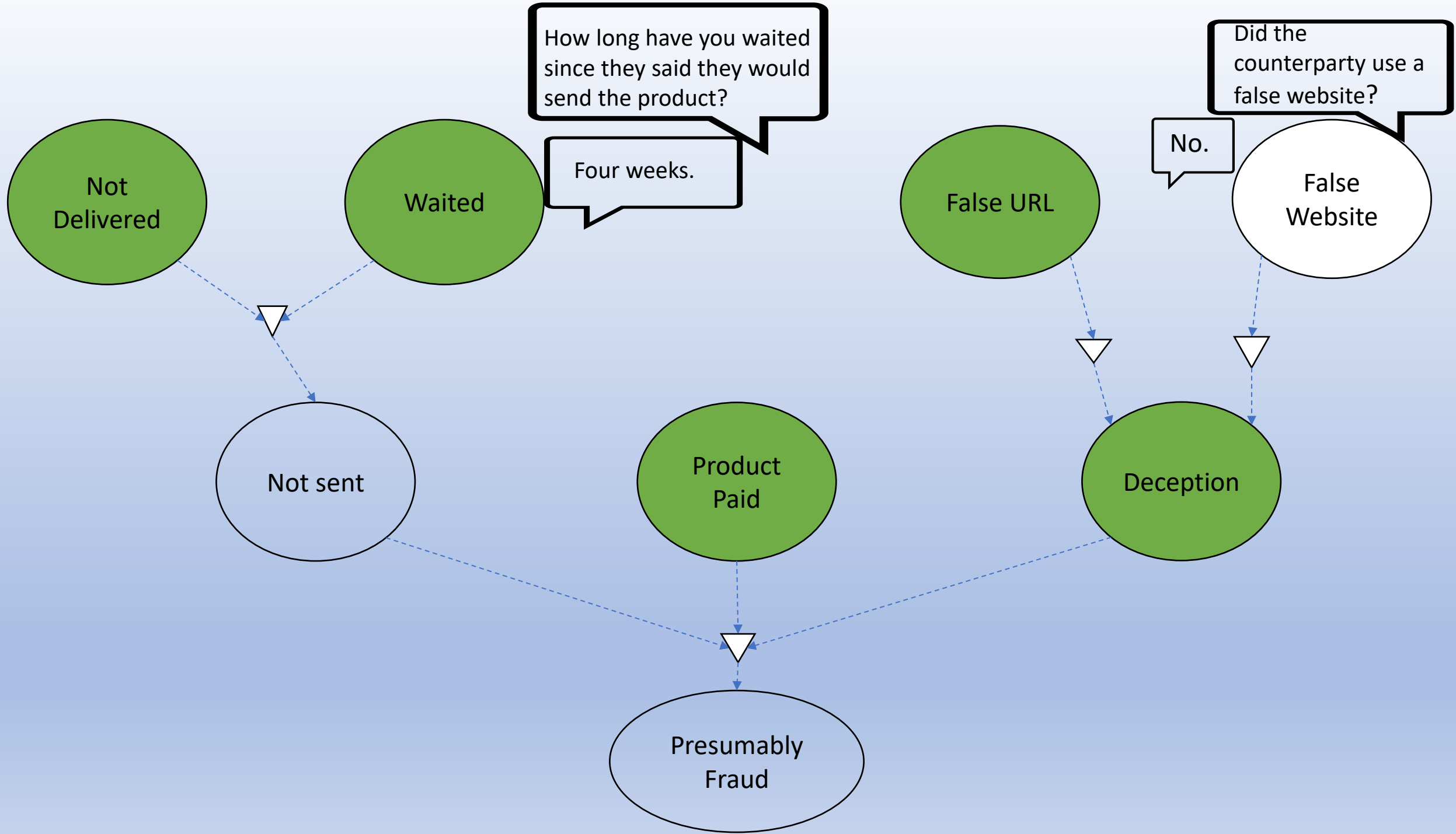


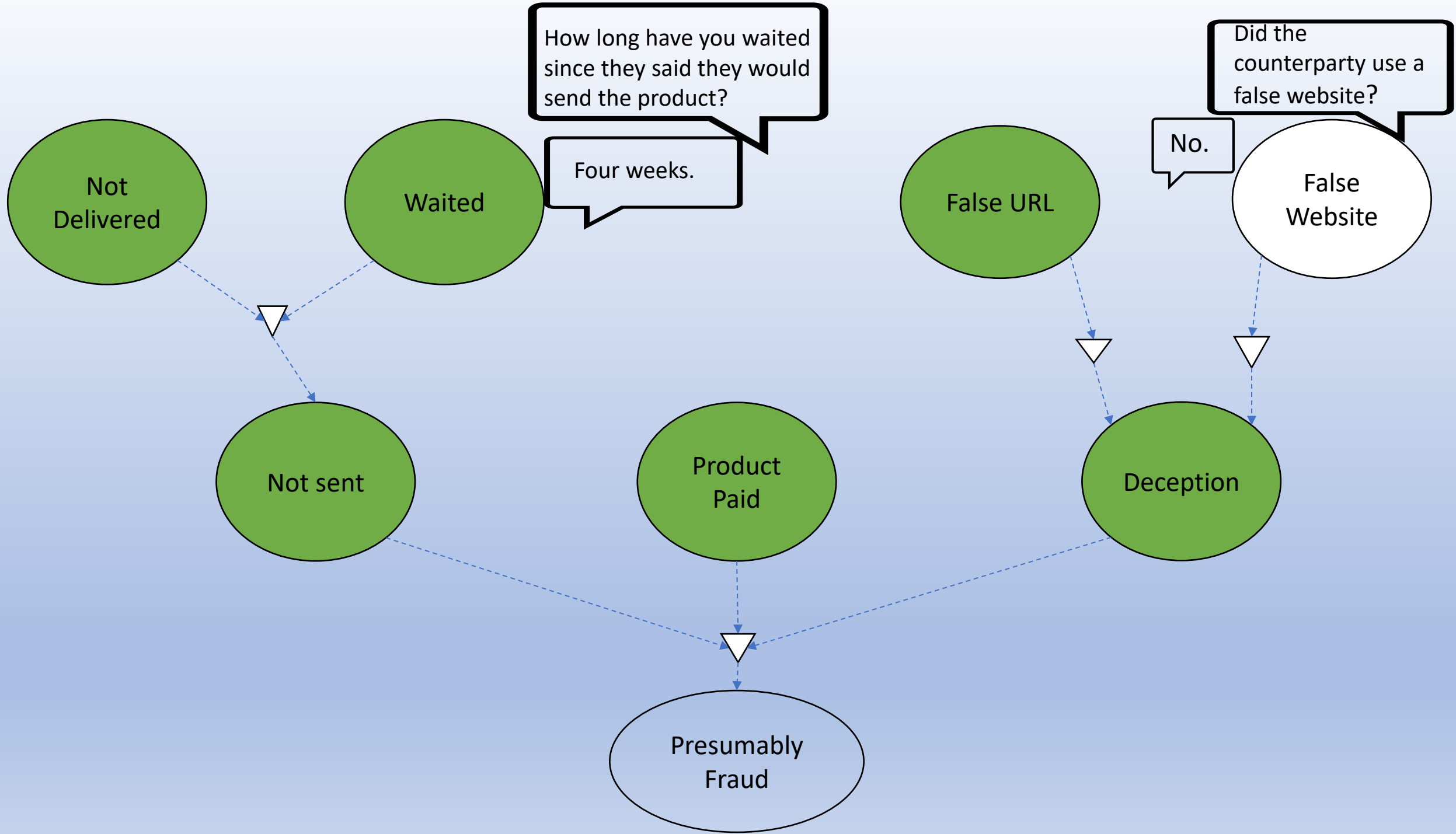
Application: Dutch Police

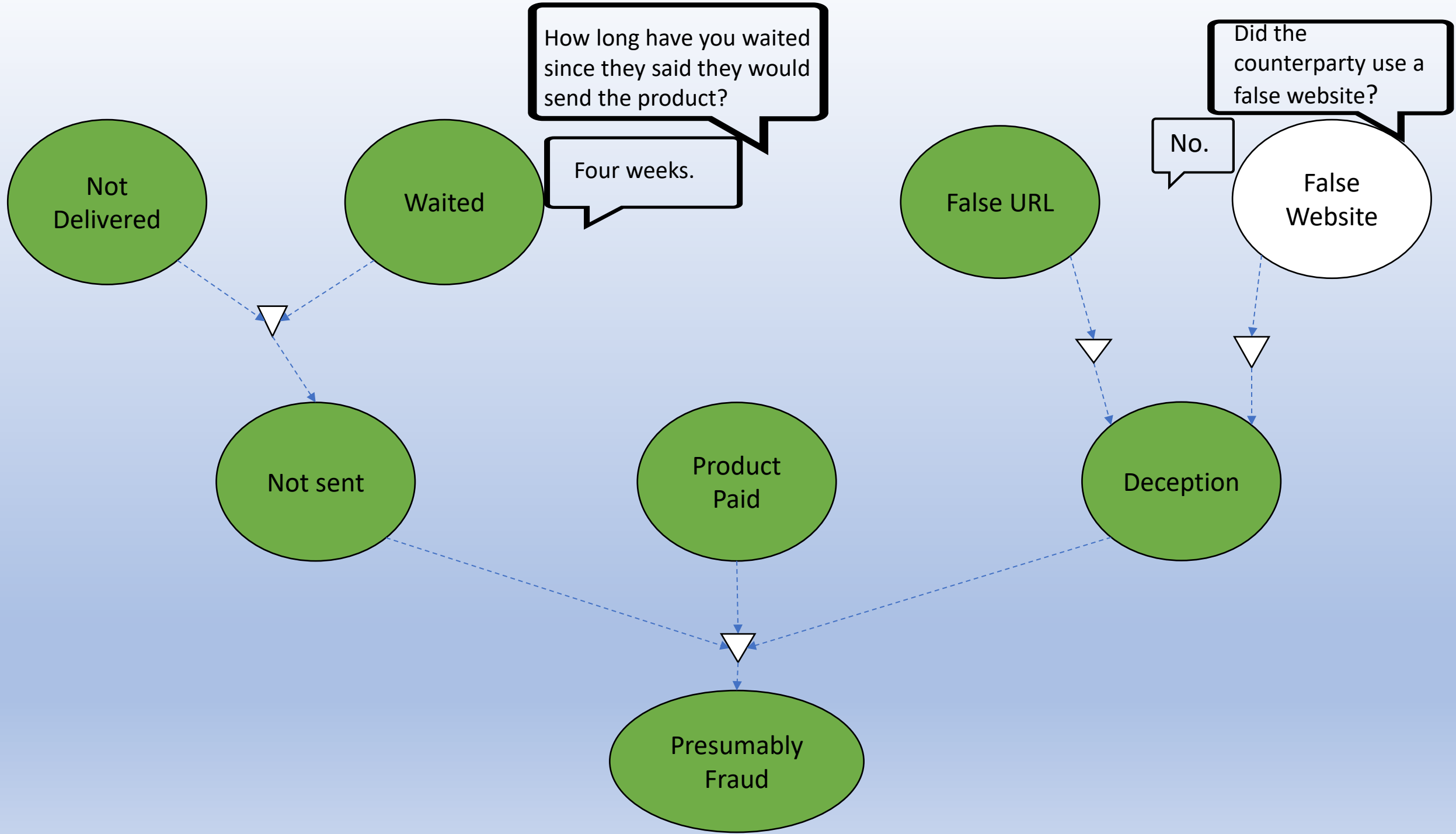


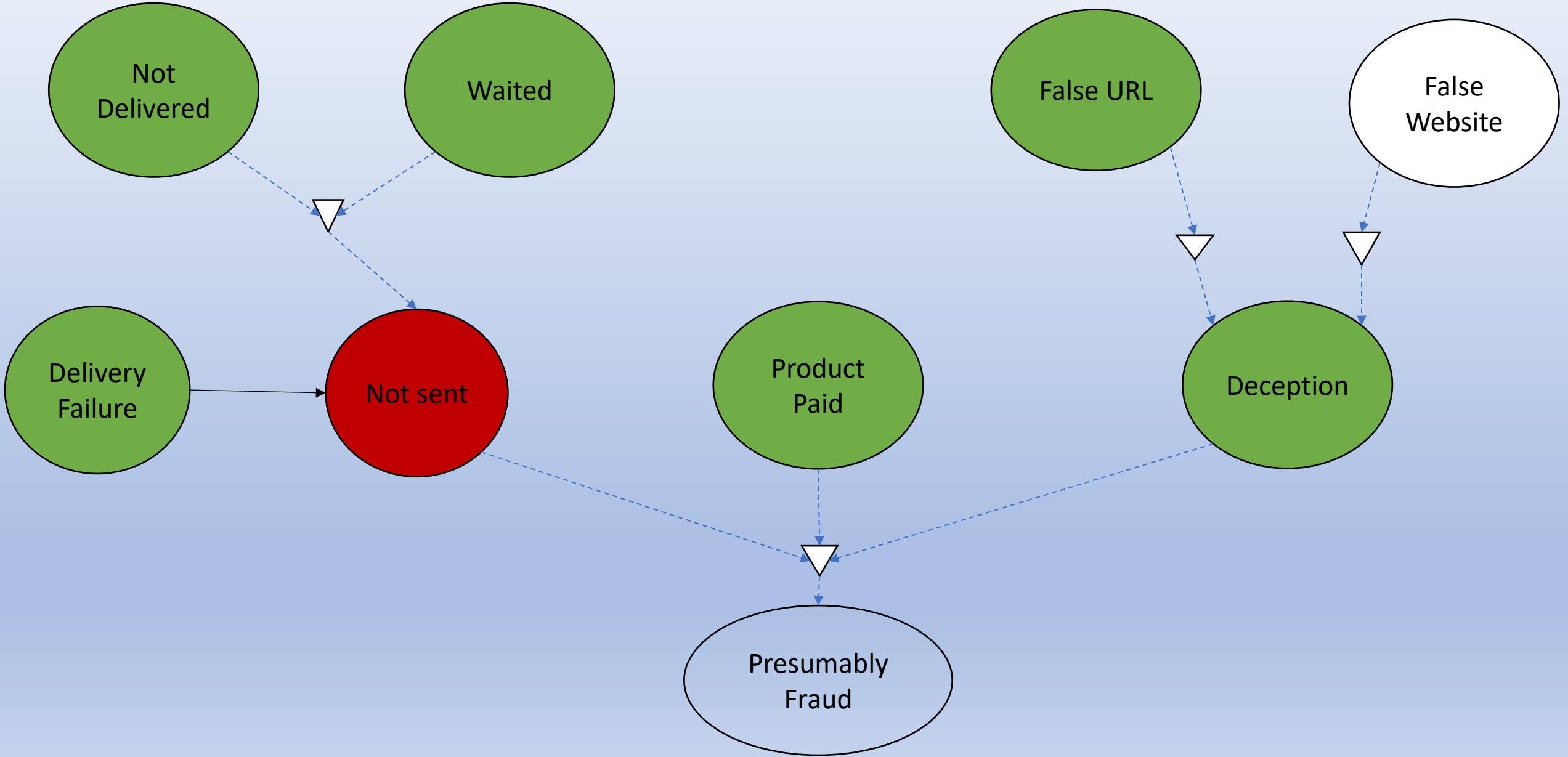

I’ve paid for this product that has not been delivered. The url is false as well.
Did the counterparty use a false website?
No.
How long have you waited since they said they would send the product?
Four weeks.

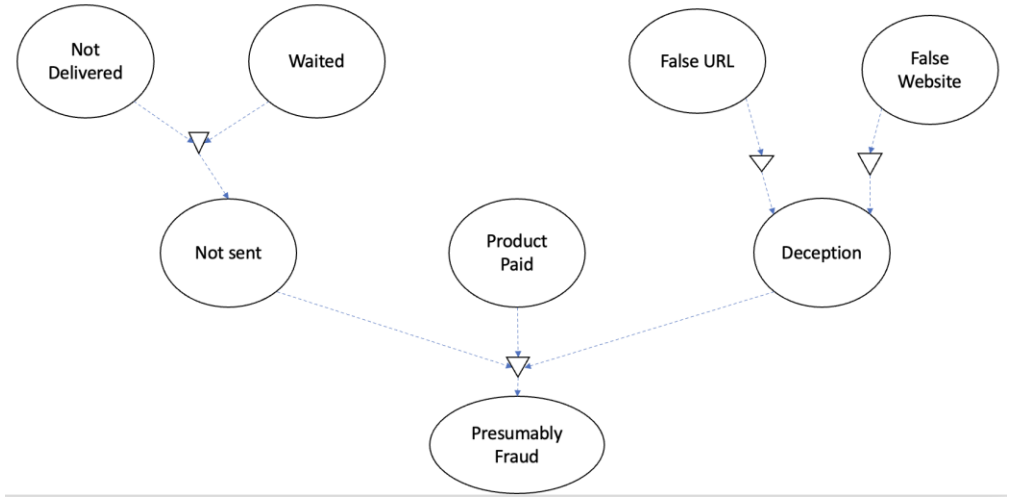

Not Delivered
Waited
Not sent
Product Paid
False URL
Did the counterparty use a false website?
No.
False Website
Deception
Presumably Fraud


How long have you waited since they said they would send the product?
Four weeks.
Did the counterparty use a false website?
No.
Not Delivered
Waited
False URL
False Website
Not sent
Product Paid
Deception
Presumably Fraud


How long have you waited since they said they would send the product?
Four weeks.
Did the counterparty use a false website?
No.
Not Delivered
Waited
False URL
False Website
Not sent
Product Paid
Deception
Presumably Fraud


Not Delivered
Waited
How long have you waited since they said they would send the product?
Four weeks.
False URL
Did the counterparty use a false website?
No.
False Website
Not sent
Product Paid
Deception
Presumably Fraud


Not Delivered
Waited
False URL
False Website
Delivery Failure
Not sent
Product Paid
Deception
Presumably Fraud



https://pyarg.npai.science.uu.nl/50-chat

Safe & Trusted AI

Humans & AI Systems

- Interaction & Communication
- Human-AI Dialogue
- Joint Reasoning

Argumentation

- Real-world Reasoning
- Justification for its claims
- Explainability & Transparency in Decision Making

Access to legal abortion improves the health and safety of pregnant people
so pregnant people should have the right to choose abortion

Argumentation for Explainable AI



Argumentation can translate the decision of an AI system in an argumentation procedure, which shows step by step how it concludes to a result



Given a set of possible decisions, the decisions can be mapped to a graphical representation, with predefined attack properties that subsequently will lead to the winning decision and will show the steps that were followed to reach it



Using argumentation for providing explanations makes an AI system friendlier and more trustworthy to the user as its reasoning mechanism for explaining its decision(s) is closer to the human way of thinking.



OHAAI

Thank you!

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